Inventory Control & Purchase Management

1. Inventory Control & Purchase Management-meaning & significance

Materials and purchase management is a body of knowledge which helps the manager to improve the productivity of capital by reducing the material cost, prevents large amount from capital turnover ratio.

Importance of Materials and Purchase Management in profit centre point of view:

Example:

I) Increase in profit by reducing material cost (total turn over Rs. 130 crore)

Subject	This year's cost (in Rs.,crore)	Next year's cost (in Rs. Crore)
Over head cost	10.00	10.00
Material cost	80.00	78.40 (2% reduction in material cost)
Labour cost	18.00	18.00
Total cost	108.00	106.40
Profit	12.00	13.60
Turn over	120.00	120.00

By reducing 2% on material cost, we get 13.33% extra profit than previous year.

II) Increase in profit by reducing labour cost.

By reducing 2% in labour cost, we get only 3% extra profit in compare to the previous case. In this case producer has to face tremendous labour problem which leads to stoppage of production.

Subject	This year's cost (in Rs.,crore)	Next year's cost (in Rs. Crore)
Over head cost	10.00	10.00
Material cost	80.00	80.00
Labour cost	18.00	17.64 (2%reduction in labour cost)
Total cost	108.00	107.64

Profit	12.00	12.36
Turn over	120.00	120.00

Materials productivity has a significant and direct effect on company's profitability (ROI) Rate of return on investment is the ratio of profit to capital which in turn for the purpose of analysis may be split up in to its basic constituents, profit margin turnover ratio.

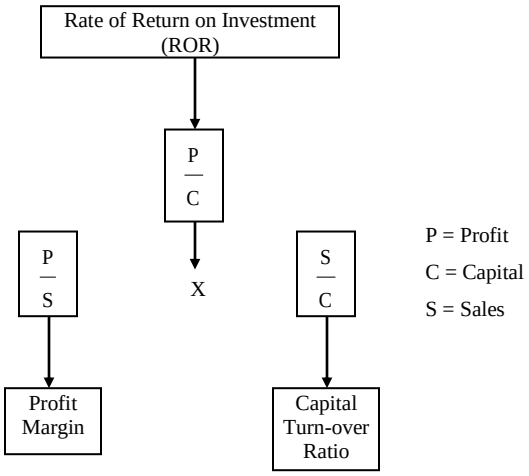


Fig: R.O.R and its constituents

Two basic ways to improve R.O.R are:

a) Increase profit keeping capital constant.
b) Reduce capital keeping profit constant.

a) Increasing profit keeping capital constant:

- Profit is the life line of an organization
- Profits give share holder its dividend
- Profits give employees the wages.
- Profit gives company to buy materials, machines, tools and other inputs.
- Profits provide greater job opportunities.

- Profits help the organization to improve quality of life and increase infrastructure.

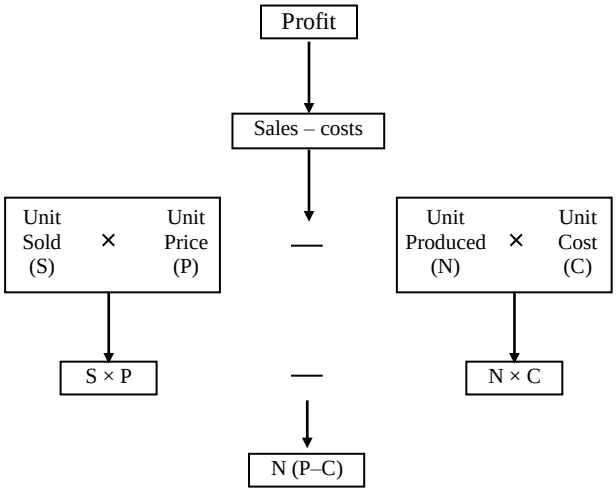
By taking following measures an organization can control and improve profit.

i) increase units sold (S)

ii) increase units produced.(N)

iii) increase unit price (P)

iv) reduce unit cost (C)



Increase units sold (S): Sales and production must be evenly balanced to ensure whatever is produced is sold, failing which the company will end up with unsold stocks.

i) Increased units sold (N): Production can be increased either by duplicating resources (adding another identical production line) or by better management of existing resources (by improving productivity).

ii) Increased unit price (P): Price is generally a function of demand and is controlled by the forces of supply and demand. A company operating in a buyer's market may find it impossible to make increase price. Thus leaving us with the second factor to deal with and that is cost.

iii) Reduce the cost (C): Reduction in cost is something which is within the control of the company. The total cost of a product consists of three main constituents: material cost, labour cost, and overheads, the material cost being the major portion of the total cost.

Materials in Indian context constitute more than half of the cost of production in most industries and projects. In some industries, 60-70% of the total production cost is due to materials. This makes the materials management the biggest single are having tremendous scope for cost reduction. A well coordinated materials management programme may lead to 15-20% of cost reduction.

A rupee saving n material cost is almost equivalent to ten rupees of sale. More over, sale is one time sale. Saving on other hand is a recurring benefit. Effective materials management, therefore, has a tremendous effect on the profitability of the firm.

b) Reduce capital keeping profit constant:

An alternative to increasing profit to improve ROR is to reduce capital. Capital can either be tied capital or working capital which in a manufacturing firm is generally tied up in the proportion of 60:40. the fixed capital –capital deployed in fixed assets like land, buildings, plant and machinery, jigs and fixture ,etc. –is fixed any how and hence a little can be dine about it . This leaves us with the working capital. The study of this portion of the capital reveals that a little over 80% of the working capital is locked up in inventory. (Raw materials, Work –In- progress, finished goods, spares, etc.) ,which can be reduced if proper materials management techniques (JIT, MRP, scientific inventory control) are employed.

Purchase objective and Functions

Some of the broad area of specific objective of purchase department are:

1) ***Economic purchase operation:*** Purchase is an economic function which should contribute to the profits of an organization so, it should have working knowledge in the field of economics, engineering, law, transport, imports, government, regulation policies.
2) ***Consistent Market survey:*** methodical collection, compilation, analysis and evaluation of collected data (primary and secondary) are vital in drawing definite conclusion.

3) ***Proper Control over financial commitments:*** Material bill and services occupy 50%- 60% of company's turnover so, proper control over financial commitment is very necessary.

4) ***To provide information and assistance to top management and other concerned department:*** forecast of price trends , market situation such as possible scarcity of materials , future investment commitments vs. working capital , are some the areas where management must be kept informed periodically and based on which certain purchase strategy and policy could be evolved.

Purchase Parameters

Or
Goals of purchase department

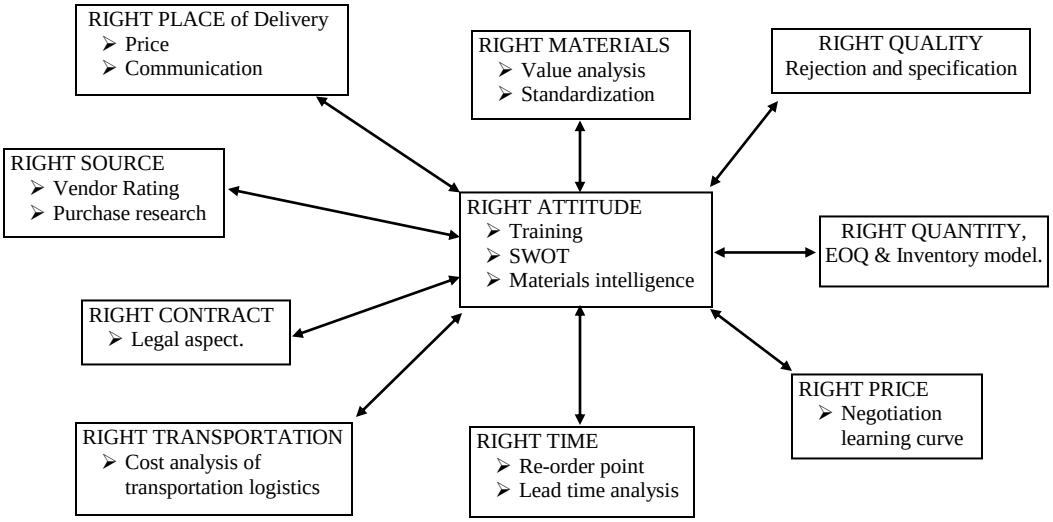


Fig: (Goals of purchase department)

Functions of Purchasing Department / Management

1) Recognition of needs: - it has to be anticipated in advanced that what materials should be needed and should buy in sufficient quantity.
2) Describe the need accurately by clear specification.
3) Selection of proper source of supply.
4) Research and Development for potential alternative materials.
5) Ascertaining or fixing price.
6) Placing purchase order.

7) Follow up the purchase order
8) Development of vendors and proper procedure.
9) Maintenance of good vendor relation.
10) Arranging transportation for material
11) Development of techniques of communication.

Purchasing policy and procedures

Policy: policy is a general guideline for decision making. An objective is the end; policy is the means to accomplish the end.

A policy can be defined as a statement either expressed or implied of those principles and rules that are set by the company as guides.

Procedures: purchasing procedures should be standardized in an organization with scope and provision of flexibility to accommodate any changes as for changes in the business environment and commercial changes.

E.g. Procedures of indent:-

- when to indent
- how to indent
- what are the responsibilities & authorities regarding scrutiny.

Should be clearly mentioned.

In this way procedures regarding material requisition, vendor's enlistment , enquiries, conditions of purchase order, follow up , inspection , delivery, guarantee & warranty , invoicing, bill passing, maintenance of records, performance appraisal of vendors should be clearly identified, expressed, and circulated.

Policy manual of materials with special emphasis on purchase

The conditions necessary for the success of materials function from the profit centre point of view, must be incorporated in the form of a manual.

The policy manual familiarizes the procedures and guidelines to be observed. Obviously the following list is only indicative and not exhaustive.

1) It defines responsibilities and authorities for purchase.
2) It intimates all concerned dept. about the systems and procedures of –
    - indenting for materials
    - procurement
    - Authorization
    - Do's and don'ts and extend
3) Buyer –vendor relationship with ethical standard.
4) Indicates the limit of purchasing power by different at different level.
5) Unambiguous policies and procedures.
6) Clearly indicate the policy and procedure to be followed for
    - indent
    - tendering
    - receiving quotations
    - Opening quotation
    - Vetting by finance and legal dept and concurrence of engineering dept. (if required) before placement of final purchase order or, better of indent
7) Promotes new ideas and suggestions for improvement of purchasing system.
8) Express the procedures of -
    - Source selection
    - Vendor development
    - Vendor's performance evaluation for rating and ranking purpose
9) Relationship to be maintained between purchase dept. and other concerned dept. for effective coordination.
10) Procedures to be followed for negotiation with vendors.
11) Follow-up and inspection system
12) Guarantees and warranties system
13) Procedures of passing bills and release of payment
14) Purchase record and record keeping
15) Legal aspects of purchasing

16) Management information system

Purchasing Systems

In organization, depending ON the size and nature of operation, the quantum of purchase varies any where between a few thousands of rupees and hundred and Crore of rupees.

Formal procedures have to be laid down in initiating purchase, selecting suppliers, placing purchase orders, follow-up, receiving materials and so on. In this chapter we will deal with various systems that are used in a purchase department.

We can classify the systems in the following manner:-

1) Pre-purchase system.
2) Ordering system.
3) Post-purchase system.

The salient features in each of the systems mentioned are as follows.

1) Pre-purchase system: initiating the purchase through requisition, requirement programmes, selection of suppliers, obtaining quotation and evaluating them are broadly the pre-purchase activities.

i) Requisition: the department concerned in need of a material usually presents a completed requisition form. It has to be countersigned by the senior officer. In any organization only a limited nos. of officers are empowered to countersigned the requisition as it amounts to authorization of the expenditure.

Note: the capital equipment cannot be procured in this manner. That decision are taken normally at the board level and they are treated differently for taxation and accounting purpose.

ii) Traveling requisition: This document is widely used for requisitioning items that are required frequently in bulk quantities over a long period. It travels from the requisitioning department to the purchase dept. often.

iii) Enquiries: Many organizations often invite suppliers to quote rates for supply of materials.

2) Ordering systems: Having selected the supplier and the rates agreed, the buyer places the purchase order on the supplier, expressing the term and condition.

a) **purchase order reference number** (which will be quoted in all subsequent follow-up measures pertaining to that order)
b) Description of the materials and detailed specification.
c) Quantity required and delivery schedule.
d) Price and discount.
e) Shipping instruction.
f) Location , where the materials are to be shipped (usually the name and address of buyer)
g) Signature of the material manager who can authorize the purchase order.
h) Detailed terms and conditions (as a common practice these are printed at the back of the purchase order.)

3) Post purchase system

a) follow-up implies commitment of time and money and therefore it has to be selective.

b) Receipt: A systematic record of the consignment received, carrier details and descriptions are to be maintained in chronological sequence.

c) Invoice checking: This step is concerned with checking invoice.

Principles of Purchasing / 6 Laws of Purchasing

6 R's of purchasing

i) Right quality: Quality can be defined as the "power of accomplish or the capability of doing certain thing"

- in other words quality means the useful value of a specific thing for a specific purpose to fulfill.
- Quality can be specified in different meaning, i.e., dimension, power, tolerance, harness, etc.
- Quality should be correctly specified.
- Right quality means neither too high nor too low.

ii) Right quantity: - purchaser must buy the materials in right quantity to ensure that there is no stoppage of production or, no extra stock piling. Right quantity is yet another important parameter in buying. Quantity decisions are influenced by "replenishment method "and "buying methods.

Replenishment method such as "re-order level", "two-bin system", "review system: optional replenishment" and "review system: compulsory replenishment" helps to provide broad guidelines.

Buying methods followed by the buyer has an influence on order quantity. For example re-order quantity under:

- ***Hand to mouth buying*** is too small**.**
- ***Scheduled buying*** can be either economic order quantity, or smaller or larger than EOQ.
- ***Forward buying*** is generally very large covering a long period of consumption.
- ***Contract buying*** is received in staggered lots, each lot at times may equal to EOQ.

Besides the above mentioned factors, consumption, market conditions, lead time, source of supply, etc influence the decision of right quantity.

iii) Right price: it is purchasers' skill to determine the optimum price for a valued product. it does mean that the lowest price but the price which minimizes the overall cost. Right price is not that easy to determine. The technique of :

- **Negotiation:** It is used when there are limited vendors, and /or time available to make purchase is short, and /or items belong to fixed price category.
- **Tender system:** It is followed in public sector organization to identify the lowest potential bidder.
- **Learning curve:** It is employed to determine the price of the item with high labor content.

iv) Right sources:- the right source must fulfill the following condition

- Must be fully equipped to manufacture and supply the items ordered.
- He must look after the interest and buyer interest.
- He must also assist the buyer in market research.

- Must be gentle in nature and improving himself to cope with the condition.
- He must be man of integrity.

v) Right time: Time is very essential components in case of any business. Right time and lead time are closely related. Right time implies that time at which the goods requested should be received while lead time refers the time that elapses between the communication of the need for the item by indenter to purchase till the item is actually received and is made available for consumption. Basic elements of lead time are:

- Time required by the indenter to communicate requirement to purchase.
- Time required by purchase to locate, select, develop qualified source of supply including agreement on contractual terms.
- Transit time for purchase order to reach supplier.
- Time required by supplier to fill buyers order.
- Time required by the supplier to route buyer's order through his administrative channels.
- Transportation time for the goods to reach buyers' destination.
- Time required by buyer's receiving department to collect materials from transporter's go down, verify receipted quantities and prepare necessary documents.
- Time required by the buyer's inward inspection to verify the quality of goods.
- Time required by the main stores to take possession of the goods, deposit them into appropriate bins and update stock cards.

vi) Right term: It concerned with legal term and condition.

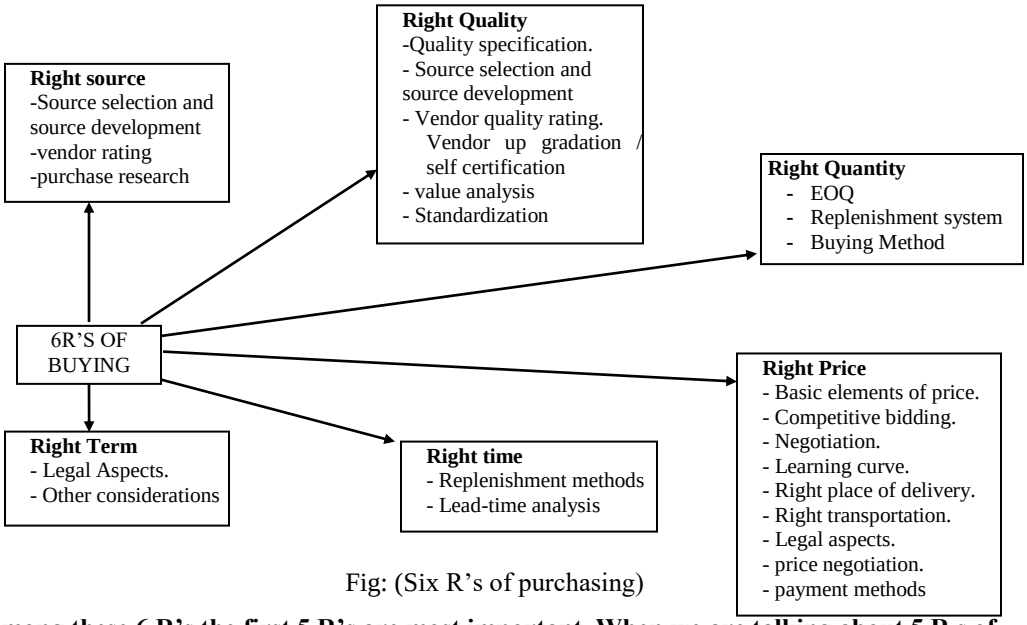


Fig: (Six R's of purchasing)

Note: among these 6 R's the first 5 R's are most important. When we are talking about 5 R s of purchasing we have to mention Right Quality, Right Quantity, Right Price, Right time and Right Sources

INVENTORY MANAGEMENT

Inventory: - inventory is a list or, schedule of article / materials held on charge of a person or, stock of articles and materials held-on behalf of an organization.

Inventory Management: - it means planning, procurement, holding & accounting and distribution of these article and materials.

Needs of Inventory Management:-

- to cope with uncertain conditions of availability
- to cater better consumption pattern.
- To counter act lead time
- To attend probable up-ward fluctuation in customer demand.
- Advantage while purchasing a lot.
- To cope with uncertainty resulting from national and international politics.

Main purpose of inventory management:

- What quantities are required?
- When should we order /reorder?
- What quantity should be ordered?

Types of inventory (According to materials)

1. Raw materials
the materials, components, fuel, etc used in the manufacture of the product

2. Work in progress
partly finished good and materials, sub-assemblies, etc. held between manufacturing stages

3. finished goods
Completed products, ready for sales or, distribution

Types of inventory (According to function)

1. Anticipating inventories
when a firm anticipating a rise in price, it may purchase in bulk. Quantities and hold the Same until price rise. In case of seasonal demand.

2. Fluctuating inventories
to cope with the fluctuation in Demand company always keep safety Stock.

3. Lot size inventories
Goods are bought in large lots to get the benefit of discount.

4. Transportation inventories
raw materials and finished goods are sent from one place To another. Some amount of inventory are always In transit.

PHASES COVERED BY INVENTORY MANAGEMENT:-

i) Maintaining continuity of production by providing continuous supply of materials.
ii) Providing satisfactory service to customer.
iii) Reducing the amount of working capital tied-up in inventory by proper planning.
iv) Ensuring that the laid down procedures are followed in inspection, recording storage etc.
v) Eliminate the wastage.

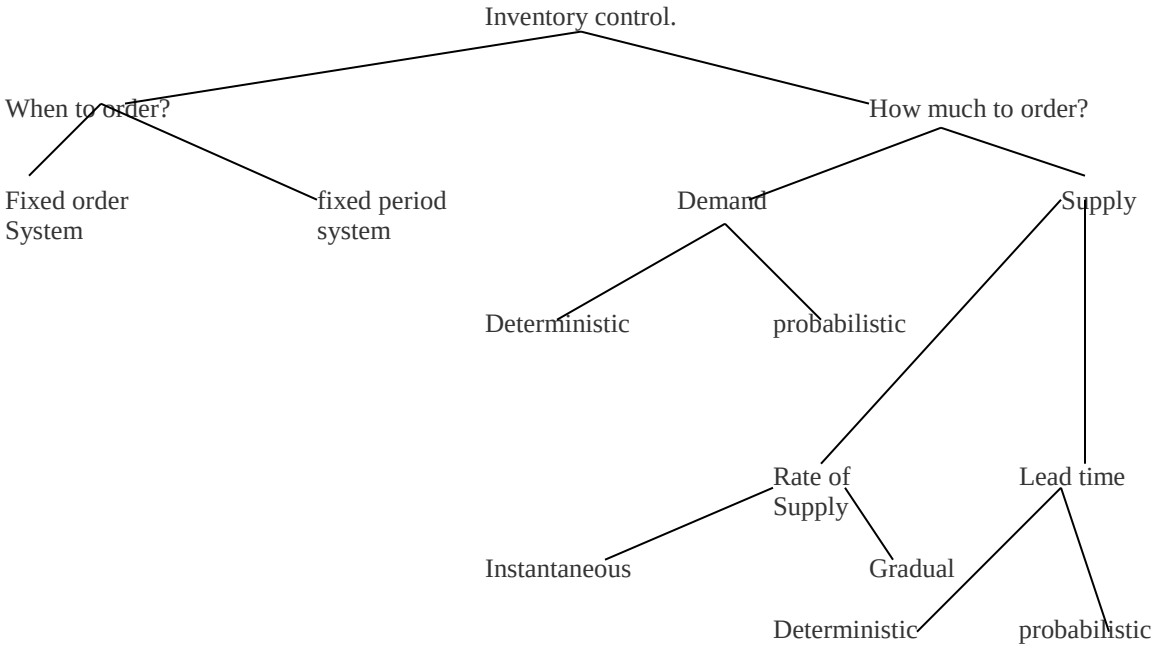
vi) Establishing timing schedule
vii) Ensuring the quality and specification of supplied materials
viii) To take care to avail the concession in purchasing material.
ix) Formulating inventory receipts, issue and storage procedures.
x) Requisitioning materials where re-order level is reached.

Inventory control:-

Inventory control is the process of deciding what and how much of various items are to be kept in stock.

Pre-requisites or, inventory control must fulfill the following conditions:-

i. Determine item to be stocked.
ii. Determine when and how much to replenish.
iii. Keep suitable records.
iv. Weed out obsolete items.



Objectives of Inventory control:-

1. To minimize capital investment.

2. To minimize losses due to obsolescence.
3. Cope with fluctuation of demand.
4. Ensure uninterrupted availability of materials.
5. Maintain records.
6. Provide scientific planning.

Advantages of inventory control:-

1. it improves the liquidity position of the firm.
2. it ensure smooth production operations by maintaining reasonable stocks of materials.
3. it facilitates regular and timely supply of materials and finished goods.
4. it protects the firm against variations in raw materials delivery time.
5. it facilitates production scheduling by avoiding shortage of materials.
6. it helps to minimize losses due to obsolescence.
7. it enables firms to take advantage from price fluctuation.

Types of cost:-

1) **Item cost (basic price)**:- it is the price of one unit.
   Components are –
   a) Direct material cost
   b) Direct labor cost.
   c) Direct expenses
   d) Overhead cost.
   e) Profit of the manufacturer.
2) **Ordering cost:-** This cost is associated with the placement of an order for the acquisition of inventories. Components are –
   a) Manpower cost – money spent in sending enquiries, receiving quotation, companies, placing and typing order.
   b) Finalizing orders and placing order cost.
   c) Transportation cost and stationary cost.
   d) Inspection cost.

3) **Inventory carrying cost or holding cost: -** it is defined as the cost of holding material inside and outside the stores. It is associated with level of inventories, components of inventory carrying out are –
    a) Interest on capital invested in the inventory.
    b) Storage space cost including rent, electricity.
    c) Handling cost.
    d) Insurance and taxes.
    e) Inventory maintenance cost.
    f) Obsolescence cost.
    g) Deterioration of quality cost.
    h) Cost of maintaining inventory records cost.
4) **Cost of shortage**: - it is an extremely important cost that never appears in accounting records.

ECONOMIC ORDER QUANTITY:-

The inventory control problem in which demand is assumed to be fixed and completely pre-determined are usually referred to as the EOQ. In the following diagram, if we plot Quantity to X-axis and Cost to Y-axis then we found inventory carrying cost curve or holding cost curve (pink in color) .it increases with quantity. If we plot ordering cost (as plotted in the diagram) we see that cost reduces with the increase of Quantity ordered. After adding different points of ordering cost and holding cost curve, we can obtain another curve which is known as total cost. In the production process there are two opposite costs, one encourages increase in order size and other discourages.

It is evident from following diagram that the minimum total cost occurs at the point where the ordering costs and inventory carrying cost are equal.

Different components of EOQ :-

D= Annual Demand. It is expressed in unit.

Q = Quantity to be ordered (units/cycle)

N = Nos. of times order placed for material. It is expressed in Nos. /year.

t = Time period between placement of orders. It is expressed in year/ month / week.

Co = Ordering cost, expressed in Rs. / order.

Ch = holding cost , expressed in Rs. / unit/ annum or percentage of total cost.

Cb= Basic price of item expressed in Rs./ unit.

Tc = total cost for inventory control expressed in Rs. (annual cost)

tl = lead time , expressed in months or weeks.

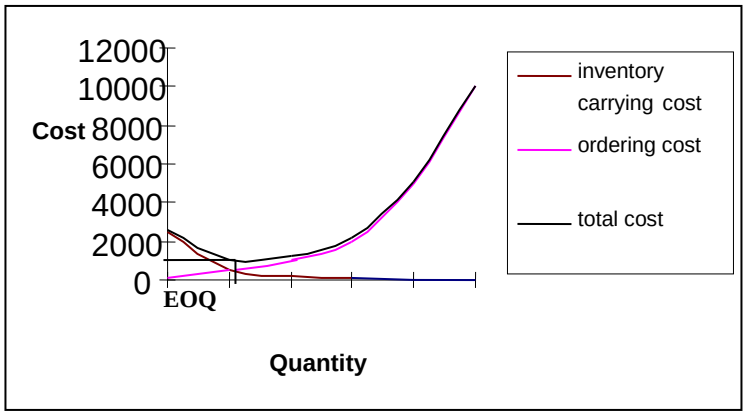
Bs = Buffer stock, expressed in unit.

ROL = the reorder is placed whenever the inventory level comes down to certain level.

This level is called ROL or ROP (Re Order Level or Re Order Point)

Derivation of EOQ :-

1) Annual ordering cost or total ordering cost.
   Toc = N*Co = (D/Q)*Co.
2) Total (Annual) holding /carrying cost.
   Thc = (Q/2)*Ch.
3) Since, the minimum to cost occurs at the point where ordering cost and inventory carrying cost are equal .
   So,
   Toc = Thc
   Or, (D/Q)*Co =(Q/2)*Ch
   Or, 2D*Co = Q^2 *Ch
   Or, Q^2 =(2D*Co)/Ch
   Or, Q* = √(2D*Co)/Ch



1) Optimum nos. of order placed = N* = D/Q* = D/(√(2D*Co)/Ch)
2) Total optimum time between two orders t* = Q*/D = (√(2D*Co)/Ch)/D
3) Total cost Tc* = (D/Q*) *Co + (Q*/2)*Ch +(D*Cb)
4) Tc* = [D/(√(2D*Co)/Ch)]*Co + [(√(2D*Co)/Ch)/2] *Ch

+ (D*Cb)

The inventory control problem in which demand is assumed to be fixed and completely pre-determined are usually referred to as the EOQ .
In the above diagram (graph), we plot quantity in x-axis and cost in y-axis. Then we find OC curve. This curve is inventory carrying cost curve or, holding cost curve. It increases with Quantity. Because, we know it consists of several cost which directly proportionate with Quantity of materials. If we plot ordering cost (as plotted in the diagram and denoted as AB), we see that cost have been reducing with the increase in quantity ordered. After adding different points of ordering cost and holding cost curve, we can obtain another curve which is known as total cost curve.
In the production process there are two opposite costs, one encourages increases in the order size and other discourages.
EOQ is that size of order which minimizes total annual (or, any other time period as specified by individual firm) cost of carrying inventories and cost of ordering.
It is evident from above that the minimum total cost occurs at the point where the ordering costs and inventory carrying cost are equal.

Establishing EOQ by problem:-

XYZ manufacturing Co. wants to determine the EOQ on the basis of the following information.

Annual usage = 5000 units.
Price/unit = Rs. 4/-
Carrying cost = Re 1/-/unit
Cost per order =Rs. 100/-

Solution

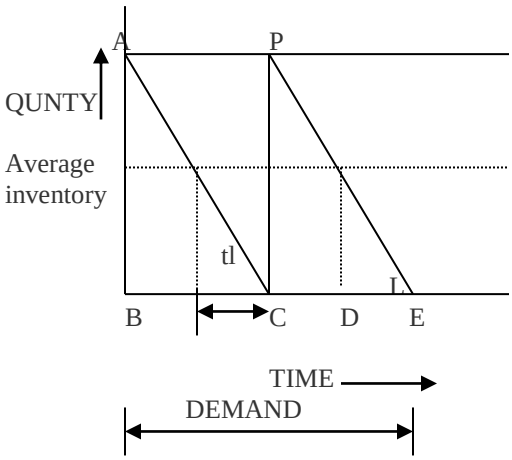
1. No. of orders/yr.	2. Order size (or) value of order	3. Average Inventory (2)/2	4.Carrying cost (Re.1/unit)	5. ordering cost @Rs.100/order (1. *100)	6. Total cost /yr. (4+5)
1	5000/1=5000	5000/2=2500	Re.1*2500=2500	1*100 =100	2500+100=2600
*5	5000/5=1000	1000/2=500	500	5*100=500	1000
10	5000/10=500	500/2/=250	250	10*100=1000	1250
20	5000/20=250	250/2=125	125	20*100=2000	2125
50	5000/50=100	100/2=50	50	50*100=5000	5050
100	5000/100=50	50/2=25	25	100*100=10000	10025

EOQ under Wilson lot size formula or, Simple situation formula or, EOQ model under infinite (instantaneous replenishment rate.

Assumption :-

1. the replenishment of stock is instantaneous (demand is fixed).

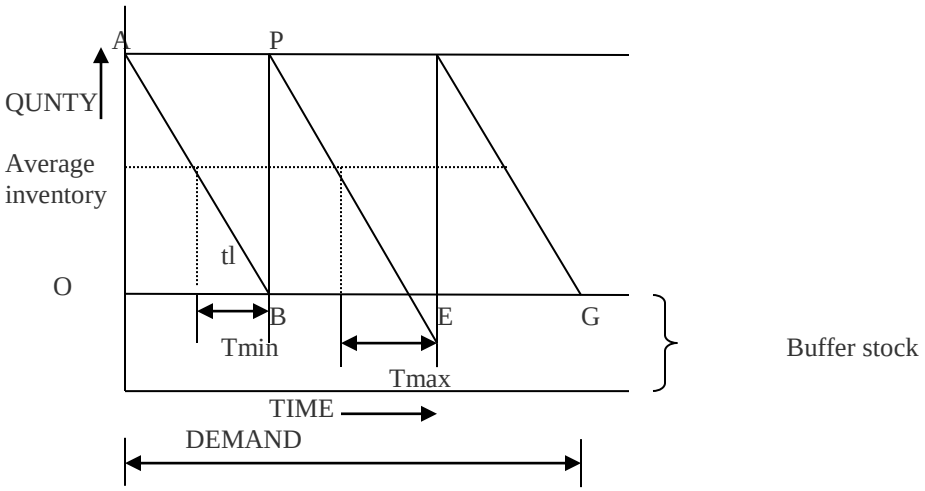
2. No shortage is allowed (no back ordering)
3. price per unit is fixed and is independent to the order quantity.



- In the above figure the materials are used in a constant rate and continuously.
- The annual consumption being D unit
- Manufacture placed the orders in such away that by the time the inventory is over, the consignment of quantity Q arrives.
- The lead time for the supply, tl time unit is constant.
- Since, lead time and consumption rate are constant, the reorder to the supplier is placed whenever the inventory level comes down to a level , called ROL (Re Order Level) .
- Here, ΔABC, ΔPCE is called inventory cycle.

Safety stock:-

- It implies the stock of inventory held as a safety measure against fluctuating in demand and lead time.
- Safety stock is the function of lead time, the greater lead time, the greater safety stock.
- Safety stock should be differentiated from working stock.
- Working stock is the stock which is generated from the orders.
- While determining the safety stock, reorder cost and reorder quantity should be considered. The cost of reorder and quantity to be reordered depend upon the following factors.
  - i) The minimum level: - the minimum level of inventory is decided by taking in to account usage value of the item, normal lead time, the availability of substitute.
    Below this level the inventory should not fall.
  - ii) The re-order point: - the reorder point should be the minimum level + safety margin, which is kept to ensure that shortage does not occur.
  - iii) The minimum level: - this level can be determined by adding up the minimum level of inventory and standard order quantity.
  - iv) Standard order quantity means EOQ.



$\overline{D}$ =Average Demand in lead time.

BS= Buffer stock = $\overline{D}$ *(Tmax-Tmin)= lead time demand

ROL =tl * $\overline{D}$ + BS

tl =(Tmax- Tmin)

Average Demand = $\overline{D}$ = Annual Demand/week or month

STEPS OF INVENTORY CONTROL:

1. List out all materials
2. Standardization / codification to verify reduction.
3. Prepare materials directory.
4. Analysis and classification of all items.
5. fixed up inventory level
6. Implementation.
7. Periodic review for upgrading the systems.

Selective Control of inventories and methods

When manufacturer divide materials according to cost , criticality, availability and consumption and then use different inventory control mechanism for each of the item in each type.

Suppose, according consumption value materials can be divided in to thee categories

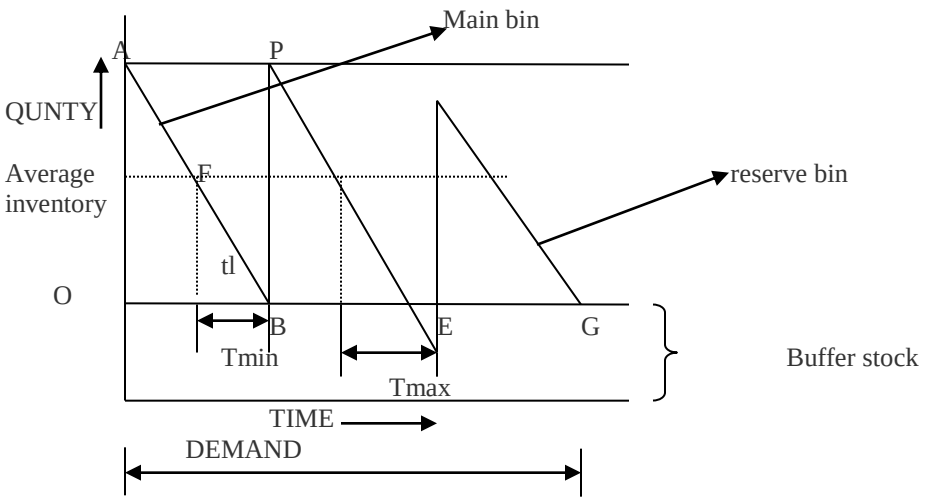
A – 70% of consumption value.

B- 20% of consumption value.
C- 10% of consumption value.
When, marketer takes different inventory control mechanism to control A, B,&C items, then it is called selective control of inventories.

Methods of selective inventory control:-

1. **Fixed order Quantity System (Q-System).**
2. **Fixed interval System or, periodic review System (P-System).**

1. **Fixed order Quantity System (Q-System):**
   - The order quantity is fixed.
   - The order is placed when inventory drops to re-order level.
   - It is most suitable when carrying cost is measurable and significant
   - It is preferred when the supplier places a minimum order quantity restriction.
   - It is suitable for A and X items having highest consumption value and highest unit cost respectively.
   - It is also known as ***two –bin system or maximum –minimum system,*** in this system re-order made when inventory falls on re-order level =lead time requirement +BS.
   - Further, the bin of an item is of two types, namely the **main bin and reserve bin.**
   - From main bin we meet the demand that occurs before lead time and from reserve bin we meet the demand during lead time.
   - When main bin is empty. Fresh order is placed which should arrive well before the reserve bin is exhausted.



$\overline{D}$ =Average Demand in lead time.

BS= Buffer stock = $\overline{D}$ *(Tmax-Tmin)= lead time demand

ROL =tl * $\overline{D}$ + BS

tl =(Tmax- Tmin)

Average Demand = $\bar{D}$ = Annual Demand/week or month

Tl, Tmax, Tmin = lead time

The above figure shows:

1) Supply equal to EOQ is received at point O and quantity in stock reaches a point A.
2) The materials are then issued and at time F, when the stock reaches the ROL, an order is placed for quantity Q=EOQ, and supply continued.
3) At point B, the supplies of order placed at point F are received and the stock reaches P.
4) At the point E there is delayed in receiving the supplies and we cut in to Buffer stock.
5) Thus for this system of ordering , we fixed up the size of the order i.e., every time same quantity 'Q' is ordered but the time of placing the order is allowed to vary depending upon the actual usage or demand.

In this case, the firm has to decide on three aspects of an item before placing the order.

a) EOQ=√2DCo/Ch.
b) Optimum Buffer Stock
c) R.O.L

2. Fixed interval System or, periodic review System (P-System).

This is also known as the FIXED-ORDER INTERVAL SYSTEM or CYCLIC REVIEW SYSTEM.

1. the reorder time is fixed.
2. the re-order quantity varies according to inventory on hand.
3. it is suitable when carrying cost in meaning less and insignificant.
4. it is preferred when the supplier will only ship at fixed date.
5. it is suitable for B and C class items.

This system has a *fixed ordering interval* but, the size of order quantity may vary with the fluctuations in demand. The system is specified for any item by

i) Review period and
ii) Replenishment level (R)

The operating procedure consists of review the inventory position regularly. once in every t unit of time . At each review period, an order is placed for an amount equal to difference between a fixed replenishment level and actual inventory level . the calculation of R is based on the formula –

R= Avg. consumption during review period + lead time +BS.

Since, the order quantity would be large than usual. When the demand has been less than the expectation.

The order quantity is variable in size from one review date to another date.

LIMITATIONS OF EOQ MODEL:-

1. **The demand for inventory is seldom constant:-** when demand fluctuates, the EOQ model will give misleading results. In a period of rising demand , EOQ model based on historic demand level will suggest smaller inventory levels than are economical.
2. **The lead time for any supplier is generally unpredictable:-** therefore buffer stocks are required to insure against changes in lead time. It is difficult to determine buffer stock as it depends upon uncertainty in the lead time.
3. **it is very difficult to determine carrying cost:-** only a rough estimate can be made of obsolescence and deterioration costs.
4. **The EOQ formula is based on assumption that no stock out will take place:-** in some cases an occasional stock-out position may be less costly than carrying excessively large stock, but, it is not easy to determine the cost of stock out.

SELECTIVE CONTROL OF INVENTORIES / CLASSIFICATION/ ABC,VED, GOLF, FSN ….ANALYSIS

Selective treatment of inventories is based on the following basic philosophy of business :- *"neither one can control everything nor one should try to do so (even if one can). Uniform control is rarely effective . Effectiveness results when important aspects of a problem are pursued more rigorously than others. A major portion of managerial time should be spent in performing more important jobs. Less important tasks- those involving routine decisions and which involves less risk – should be delegated to a lower level .*

this law can effectively be applied in inventory function so to identify items which are more important than others. The classification enables managerial time being spent according to the importance of the item. Selective control is essential since uniform control of all items:-

1. is expensive
2. gives diffused effect.

Selective control means variations in method of control from item to item based on selective basis. The criterion used for the purpose may be cost of item, criticality, availability, consumption, etc. various classification are employed to render selective treatment to different types of materials, each classification emphasizes on a particular

aspect. For example ABC analysis emphasizes on usage value (i.e., consumption of the item in terms of money), VED analysis considers criticality, SDE analysis is based on procurement difficulty or availability.

Analysis criteria	Classification
1. COST	**a) ABC (annual usage value of an item expressed in monetary value. i.e., annual quantity * price /unit)** **ABC stands for Always Better Control.**
	b) HML (it is based on unit cost of materials). **HML stands for –** **H- High unit cost.** **M-Medium unit cost.** **L-Lower unit cost.**
	c) XYZ (it is based on stock value)
2. CRITICALITY	**VED items (it is based on criticality in usage)** **VED stands for –** **V- Vital** **E- Essential** **D-desired**
3. PROCUREMENT DIFFICULTIES/ AVAILABILITY /SEASONALITY	**SDE (it is based on procurement difficulties and availability)** **SDE stands for** **S- Scare item , which is not easily available in the country.** **D- Difficult items.** **E- Easy available items.**
	S-OS analysis. **S-Seasonal.** **OS- Off –seasonal**
4. SOURCE OF PROCUREMENT	**GOLF analysis (it is based on source of procurement or supplier)** **G- Government.** **O- Ordinary** **L- Local** **F – Foreign**
5. CONSUMPTION	**FSN analysis (it is based on movement of the item)** **F- Fast moving goods.** **S- Slow moving** **N- Not moving item**

ABC (ALWAYS BETTER CONTROL) ANALYSIS

ABC analysis is basic analytical management tool which enables management to place the effort where the result will be greatest. This techniques, popularly known as **A**lways **B**etter **C**ontrol.

Characteristics of ABC items:-

A – no. of items less than 5-10% of total items but, consumption value to procure these materials 70 -75% of total amount of procurement. These items require detailed and rigid control and need to be stocked in similar quantities. These items should be procured frequently, the quantity per occasion being small.
B – These items are generally 10-15% of the total items and represent 15-20% of the total expenditure on the materials. These are intermediate items.
The control on these items need not be a as detailed and rigid as applied to A items.
C – these are numerous (as many as 70-80% of the total items), inexpensive (represent hardly 5-10% of the total annual expenditure on materials) and hence insignificant(do not require close control) items.
The procurement policy of these items is exactly reverse of A items. C items should be procured infrequently and in sufficient quantities.

OBJECTIVE OF ABC ANALYSIS

1. It enables materials manager to exercise Selective control when he is confronted with a large no. of items.
2. Degree of control should be rigorous for A and minimum for C.
3. The process should be accurate for A.
4. It also helpful to rationalize the number of orders and reduce the overall inventory.

Example: - 3 items accounting for a consumption of Rs. 60,000/-, Rs. 4000, and Rs. 1000/- respectively. Each being ordered four times a year. The following pattern would then emerge.

Item no.	Annual consumption value	Nos. of orders	Value per order	Average inventory.
1	60,000	4	15,000	7500
2	4000	4	1000	500
3	1000	4	250	125
Total inventory				8125

If ABC analysis is done , then it implies that ‘A’ item should be ordered more, say 8 orders, 3 in case of ‘B’ and 1 for ‘C’ item.

Item no.	Annual consumption value (Rs.)	Nos. of orders	Value per order (Rs.)	Average inventory. (Rs.)
A (due to highest consumption value)	60,000	8 *	7500	3750

B	4000	3	1333	667
C	1000	1	1000	500
Total inventory				4,917

(*)it is made arbitrarily.
If it is made according to formula then it will reduce Average inventory cost.

Item no. (1)	(2) Annual consumption value (Rs.)	(3) = √ (2) or = √Annual consumption value	(4) Nos. of orders = Total orders*{ item wise (3)/Total value of (3)}	(5) Value per order (Rs.)	(6) Average inventory. (Rs.)
A	60,000	√60,000=244.9=245	12*(245/340)=9	6667	3334
B	4000	√4000=63.2=63	12*(63/340)=2	2000	1000
C	1000	√1000=31.6=32	12*(32/340)=1	1000	500
		Total =340	Total orders = 12		4834

Comparison between A, B and C items

Characteristics	**A items (high consumption value)**	**B items(moderate consumption)**	**C items (low consumption value)**
1. Control	Strict	Moderate	loose
2. inventory control System	Fixed order quantity System.	Fixed interval or periodic review system	Same as B
2a. Control Report	Weekly	Monthly	Quarterly
3. Safety stock	Very low or no	low	High
4. ordering	Frequent	Once in three	Bulk, once in a 6 months.
5.Follow-up	Weekly	Monthly	Quarterly
6. Value analysis	Rigorous	Moderate	Minimum
7. Sources	As many as possible	Two or more	Two source for each item
8. Forecast	Accurate	Based on past data and present plan	Rough estimate for planning.
9. Review of waste, obsolete and surplus.	After 15 days	Quarterly	Annual
10. Authority of purchase and storage	centralized	Combination	Decentralized
11. Effort to reduce lead time	Maximum	Moderate	Minimum
12 handling Authority	Senior Authority	Middle Management	Fully Delegated

Steps of Conducting ABC analysis

i) Obtain unit cost of each manufactured or purchased item in inventory.
ii) Obtain the usage in units for each item or estimate the usage (future) over a period of time).
iii) Obtain the net value of the usage by multiplying the unit cost and the usage (unit cost *no. of units)

iv) Average the items in descending order of the usage value.
v) The number of items and their values are accumulated on a percentage of total basis.
vi) Roughly divide the list of values in to three (3) groups-A –accounts for 70-75% of total account.; B- which account for 10-15%and rest is C item- accounts for 5-10% .

V – E – D ANALYSIS

While in ABC analysis items are classified according to their consumption value, in VED classification they are classified according to their criticality.
V – Vital item – are those items whose no availability can cause havoc and can stop the wheel of organization.
E- Essential – are those item whose non availability causes temporary losses/dislocation of production.
D- Desired – which are necessary but their non-availability does not result in immediate loss of production or stoppage.

Relationship between ABC and VED item

	V items	**E items**	**D items**
A items	Low stock and constant follow –up and control	Medium stock and no follow-up	Nil stock
B items	Medium stocks with little follow-up and control	Medium stock	Low stocks
C items	High stocks	Medium stocks	Low stocks

F-S-N ANALYSIS

F-S-N analysis is based on the consumption figures of the items. The items under this analysis are classified in to three groups : F- Fast moving , S – Slow moving , and N – non- moving.
To conduct the analysis, the last date of receipt or the last date of issue whichever is later is taken into account and the period, usually in terms of number of months, that has elapsed since the last movement is recorded.
Such an analysis helps to identify:

i) active items which requires to be reviewed regularly.
ii) Surplus items whose stock are higher than their rate of consumption;
iii) Non-moving items which are not being consumed. The last two categories are reviewed further to decide on disposal action to deplete their stocks and thereby release company's productive capital.

Further detailed analysis is made of the third category in regards to their year-wise stocks and items can be sub-classified as non-moving for 2 years, non-moving for 3 years, non moving for 5years and so on.

X-Y-Z ANALYSIS

X-Y-Z analysis is based on value of the stock on hand.(inventory investment). Items whose inventory values are high are called X item, while those whose inventory values are low are called Z items and Y items are those which have moderate inventory stocks.

Usually X-Y-Z analysis is used in conjunction with either ABC analysis or HML analysis.

XYZ analysis when combined with ABC analysis is used as under:

Class of items	A	B	C
X	Efforts to be made reduce stocks to Z category	Efforts to be made to convert them to Y category	Steps to be taken to dispose off surplus stocks
Y	Efforts to be made convert these to Z category	*	Control may be further tightened
Z	*	Stocks levels may be reviewed twice a year	*

X Y Z analysis when combined with F S N analysis helps to formulate more specific strategies as under:

Class of items	F	S	N
X	Tighten Control	Deplete the stock to very low level	Disposed off immediately at optimum price.
Y	*	Deplete the stock further at good price	Disposed off as early as possible.
Z	Liberalize control (to reduce clerical cost)	*	Dispose off as early as possible even at lower price.

(* items are within control. No further action is necessary.)

XYZ, therefore helps to identify a few items which account for large amount of money locked up in stock and take steps for their liquidation/ reduction.

XYZ when combined with FSN analysis helps to classify non-moving items into XN,YN,ZN group and thereby identify a handful of non moving items which account for bulk of non –moving stock. These can be stidied individually in details to take decision on their disposal or retention

Just In Time (JIT) Concept

i. This is an acronym for Just In Time or Toyota product system and is also known as Zero Inventory System.
ii. The Zero Inventory system does not mean literally nil inventory , small amount of inventories are maintained ,which is required to sustain production activity between two operation in an organization.
iii. This philosophy initiated in Japanese industry for industrial growth and captured a major share of world market.
iv. The aim is elimination of all waste, being anything not essential such as scrap etc.
v. Production part, component or sub assembling must occur Just In Time, not earlier or not later.
vi. This is accompanied by a "pull" strategy instead of "push" strategy.
vii. "pull system works without the need of costly and integrated procedures, thereby minimize the WIP inventory.

Competitive study between Traditional buying and JIT

Characteristics	Traditional	JIT
1. Supplier /Buyer Relationship	Adversarial	Partnership
2.Contract Period	Short or long term	Long term
3.Communication	Written purchase order	Verbal or telephonic
4.Quality	Inspection	No inspection
5. Quantity	Bulk/large	Small
6.No. of supplier	Many	Few
7. Design process	Print then quote	Ask for supplier ideas.
8. Ordering cost	Negotiation+ open order cost + expediting cost +Receiving cost + Inspection cost + Transportation cost.	Negotiation + transportation cost.
9. lead time	High	Low
10. Transportation	Full truck of single item	Full truck with multiple item

JIT WILL ONLY WORK IN COMPANIES WHERE :

1. The top management knows the price.
2. Company is looking for confidentiality rather than competitiveness.

Kanban-an Integrated JIT System

INTRODUCTION

Japanese are good at manufacturing. Just ask any global producers of automobiles, copiers, or personal electronics what happened in the 1980s. They will probably tell you how the Japanese captured a large share of the global-market by creating world-class standards in design, materials, and management. What is often overlooked is the attempt to understand how the Japanese industry succeeds at the services that support the manufacturing process (Krajewski et al, 1987: 40). Within the production field, the Kanban process is the most significant of these services.

The concept of time-based management is nothing new for managers outside of Japan and has been in practice for many years. However, the Kanban process involves more than just in time deliveries and inventory control. Briggs (1993: 29) notes that Kanban process components are the most 'exportable' of Japanese techniques, but the complete process itself has not yet been successfully adopted outside Japan.

THE ORGANIZATION OF THIS REPORT

This report will focus on the interlinked components and features which constitute the Japanese Kanban process of time-based management. In addition, it will examine the potential for the successful implementation of the process into Australian manufacturing firms. Experience from the adoption of Kanban theories in North American manufacturers will serve as the foundation on which the Australian case is built upon.

THE KANBAN

The Japanese refer to Kanban as a simple parts-movement system that depends on cards and boxes/containers to take parts from one work station to another on a production line. Kanban stands for Kan- card, Ban- signal. The essence of the Kanban concept is that a supplier or the warehouse should only deliver components to the production line as and when they are needed, so that there is no storage in the production area. Within this system, workstations located along production lines only produce/deliver desired components when they receive a card and an empty container, indicating that more parts will be needed in production. In case of line interruptions, each work-station will only produce enough components to fill the container and then stop (Roos, 1992: 112). In addition, Kanban limits the amount of inventory in the process by acting as an authorization to produce more inventory. Since Kanban is a chain process in which orders flow from one process to another, the production or delivery of components are pulled to the production line. In contrast to the traditional forecast oriented method where parts are pushed to the line (Roos, 1992: 113).

The Kanban method described here appears to be very simple. However, this "visual record" procedure is only a sub-process in the Japanese Kanban management system.

ADVANTAGES OF THE KANBAN PROCESS

Roos (1992: 115) notes the following advantages of Kanban over the traditional push system:

- 1 A simple and understandable process
- 2 Provides quick and precise information
- 3 Low costs associated with the transfer of information
- 4 Provides quick response to changes
- 5 Limit of over-capacity in processes
- 6 Avoids overproduction
- 7 Is minimizing waste

 8 Control can be maintained
 9 Delegates responsibility to line workers

He further indicates that "Kanban represents an efficient tool to continuously rationalize the production process and find the source of problems" (Roos (1992: 115). Since the circulation of Kanban will stop if there is a production problem on line, it is easy to both spot and correct the problem instantaneously.

THE KANBAN PROCESS- MORE THAN INVENTORY CONTROL

To managers outside of Japan, Kanban may look only like a pure production method having little or nothing to do with the surrounding environment. This is a fallacy. Instead, the concept takes form on the shop floor, in close interaction between the work force and management, and more importantly, involves both internal and external customers. Kupanhy (1995: 62) identifies Kanban as a production system which draws many of its elements from two primary sources: industrial re-engineering, and work force (Japanese) Kanban management.

2-3 INDUSTRIAL RE-ENGINEERING AND KANBAN

Industrial re engineering which goes hand in hand with Kanban consists of elements such as:

 1 Modular/cell production. Flow-of-products-oriented layout of processes and machines layout.
 2 U-shaped production/processing lines
 3 Total preventive maintenance
 4 Mass production of mixed models

The interrelationship between the Kanban concept and industrial re-engineering is clear.

Modular/cell manufacturing, which is sometimes referred to as group technology involves organizing machinery so that related products can be manufactured in a continuous flow (Kupanhy, 1995: 62). Here, products flow smoothly from start to finish, parts do not sit waiting to be worked on, and forklift trucks do not travel kilometers to move parts and materials from one part of the plant to another. This can be contrasted to a typical production system, where machines are grouped by function and products move from function to function from one end of a facility to another and back again. This results in long waiting times between procedures. Kanban will not work effectively without efficient logistics systems and process-oriented plant layouts. Kanban controlled production and the Kanban itself must be able to flow smoothly between processes (Kupanhy, 1995: 63). Modular/cell manufacturing can be realized by U-shaped processing lines, which integrate the manufacturing process into a continuous flow and increase supply accessibility to the lines. It would be impossible to join different processes to form a U-line if processes are not integrated. In addition, Total Preventive

Maintenance, which prevents machines from breaking down or malfunctioning during the production time, also contributes to the efficiency of Kanban.

Toyoda Gosei Co. advisor Taiichi Ohno, architect of the Toyota Kanban system believes that the real benefits of Kanban probably will not be realized until the auto industry moves into a mixed production mode in which modular production methods are employed (Schreffler, 1987: 74).

2.Purchasing & procurement:
Principles of sourcing, purchase methods & procedures, legal aspects of purchasing. Reference to Contract Act, Sale of Goods Act, Drug Control Act in respect to purchase activities. Import substitution

Purchase methods & procedures

Methods of Buying

Introduction

The buying department of the company is responsible to provide goods and services required by the company at least cost to the company. The request to procure may be received either from stores department or from one of the functional departments.

Such requests may be received:

i) *Either for direct materials or for indirect materials;*

ii) *Either for production items or for non-production items;*

iii) *Either for low priced items or for expensive items;*

iv) *Either for the items which are controlled by the forces of supply and demand or for the items which are available off the self;*

v) *Either for items to be procured from manufactures or for items to be bought from middlemen;*

vi) *Either for the seasonal items or for the non-seasonal items;*

vii) *Either for items produced to buyer's design or for items produced to commercial standards;*

viii) *Either for items sold at premium or for items sold at discount;*

ix) *Either to meet immediate needs or it may be to satisfy needs at a later date;*

x) *Either at time when prices are at their peak or at a time when prices are stable;*

The above discussion, therefore, implies that there can be wide variations in the practices to be followed for the purchase of different types of items. This chapter deals with such practices (i.e. different buying methods)

Factors Influencing Selection of a Buying Method

A number of factors influence selection of a buying method. They are:

i) Nature of the item;
ii) Regularity of its demand;
iii) Quantities required;
iv) Susceptibility to price variations

Some organizations decide upon different methods of buying on the basis of market condition where decision may be –

1) Hand to mouth buying.
2) Speculative buying
3) Hedging
4) Forward buying / Market purchasing

Apart from these main buying methods there are other buying methods.

5) Scheduled buying.
6) Contract buying
7) Blanket orders.
8) Tender buying.
9) Seasonal buying.
10) Group purchasing.
11) Sub-contracting.
12) Central purchase organization.
13) Directorate general of supplies and disposal.

Different types of Buying Methods

1. Hand to Mouth Buying

Hand to mouth buying also called **"buying according to the requirements"**. It refers to the frequent purchases of an item in small quantities.

Important characteristics of hand to mouth buying are undertaken:

a) When demand arises.
b) To cover immediate requirements.
c) When the quantity purchased is small.
d) When there is emergency or urgency.

Advantages:

i) Inventory investment Low.
ii) Carrying charges is Low.
iii) Losses will be lesser when price declines.
iv) Reduced deterioration and obsolescence of materials.

Disadvantages:

i) Higher clerical costs.
ii) In emergency case quality may be degraded.
iii) Increase in prices may lead to loss.
iv) In urgency buyer may have to buy the materials with high price.
v) Production schedule may be interrupted for the shortage in the time of urgency.

Criteria:

This method applies to:

a) Items whose prices are expected to fall in the near future;
b) Items which are perishable;
c) Items required for under development products in product development phase;
d) Items which are used infrequently and would not be required to stocked.
e) cover immediate requirements of a stock items caused either due to delay in delivery from regular supplies or due to increase in consumption;
f) Replacement spares;
g) bulky Materials

1) Speculative buying

Definition: *Speculative buying refers to the buying large requirements of an item when its price is low with the intention to sell bulk of it at a higher price for speculative profits.*

Characteristics:

i) An item which is not required for production may be purchased.

ii) Its single aim is to make speculative profits when price of the commodity increases.

iii) The quantity purchased is high .

Advantage:

Earning speculative profits.

Disadvantages:

- Investing large amount of capital,
- Storage problem.
- Risk of obsolescence.
- Inventory holding cost higher.

3) Hedging

One of the techniques used to minimized the risk associated with fluctuation of prices is Hedging.

Hedging is used to protect an organization , or business from an open exposure in the foreign exchange market. By using hedging, the trading party is protected from loss, - the risk of advance changes in the price of an asset.

Hedging is used by both traders to eliminate foreign exchange risk in international and by multinational corporations to avoid foreign exchange risks in the translation of their financial statement into dollars.

In order to protect their business traders have got two options **(i) hedging in Spot exchange rate and (ii) hedging in Forward exchange rate.**

(i) Hedging in Spot exchange rate

If the contract with the customer to buy or sell foreign currency is agreed upon and executed immediately it is known as **spot transaction** and the rate quoted is the spot rate. **Example**: purchase of an export bill. Here the customer tenders the bill for which the bank quotes a rate . if the rate quoted by the bank is acceptable to the bank , the bill is purchased by the bank and the account of the customer is credited with the amount.

(ii) Hedging in Forward exchange rate.

A "future" transaction takes place where the purchase is made now at the quoted price, while the actual delivery will take place at a future date to be specified in the contract.

Forward rate:

The N-day *forward rate* is the rate which appears in a contract to exchange a currency for another N days in the future. It is distinguished from the *spot rate*, which is the rate used in agreements to exchange one currency for another immediately. No currency changes hand between the parties in a forward contract at the time it is signed; the currency is exchanged at the maturity date of the contract N days in the future.

A forward exchange contract or simply a forward contract is one where a banker and a customer or another bank enters into a contract to buy /sell a fixed amount of foreign currency at a specified future date at a predetermined rate of exchange. the rate quoted for the transaction is called **forward rate.** Example: Bank of India agrees to buy one month forward US$ 1million from Bank of Baroda. That means the delivery of foreign exchange by Bank of Baroda to Bank of India will take place after one month from the date of contract

Forward premium (forward discount):

A forward premium (forward discount) is the proportion by which a country's forward exchange rate exceeds (falls below) its spot rate. Premium and Discount Forward Rate may be the same as the spot rate . then it is said to be ***'at per'*** with the spot rate. But, it rarely happens. More often the forward rate may be costlier or cheaper than the spot rate .

The difference between forward rate and spot rate is known as **'forward *margin'*** . The forward margin either may be either ***premium*** or ***discount.***

When the foreign currency is cheaper under forward rate than under the spot rate , the currency is said to be at a ***premium.*** Under direct quotation , more Indian rupees will be needed for a unit of foreign currency under the forward rate than under the spot rate. **Therefore, premium is added to the spot rate to arrive at the forward rate.**

Under indirect quotation , for a Liven unit of Indian rupees the customer would get lesser units of foreign currency than he would get by applying the spot rate. **Therefore premium is deducted from the spot rate to arrive the forward rate.** *The addition and deduction of premium is applicable both for purchase and sell transaction.*

When the foreign currency is cheaper under forward rate than under the spot rate , the currency is said to be at a discount . Under direct quotation , fewer Indian rupees will be needed for getting unit of foreign currency under the forward rate than under the spot. . **therefore , discount is deducted from the spot rate to arrive at the forward rate .**

Under indirect quotation, for a given unit of of Indian rupees, the customer would get more units of foreign currency than he would get under the spot rate. **Therefore, discount is added to the spot rate to arrive at the forward rate.**

Determinants of the forward premium:

The forward premium (or forward discount if the number is negative) is determined by the interest rate differential between the United States and Canada. According to the Interest Rate Parity theorem, the expected appreciation of the Canadian Dollar is equal to the difference between the U.S. and Canadian interest rate. For example, if the interest rate in Canada is one percent higher than in the United States, over a period of one year the Canadian Dollar will tend to depreciate by one percent. Note that a raise in Canadian interest rates first lifts the exchange rate, and only then the CAD starts to depreciate. The

interest rate differential is based on comparable assets (with risk premium already factored in), for example, Canadian and U.S. 90-day or 1-year treasury bills.

Condition of forward rate :

Insurance against exchange rate risk can be obtained through contracts in the forward market. Such activity is called hedging. A hedge is the offset of a given position in a separate bu parallel market by an equal and opposite position in which the effect of the offset reduces or eliminates the effects of a value change in both positions. In simple terms, a hedge locks in the current value in a contract. The instrument for a hedge is often a *currency swap* in which a spot contract is offset by an equal-amount forward contract.

ILLUSTRATION :-Dollar is quoted in the inter bank market as follows: (simulated value)

Spot US$ 1=Rs.31.35 -31.40

1 Month forward 20 /30 pm

2 Month forward 40/50 pm

3 month forward 60/70 pm

Calculate 1month , 2Month and 3 Months forward rates for dollars.

This is a direct quotation. Premium should be added to spot rate to obtain forward rate.

Buying :-	**1 Month (Rs.)**	**2 Months (Rs.)**	**3 Months (Rs.)**
Spot Rate -	31.35	31.35	31.35
Add: Premium	00.20	00.40	00.60
Forward Rates	31.55	31.75	31.95
Selling :-	**1 Month (Rs.)**	**2 Months (Rs.)**	**3 Months (Rs.)**
Spot Rate	31.40	31.40	31.40
Add: Premium	00.30	00.50	00.70

Forward Rates	31.70	31.90	32.10

4) Forward buying / Market purchasing:

Definition: In this case a buyer may enter in to a long term contract with a vendor, say for one in this one year vendors, and say for one year. In this one year time spam the delivery only a single delivery.

Example: - suppose buyer enter in to contract in March for one year and seller supply the material in September and it will be single delivery. But, the purchase contract inclusive of price, delivery date and other terms concluded in January

Reasons:

i) It is a safeguard in regard to continuity of supplies.

ii) It protects the price over a long period.

iii) It is a safeguard in respect to standard of quality.

iv) For seasonal product forward buying can be seen.

5) Scheduled Buying

Definition: *Scheduled buying is the process of procuring an item in slower deliveries according to the delivery schedule furnished to the supplier by the buyer.*

The salient characteristics of scheduled buying are:

i) A purchase order covering annual requirements is placed to the supplier.

ii) The supplier is given the estimate of the procurement needs covering a mutually agreed period of time.

iii) When fresh delivery schedules are given to the supplier prior to completion of the previous schedule. Fresh schedule supersedes the previous schedule.

iv) Monthly deliveries are usually specified except for perishable materials, bulky items and others required in large quantities

Advantages:

i) This is an win-win situation where both buyer and seller enjoy the savings resulting from regularity of production and smaller inventories.

ii) Buyer is assured of supply of goods while supplier is assured of business.

iii) Supplier can effectively plan his factors of production while buyer can plan his requirements of finance

Ctriteria:

Scheduled buying is best suited for –

- Items of regular use, Proprietary items and items produced according buyer's design.

7) Contract Buying

Definition (According to Spiegel): *"Contract buying is the purchasing made under contract, usually formal, of needed materials, the delivery of which is frequently spread over a period of time."*

Characteristics:

a) Contracts are given to suppliers for large amount of future requirements or for a certain period.
b) Quantity received per occasion is small.
c) The cycle time between two consecutive receipts may be any period considering the value of requirements, distance and the mode of transport;
d) The buying department usually finds sufficient time to secure competitive bids and negotiate terms of contract.

Advantages:

i) It saves time and money of the company from the inviting quotations, preparing, comparative statements, placing of orders, etc.
ii) it ensures regularity in supply .
iii) The buyer needs to keep very little working stock and safety stock. So that capital blockage is minimum.
v) The buyer can plan their financial requirement as they have a prior idea about payment to vendor.

Types of contracts:

Contract buying is of three types:

- **Rate contract:** In this case rate is fixed but, quantity is not fixed.
- **Running contract: H**ere the rate and the quantity both are fixed for the contract period.
- **Service contract**: in this case various services are obtained periodically.

Criteria:

Contract buying is suited to the procurement of materials and production items of regular use.

8) Blanket Orders

Definition: *Blanket orders refer to the purchase of variety of items from single source, usually a middleman.*

Characteristics:

a) A blanket order specifies the categories of items covered by the order;
b) The supplier is given requirements on phone who supplies at the prevailing prices with discount.
c) The items covered by the order generally have low unit price;
d) More than one middlemen may be selected ensure right time of delivery

Responsibility of the buying department:

i) The middleman who is selected should be located nearer to buyer and should have large stock facilities (warehouse) and who are known for their honesty and reliability.
ii) The buying department must watch the items in the groups and periodically check up whether the demand for any item has risen considerably.

Criteria:

The method is best suited to general hardware, electrical supplies, stationery.

9) Tender Buying

Generally government departments and public sector undertakings in India follow this method of buying.

Characteristics:

i) The buying department establishes a bidder's list and invites them to submit bids (quotation)

ii) Bids on receipt are evaluated by comparison and the right supplier is selected. Mainly lowest price is the criterion of this method. But, when supplier quoting the lowest price has questionable delivery time, quality, reliability or financial stability they may be evaluated under different criteria.

Advantage:

i) By tender buying is the purchaser can select qualified supplier on the basis of competitive price.

ii) It eliminates favoritism, patronage and personal preferences.

Disadvantages of the method:

Tender buying is costly and time-consuming.

Types of Tenders:

Tenders are of four types:

i) **Single tender** refers to the system of tendering wherein the *details of the requirements are communicated only to one firm.*

ii) **Limited or closed tender** refers to the system of tendering wherein *enquiry is sent to a limited number of suppliers who are on the approved list of suppliers and bids are received in response.*

iii) **Open tender** system is the system of tendering wherein *the enquiry is advertised in the newspapers or periodicals. in response.* It may be termed as 'open tender', 'advertised tender', or 'unlimited tender' is used.

iv) **Global tender** is the system of tendering wherein the *enquiry is advertised in the newspapers and trade journals of not only of the home country but also in the foreign country.*

10) Seasonal Buying

Definition: *"buying of the annual requirements of an item during its season"*. This method is used for items available in particular season only. **Example**: Food processing industry.

a) The items involved are seasonal and therefore need to be purchased and stocked in sufficient quantities till the next season (e.g. oranges, sugarcane, apples, mangoes etc.).
b) Usually purchases are made directly from producers / farmers of the goods.
c) The items covered may be small in size but they are required in large quantity.
d) Market price is the lowest during the season. Therefore, the items can be purchased at the cheapest rates:

Criteria:

a. Purely seasonal products
b. Items which are having high price in off season.

11) Group Purchasing

Definition: *'buying of items in a single purchase order'.*

Characteristics:

a) Items required in small quantities are classified into few basic groups. These basic groups depend on the source of purchase.
b) Inventory levels are fixed for each item within each classified group.
c) One purchase order – one for each group – covering a number items within its group.
d) Stocks-on-hand are reviewed periodically.
e) Replenishment is also done

Advantages:

Group purchasing reduces clerical and delivery costs, because, one order is placed for a number of varied small items instead of an individual order for each item.

12) Sub-Contracting

Definition: Sub-contracting is the hiring of another firm to perform some of the manufacturing operations or assembling or producing certain parts and sub-assemblies to be incorporated into the buyer's end product.

Types of sub-contracting:

Sub-contracting is of two types:

i) The company makes some units of the final product on their own shop and buys balance from outside.

ii) The company buys some assembly form outside.

iii) The company can do certain operations like electroplating, heat-treatment, rough blanking etc. done from others

Conditions :

Sub-contracting is desirable when –

i) The product involves number of components requiring different types of machines.

ii) Special expertise is required for certain operations and this is not available in the buyer's firm.

iii) There is lack of capacity at the home plant

iv) If the cost of buying is less than cost of production then company may buy materials from outside.

Responsibilities of the buying department:

i) The buying department has to locate, select and develop qualified sub-contractors who can supply parts of right quality, in right quantity, at right cost in right time

ii) Sub- contractor should be located within close proximity. Because,

- Transportation cost, and time is less.
- Follow up is easy and less costly.
- Buyer can have better control on supplier's quality

iii) More than one sub-contractor should be selected for each component.

iv) Maintain supply of goods from alternate source in the event of any production hold-up with one of the sub-contractors.

v) Buyer should ensure the quality of the materials which are required to make final product.

vi) Delivery scheduling is the most important aspect of sub-contracting.

vii) Buyer can give financial, marketing and administrative support to the supplier to ensure time, price, delivery of goods (in the time of natural condition and in the time of emergency)

Central Purchase Organisation

A large firm in the public or private sector may have section-wise stores at different places. The requirements of these stores can be satisfied by either of the following two methods:

i) Each store to make its own purchases.

ii) A central stores to make purchases and supply material in turn to section wise stores.

The advantages of central stores purchases are:

i) The central purchase organisation (CPO) can obtain quantity discounts, lower rate and better contract terms due to large purchases made possible due to consolidating the requirements of individual stores.

ii) Malpractices by individual purchase officers by some kind of understanding with the local dealers which can result in purchase at higher prices are avoided.

iii) The Central Purchase Organization can exercise strict control on consumption thereby minimizing the risk of malpractices.

iv) The central purchase organisation can contract directly with the manufacturers and obtain items as per specifications. Individual section stores may not be able to do this.

Examples:

Reserve Bank of India, State Bank of India, etc.

Directorate General Suppliers & Disposal (DGS & D)

The DGS and D is the Central Purchasing Organisation for the various government departments. It enters into contract with various firms for the supply of certain materials to the government departments during the year at the agreed rate. A formal document raised for the purpose is called “rate contract”.

Meaning of Purchase Cycle

Purchasing activity plays a vital rile in all the firms in general and in the manufacturing firms in particular. Purchasing, is not merely “buying to satisfy the indentor’s requirements”, but “buying goods of right quality, in the right quantities, at the right time and at the right price”.

Purchase cycle consists of following eight major activities (Fig 2.3)

1. Establishing and communicating the need for procurement.
2. Scrutiny of the purchase indents.
3. Market study and selection of sources of supply
4. Order preparation
5. Follow up
6. Receiving and inspection
7. Storage and Record keeping
8. Invoicing and Payment.

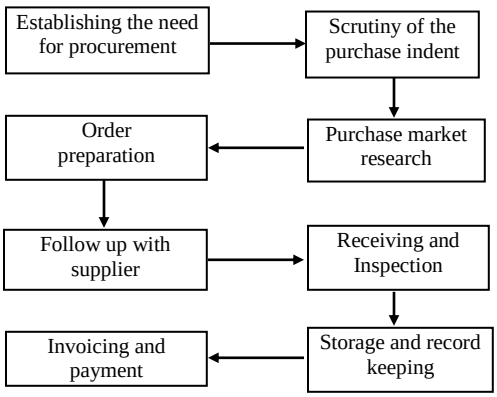


Fig: Major activities of purchase cycle

Each of these activities is described below and their major aspects are summarized below:

Elements of Procurement Cycle / Steps of purchasing cycle

1. Establishing and communicating need for Procurement

The need for purchase originates in one of the firm's operating departments or its inventory control section. The demand may be for raw materials, such as steel; or it may be for semi-finished goods such as castings, forgings, semi-matched parts; or it may be for bought out parts; or for cutting tools such as drills, reamers, cutters, etc.; or it may be for supplies or for spares. The need is communicated to the purchase department through a formal document called "*Purchase Indent*" or a "*Bill of Material*"

a) Purchase indent: Purchase indent, also called purchase requisition, is a formal request made to the purchase department to purchase materials or services specified, time when required etc. A purchase indent originates either from the firm's inventory control section, production control department, or from one of the operating departments.

1. ESTABLISHING THE NEED FOR PROCUREMENT

1. Recognising the need for procurement
2. calculate the requirements
3. jotting down the specifications
4. informing requirements to purchase
   (i) Purchase indents / Bill-of-material

2. SCRUTINY OF PURCHASE INDENT

1. Completeness
2. Appropriateness
3. Passing the indent through stores
4. Logging of indents into indent register

3. MARKET RESEARCH

1. Source selection & source development
2. Advertisement.
3. Telephonic quotations
4. Written quotations

4. ORDER PREPARATION

1. Scrutiny of quotations
2. Negotiations
3. Placing orders to suppliers
4. Obtaining suppliers' acceptance.

5. FOLLOW UP

1. Pre-delivery follow up
2. Shortage chasing
   (a) Reminders
   (b) Personal visits
   (c) Telephones
   (d) emails
   (e) Faxes / Telexes
   (f) Posting of personnel at suppliers' works

Constituent activities of a purchase cycle (Contd.)

6. RECEIVING & INSPECTION

1. Receiving dispatch details (RR/LR) and logging them into the consignment register
2. Collection of material
3. Inspection for physical damages to the packages and number of packages
4. Entering consignment details into GR register
5. Uncrating of goods
6. Quantity certification
7. Raising of GRR
8. Intimating receipt of materials to the indentor
9. Inspection of goods

7. STORAGE & RECORD KEEPING

1. Movement of materials to concerned store / rejection store
2. Quantity certification
3. Application of protective coating / marking
4. Storage of materials into appropriate racks
5. Posting of receipt into stock card.

8. INVOICING & PAYMENT

1. Receiving GRRs in Accounts department
2. Receiving suppliers' bills
3. Linking of GRR and suppliers' bills
4. Posting of purchase register
5. Passing of bills
6. Effecting payments

Fig: Constituent activities of a purchase cycle

- The indent alternatively rose either by plant engineer, or by maintenance engineer, or by office manager, or by other responsible persons authorized to request these items.
- Organization permitted different person to sign the purchase indents for different kinds of items are issued by the top management.
- **Stock items (**items of regular use**)**: for these items the purchase indents will be given by inventory section or stores are generally. When the stock level drops to or nears the re-order level, an indent is sent from stores to the purchasing department to replenish the stock.
- The buying department may also receive indents for capital goods, the purchase of which has already been approved by the management.
- **Non-standard items**: The indents for these items are filled in by the operating departments. The requests for items may come from planning department. Indents are raised by all indenting departments. It should have three copies. **Original** is sent to the purchase, **second** goes to the stores, and **third** copy remains with the originating department.
- Purchase indents from inventory section or the store is sent directly to the purchase department. **Purchase indents from all other persons must be approved by the authorized signatory and should be routed through inventory section or stores to see whether required items are carried in the stores.**

b) Bill-of-material:

- Bill-of-material, also called parts list of building list,
- Bill of material is prepared by the engineering department, production and material dept.
- The final assembly is broken into major assemblies. Major assemblies are divided into parts. The individual parts comprising each assembly are arranged, as far as practical, in the manner in which each part is assembled.

The main advantages of bill-of-material are:

i) It eliminates the unnecessary clerical activities regarding purchasing of goods. This saved time and energy.
ii) It can minimise clerical error.
iii) It can reduce wastage of materials and also reduces unnecessary inventory cost.

2. Scrutinising Purchase Indents

a. A purchase indent may describe the items either by its brand name, or by commercial standard, or by its performance standards. Sometimes the indents may be forwarded sample of the item.
b. Purchase departments must scrutinize the accuracy and completeness of quality description.

The scrutiny of the indents is a routine activity of the purchase department. The indent is scrutinized to see whether –

i) It should be signed by the authorized signatories to avoid irresponsible and unethical purchases;
ii) It should be coming through stores department to certify non-availability and stock history of the item in stores;
iii) Quantity and quality specification of the item should be correctly and clearly written.
iv) The description of the required item should be clearly given;
v) quality specification of the item may be correctly and clearly written

Each purchase indent after scrutiny is recorded in the purchase indent register and then given to the concerned buyer. The second copy is initialed, dated and returned back to its indentor.

3 Identification, Selection, development of Sources of Supply or Survey

- After the scrutiny of the indents the next stage is market survey and selection of the sources of supply.
- This stage involves differentiate and sorting of items into item groups, review of available information, and selecting potential source(s) of supply who can supply goods -

i) of the right quality;

ii) at the right price; and

iii) Meet buyer's quantity requirements, and make reliable delivery promises.

a. Buyers can select suppliers by virtue of their experiences and data collected from their various internal sources such as catalogues and price lists, etc.

b. In general, the work involved in market search and source selection is as under:

i) For the items purchased to commercial standards, telephonic quotations are obtained from vendors and verbal order is given to a supplier whose terms are most suitable.

ii) *For highly price fluctuating products and items an inquiry is sent to probable sources and quotations are received*. A comparative statement is prepared from the quotations received from the suppliers. Then company select one or more than one suppliers according to their suitability.

iii) For the regular use items buyers send design specification along with blue print or sample and delivery schedules are released to the suppliers

iv) **For non-stock items**, the sources are selected as per previous points depending upon the nature of the item

v) For **capital equipment**, enquires are either mailed to the machine tool manufactures or the enquiries are advertised (open tender) in the paper of this country. Company can float global tender by advertised the enquiries in the news paper of other country.

vi) *If lesser amount is involved and the item is easily available, the buyer may buy with petty cash*.

vii) For new items, the buying department have to explore information to identify sources and undertake the activity of selection and development of suppliers.

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4. Order Preparation

- After selection of source of supply, the next step is to authorize the selected suppler to supply material, which is done by placing the purchase order.

- **A purchase order** is a formal document prepared by the buying department on behalf of the company to authorize (request) the supply of the goods and services in the quantities, at the time and at the price specified there in the document.
- A purchase order is a legal document and serves as an evidence of the contract between the buyer and the seller.

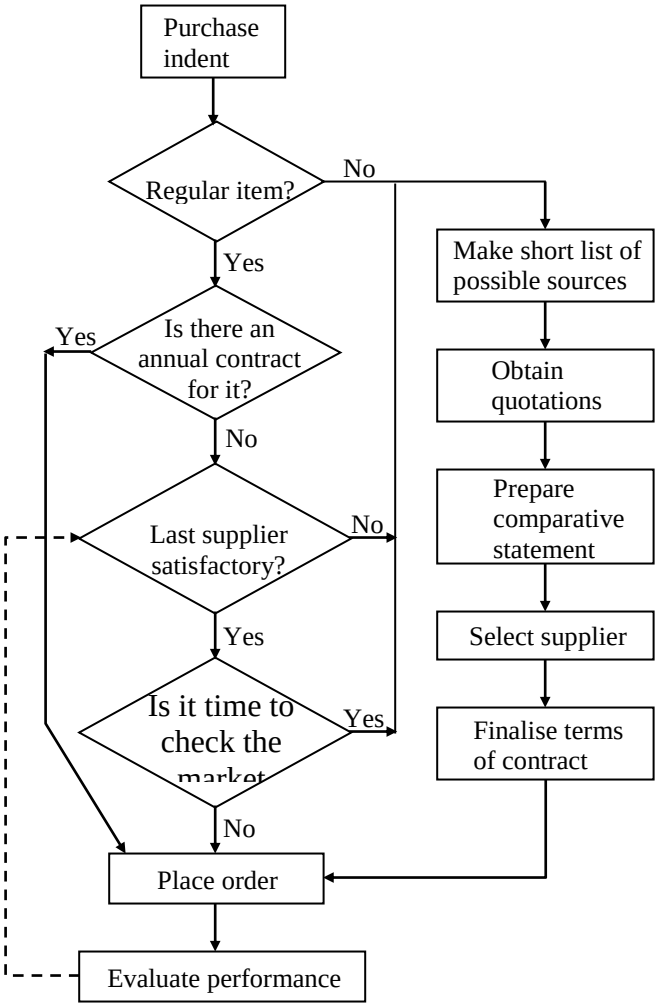


Fig: Flow chart of guidelines to select a supplier

After getting the orders typed, the buyer has to check for its correctness and sign it. Once the purchase order has been signed, it requires to be entered in the purchase register. The individual copies of the purchase order, materials required urgently are purchased on verbal or telephonic orders which are covered immediately by written purchase orders.

A written purchase order serves the following objectives:

i) It gives full details of the materials to be supplied to avoid ambiguities.
ii) It serves as a future reference for placement of orders.
iii) It helps buyer's accounts department to link goods-receipt-reports with supplier's invoices.
iv) It helps buyer's receiving department to verify that the materials received are in accordance with those ordered.

The copies of the purchase order go to the supplier, the accounts department, the receiving department, the indenting department, the reference file, and the follow-up section.

The placement of an order cannot be considered complete until the acknowledgement of the purchase order is received from the vendor.

5. Follow-up with Suppliers

The follow up function now-a-days has become the foremost function of the buyers. Basic rules of follow-up are:

i) Follow-up should be based on market condition and buyer's experience with the vendor's delivery performance.
ii) Buyer should keep a constant himself up-to-date with latest progress on each order.
iii) Mode of follow-up should be based importance of the item, reliability or otherwise of the supplier, number of suppliers, location and so on.

Purchase follow-up is made in two stages**: pre-delivery follow-up** and **shortage chasing**.

Pre-delivery follow-up is intended to remind the supplier of the due date and obtain advance information of expected delays. Pre-delivery follow-up enables buyer –

- If the supplier is expected to fail in his delivery commitment; buyer has to develop alternate arrangements to ensure that delivery should be on time. In this case buyer may identify another source of supply to make delivery on schedule. Typical methods used are:
  i) Phone call to local suppliers;
  ii) Letters to outside supplier typed and signed by the buyer;
  iii) Regular visits –to new suppliers to review progress
  iv) Delivery confirmation letters from supplier intimating that the delivery shall be made on promised date. It is one sort of guarantee card;

Shortage chasing is the universally accepted most vital part of purchase follow-up. purpose:

a. First to obtain the shortage materials as soon as possible and
b. Second to intimate suppliers that delivery should be made on time, if not then shortage chasing will be done by company. It prevents future recurrences.
c. The approach depends upon the type of supplier:

i) In the time of vendor evaluation buyer may warn supplier about the delivery and shortage of supply.
ii) Supplier's delivery ratings should be communicated on periodical basis. Suppliers with A and B ratings should be encouraged. Those with C and D rating should be warned to improve on their delivery performance.
iii) New sources should be developed and schedule on existing suppliers who are in C and D position should be gradually rejected to pressurize them to improve.
iv) Meeting between the senior representatives should be arranged to discuss the problems.

Example: the follow-up system in operation can be done in following manner

d. Once the date is marked on the follow-up copy, the staff has to segregate all order copies and puts them into date compartments.

The buyers are asked to adopt the following method for follow-up:

i) Telephone or email the party;

ii) Faxes: when urgency exists

iii) A personal visit should be made to local supplier in case of extreme urgency and in case of failure to get the supplier on phone;

iv) Telephone and also write a reminder on a printed form indicating the delivery required, the quantity pending and so on;

v) Written letters, duly typed and signed by the higher authorities must be issued to distant supplier;

6. Receiving and Inspection

a. After receiving the purchase order the supplier fills up the buyer's requirements and arranges delivery of the materials according to instructions regarding quantities, time, route, mode of transport, etc.

b. In a small company "Receipt" or "Receiving" department/section will receive the materials and inspect. In a large company, receiving materials is looked after by the stores department.

The activities involved are as under:

(a) Receiving:

Materials in the receiving department are received against a specific document depending upon mode of dispatch and geographical location of the supplier.

Materials from local supplier

- Responsibility of delivery of materials depends on the terms of contract. Materials to be delivered by the suppliers
- The original copy of the delivery challan is retained by the receipt department. The duplicate copy is stamped *"Subject to physical count and inspection"* and is signed by the receipt clerk and is handed over to the supplier's representative.

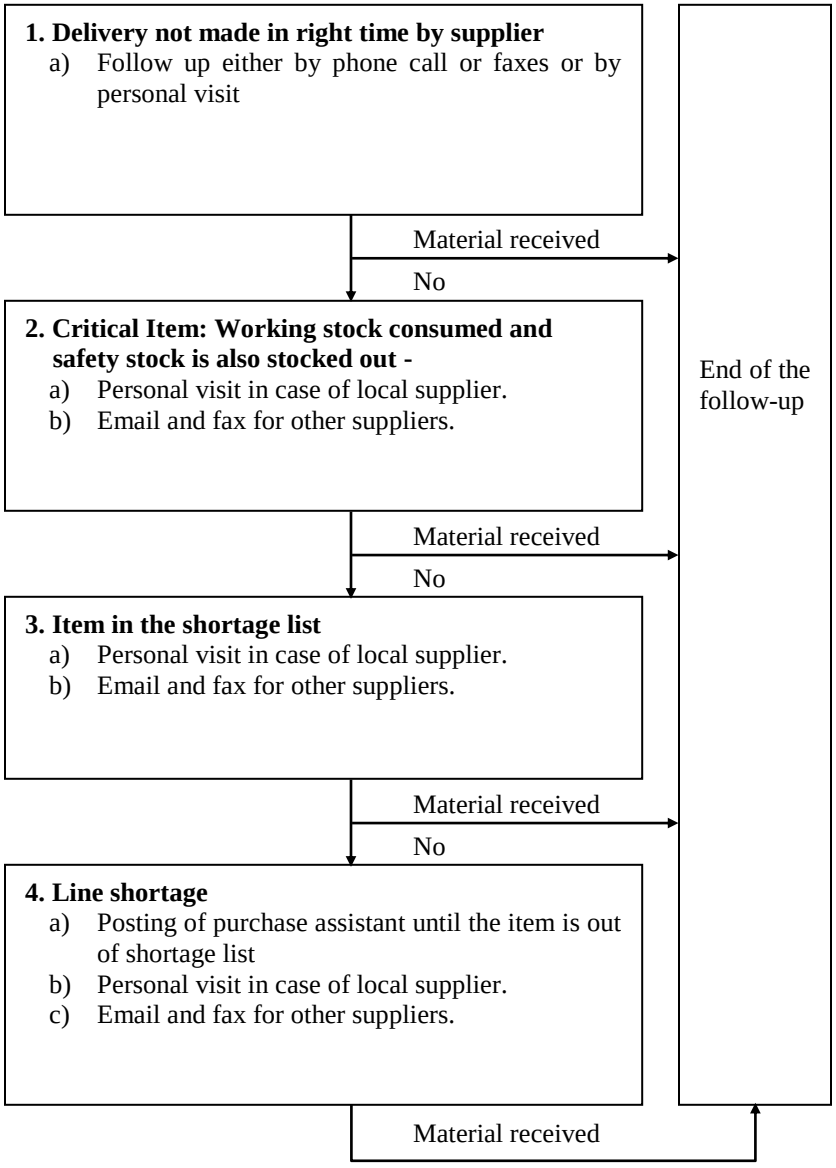


Fig: Pre-delivery follow-up and shortage chasing

Different methods of receiving Materials from outside supplier:

i) ***Post parcel:*** Post parcel is the convenient method of receipt of materials from outstation suppliers. No information is received from postal authority regarding receipt of materials but postman personally delivers the parcel. Only light weight materials such as samples, spares, and cutting tools are sent by the post parcel.

ii) ***Road transport:***

- If the materials are sent through road transport, the receipt of the transporter, called "Lorry receipt". It is sent by the supplier.
- The lorry receipt is numbered and contains description of the items packed in the boxes, the number of Containers.
- Freight paid or to be paid and the type of delivery has to be mentioned.
- If the delivery is mentioned as "Door delivery", the transporter has to deliver the goods at the buyer's works.

iii) ***Rail transport:*** This is similar to road transport except the following differences:

- the railway receipt is received by the buyer in duplicate;
- the railway does not inform the consignee of the arrival of the material;
- The materials require to be collected by the consignee.

iv) ***Air:*** Air consignment (AC) note must be sent as soon as material is booked. The receipt department on the receipt of A.C. note visits airlines office to collect materials.

Since, it takes a long time for clearing; a register is maintained to keep a proper control on the documents.

The register mentioned above serves many purposes:

- It enables clearing department to do follow up with the transport agency.
- It enables buyer to know the time when the materials to be received.

- It enables clearing department to plan their administrative work load as well as achieve optimum utilization transport facilities.

(b) Verification and logging of delivery challan of materials:

When materials are received with supplier's delivery challan, receiving department takes out the copy of the relevant purchase order and verifies to ensure:

i) Right materials
ii) They have supplied materials as per delivery schedule.
iii) Right quantity.
iv) Purchase order number, part name, part number, broad purchase category etc. are mentioned clearly and correctly.
v) Right time.

After having satisfied, the receipt clerk makes the entry of the receipt materials into a register called "Goods Receipt Register".

(c) Preparation of Goods-Receipt-Reports (GRR):

i. Goods receipt register cannot be made available to different departments.
ii. Other departments have to be informed. That's why information from the goods-receipt-register is transformed on "Goods Receipt Report" (GRR). It is also known as "Goods Inward Note" (GIN), or Receipt-cum-Inspection Advice (RCIA), or Materials Inward Note (MIN) etc.

The need for preparing a separate document such as GRR arises because of the following reasons:

i) Only one copy of the delivery challan is received. But, concerned department should be informed.
ii) Preparation of GRR helps to combine information on delivery challan and results of inspection.

(d) Communication of receipt of material:

- Moreover, the personnel who are engaged in receiving may not know whether a particular material is required urgently.
- Therefore, they have to inform the concerned departments as soon as materials are received. First communication can be made on intercom. But, after that written communication should be done.

Physical count of the received material:

Materials received should be verified for quantities.

The receipt clerk has to attach receipt tag containing all details about the delivery challan to the part or to the package. This tag also contains challan numbers, description of materials, code numbers of materials, name of the suppliers, number of packages received, quantity received and date of receipt.

The verification method of quantities may be one or more of the following types:

i) Counting by number (quantity).

ii) Weighing.

iii) Measurement of length.

After quantity verification material is kept ready for quality verification by the inward inspection department.

If the verified quantity is less than claimed quantity then authority should issues **Discrepancy Note.**

(e) Inspection of goods

All supplies are subjected to inspection and testing.

Critical items: These are inspected at vendor's plant prior to deliveries by the vendor.

All remaining items are inspected by the inward inspection at the buyer's premises to conform the following criteria

- Dimensions;
- Materials specifications, and
- Performance.

7. Storage and record keeping

- After inspection The goods are segregated in to accepted and rejected categories.
- Only fully accepted quantity is forwarded to the stores. The quantity is physically verified and entered into the ledger or bin cards and only thereafter the issue is allowed.

8. Invoicing & Payment

Receipt of supplier's invoice:

When the supplier supplies goods, he immediately prepares invoices. Sometimes, both buyer and supplier have discussion and supplier agrees to raise invoices after the receipt of goods-receipt-reports. Such a system has a number of benefits namely:

i) Reconciliation statements are very easy to prepare

ii) Invoices indicate the exact quantity of materials accepted by the buyer.

iii) Paper work is cleared faster and payment wise there is no dispute.

iv) Rising of credit or debit notes can be avoided for discrepancies in quantities or rejection.

Scrutiny of invoice:

Supplier's invoices on receipt are sent to the accounts department. The personnel of this department again scrutinize the invoice in accordance with GRR.

Journal entries:

Verified invoices are entered in a purchase register, called "purchase journal". A purchase journal is a register wherein invoices are entered supplier wise.

Payment:

The purchase journal is reviewed periodically and the cashier is informed of the invoices which are due for payment. Cheques are drawn to effect payment on due dates

Principles of sourcing (Vendor Development, rating and evaluation)

Vendor Rating and Source Selection

- Reputed suppliers are intangible asset of any organization.
- They are not only supplier of materials, but they also extremely important source of information regarding market condition , price trend and general industrial climate.

- Many organization have accepted source selection as a corporate policy. This brings fair competition among the suppliers and supply failure are minimized.
- Improper selection causing inferior goods and servicing has resulted in authority being shared with the using department.
- Therefore, proper source selection must b exercised by the purchasing department to avoid possibilities of interdepartmental conflict.

Procedures:

Procedures of source selection involves the preparation of an exhaustive list of prospective supplier and the successive elimination from this list on various grounds until the number has been reduced to the one or , few to be forwarded wit the business.

The stages of procedures as follows:

1) ***Searching stage:*** The need for a materials or product is starting point. The search process begins with the finalization of specification in consultation with technical department.

   It involves identifying the source of supply.

   It can be obtained from various journals and news paper advertisement.

   Supplier representative also be a good source of information.

2) ***Selection stage:*** The selection of supplier starts with floating of enquiries by the purchaser.

   Certain progressive suppliers often contact buyers with a request to be included in the buyer's list of approved suppliers.

   The buyer can also visit and inspect the suppliers factory to know the details.

3) ***Negotiation and trial order:*** After selection the process of negotiation will be started for placing the trial orders. Correct and cordial relationship with the vendors are essential for mutual cooperation.

4) ***Rating and experience stage:*** After the trial order are executed, it becomes necessary for the buyer to rate the vendors to enable him for determining how he should apportion his requirements among vendors.

Special Aspect of Source Selection

Local source:

- Generally local source are preferred, provided they meet the requirements of buyer in terms of quality, delivery and price.
- A personal follow-up or a local "phone call" is enough for quicker delivery.
- In fact, some organization prefers local suppliers even though they may not compare favorably with outside supplier.
- In India may big organization encourage ancillary suppliers in and around their plants through technical and financial assistance.

Subcontracting: Selection of source for sub contracting is another special issue. The manufacturer may decide to "off-load" some product to suppliers s as to facilitate scheduling of production in the plant. It also enables better utilization of manpower.

Small or big? If the requirement is small, then obviously small suppliers are prime choice.

A personalized contact can be established and because he is small, he is likely to have fewer customer, will be more loyal and will meet even unusual request.

Big suppliers are equipped to meet larger requirements they usually have reserve facilities and may thus be able to meet additional requirement.

Development of Vendors – Necessity and Importance

1) Sometimes it can happen that there is no supplier of an item or, the item is a new one. In this case development of vendor is very much essential.
    - ***a)*** first , is to locate a manufacturer who produces an item almost similar to the one required.
    - ***b)*** He may be encouraged t take up the new line of production.
    - ***c)*** Often development costs are shared by the purchasing company, and even expertise is shared by two company.
2) To facilitate procurement of materials need all organizations require to develop source of supply for the material requirement.

3) Vendor development helps to develop sources for new material and enlarge/ increase source for other materials.
4) Revolution in the field of materials management field such as , MRP, KANBAN, JIT, etc. necessitates structured discipline of material flow which leads to development of new sources.
5) Establishment of indigenous sources of supply to substitute imported materials
6) To incorporate functional improvement.
7) To reduce overall cost of a product.
8) In case of monopoly situation the price of the goods are frequently changed without any reasons, even commercial terms and conditions like payment terms, liquidated damage, performances guarantee, etc are not accepted by all of the monopoly vendors for different items. If few more parallel sources can be located, then better price , improve delivery , better quality , etc. may be obtained.
9) Due to delay of the materials by the existing vendors, as a result stock out situation arises which leads to stoppage of production.
10) When the vendor is not at all interested to reply or, provide acceptance letter of an order, within stipulated time frame, then others vendors can supply the product.
11) Proximity of vendors is necessary. Because, a local buyer is always welcome than the buyer of distant area.
12) Diversification will force the requirement of vendors.
13) When capacity of current vendors do not meet with the demand of the buyer, and then buyer needs other vendors.
14) To ensure the availability from more than one sources.

Factors influencing Source/ vendor selection

1) Price: this is the most obvious factor, along with any discount offer, although it may not be the most important. Because firm spend a large amount of money of their total income on purchasing items, finding supplier who charges *low prices* is a key objective in vendor selection.
2) Quality: quality of a supplier's materials also is important because a company may be willing to spend money to obtain materials of high quality.

3) Delivery: shorter lead times and on-time delivery help the buying firm maintain acceptable customer service with fewer inventories. The benefit of fast, on time deliveries applies to both service and manufacturing sectors. Many manufacturers demand quick, dependable deliveries from their suppliers to minimize inventory level.
4) Services: special services can sometime be very important in choosing a supplier. Replacement of defective item supplied, instruction in the use of equipment, repair of equipment and other similar services can play a key role in preferring one supplier over another.
5) Location: location of supplier can have impact on shipping time, transportation cost and response time for rush orders or emergency service. Local buying can create goodwill in the community by helping the local company.
6) Inventory policy of the supplier: if a supplier maintains an inventory policy of keeping spare parts on hand, this could be helpful in case of an emergency equipment break- down.
7) Flexibility: the willingness and ability of a supplier to respond to changes in demand and to accept design changes could be important considerations.

In addition to the above major factors several other factors are also considered in supplier selection.

1. ***Reserve capacity:*** A supplier with an adequate reserve of productive facilities can respond to increased requirements of customer.
2. ***Internal factors and quality control procedures:*** the stage of a supplier's technological development and its ability to keep up with current methods are also considered to choose an effective source.
3. ***Labour Relation:*** A possible interference with the continuity of production in a supplier's plant may originate with the workers themselves. Strikes or production slow down are directly related the labour relations of the supplier.
4. ***Warranties:*** Service may include the kind and form of warranty that accompany the supplier's products, i.e., installation, maintenance, replacement of parts etc.

5. ***Plant visits:*** it is often desirable for a representative of the production or engineering departments to accompany the buyer on such visits, especially if highly technical products are involved.

Types of Vendor Development

1) The vendor is given a plot outside the buyer's premises:

- The buyer also gives them all assistance with tool and gadgets.
- The entire product of the vendor may not be consumed by buyer itself and vendor has the liberty to sell the balance materials in the market.
- The vendor is assured of the fixed sales of his production by the buyer's firm.
- The buyer may give them managerial, financial, and technical help.

These are two types:

a) Local source (nearby source): The source of supply may be inside the state where the buyer's plant is situated. The buyer develops them with technical, managerial and financial assistance so that they get the product on desire.

Advantages:

- Closer cooperation because of proximity.
- Delivery dates are more certain as transportation sonly the minor factor.
- Rush orders are likely to be filled faster.
- Saving in entry tax.
- Implied social responsibilities to the community are fulfilled.
- Shorter lead time can frequently permit in inventory. In effect the seller carries the buyer inventory.
- Disputes are more easily resolved.

b) Far away source:

The vendor can be located any where outside the state of buyer.

Advantages:-

- As a result of economies of scale the vendor, in some situation can be more efficient and offer higher quality of better services at a lower price.
- Sales tax – since, the central tax is applicable which is quite lower than the state sales tax.

2) captive development:

In this type, the vendor's plant is inside the buyer's premises. An agreement is made with the vendor that a section of the product will be solely manufactured for the buyer and buyer has the first option to get it.

Advantages:-

- Buyer can get it at lowest possible cost.
- Saving in transportation (time & cost)
- Delivery is assured. This enables the buyer to operate at low inventory.
- Considerable savings in sales tax.

Techniques and Procedures of Vendor Development

If the current vendor(s) is not capable to supply goods in right time in right manner then company has to develop some other vendors. the techniques & procedures of vendor development are as follows.

1st step: identify the problem of current vendor.

2nd step: investigate the genuinity of the problem.

3rd step: in case of a genuine problem the buyer has to identify the area problem after identification buyer may give assistance to the vendor.

4th step: sit with the supplier with the problem.

5th step: if the vendor posses any managerial incapability, then buyer may assist them with managerial support. Else, buyer proposed them to be sole vendor of them or, negotiate with other vendor apart from this vendor.

Vendor Evaluation

Purpose:

Periodically vendor analysis is a must for such analysis, which is essential to build up efficient source of supply. This analysis seeks to evaluate each vendor, past performance and overall worth to buying company.

Vendors are generally evaluated on the basis of the following attributes.

i) Delivery as per schedule.

ii) Reliable quality.
iii) Quick replacement of rejection.
iv) Supplier with out asking for any financial consideration.
v) Answering queries readily.

Performance will be determined by applying decision matrix system. In this system weightage is determined which indicate the relative importance of each attributes and this is known as Attribute Weightage Coefficient (AWC).

During course of inspection of suppliers premises all the following factors are also to be ascertained –

2) Technical know how.
3) Financial capability
4) Organization set-up and man power.
5) Records of past performance of supplier in delivering materials in right time with right quality and quantity.
6) Own manufacturing units.
7) Staff and line capacity of firm.
8) Service level.
9) History of labor relation.
10) Quality control and testing facilities.
11) Financial adequacy and stability
12) R&D facilities.
13) After sales service.
14) Reputation in the market regarding price, quality, quantity and behavior.
15) Whether he is vendor or agent of vendor.

Levels of Vendor Evaluation:

i. **Product level**: At this level, the focus is on establishing and improving the vendor's product quality. Incoming inspection and quality inspection are carried out to establish the degree of quality conformance of incoming materials.

ii. **Process level**: At this level the production process is closely investigated rather than focusing on product inspection. It is because the quality of product depends on manufacturing process.

iii. **Quality assurance system Level**: quality assurance means checking the way in which quality inspection procedures are developed, maintained and improved.

iv. **Company level:** In this approach, not only quality aspects are focused, but also financial aspects are taken into consideration. The quality of the management of the vendor firm is assessed. In this way the buyer tries to establish how competitive that a particular supplier will be in future.

Different Methods of Vendor Evaluation

1) Categorical method:-

- In this method the buyer makes out a list of all the factors, which he considers necessary for evaluation, and at periodic intervals he makes out a performance report.
- The buyer may also seek the help of all dept. i.e., production or, quality control dept. in order to determine the grading given.
- A performance standard may be decided upon the points.

  i.e., 80-100= excellent
  70-80 = very good
  60-70 = good
  50-60 = satisfactory
  40-50 = poor.
- On the basis of points meeting between buyers and supplier should be held.
- Low rated suppliers should be warned and given chances for one or more times, if they do not show any improvement, they will be removed.

Merits:

1) It depends upon the performance of the vendor.

2) It is a very inexpensive method.

Demerits:

1) It totally depends on experience and ability of the buyer.
2) If buyer is lazy and does not have ability to do it periodically then it loose importance and ability.

2) Weighted point method:

Suitable point should be allotted on different criteria such as, quality conformation, delivery scheduling, total price, innovative attitude, etc.

Depending upon the prime use of an organization, the point of each factor can be earmarked for some users.

Point should be allotted according to preference lead down by the top management to achieve the corporate objective and mission.

Example: (A company for which quality is the key factor)

Quality – 50 points

Delivery – 30 points

Price – 15 points

Service – 05 points

For vendor 'X'

i) **for quality**, we calculate Quality Performance Ratio (QPR) = No. of Delivery without rejection / Total No. of deliveries.

Say, total no. of deliveries =100.

Deliveries without rejection =70

Therefore, QPR = 70/100 = .7 = 70%

ii) **for Service,** we calculate Service Performance Ratio (SPR) = No. of calls Attended / Total No. of deliveries.

Say no. of calls attended = 10

SPR = 10/100 = .1

iii) Price Performance Ratio (PPR) = lowest offer received / price offered by vendor

Example:

Vendor 'X' quoted – Rs. 40/- per unit.

Vendor ‘Y’ quoted = Rs. 50/- per unit.

Vendor ‘Z’ quoted = Rs. 60/- per unit.

PPR for –

Vendor ‘X’ = 40/40 = 1

Vendor ‘Y’ = 40 /50 =. 8

Vendor ‘ Z’ = 40/60 = .67

Note : in ideal situation the ratio should be =1

iv) Delivery Performance Ratio (DPR) =

Nos. of delivery on schedule / total no. of delivery = 60 /100 =. 6

Total weightage = (50 * QPR) + (30 * DPR) + (15 * PPR) + (5* SPR)

For vendor ‘X’ –

TW = (50*. 5) + (30*. 6) + (15 * 1) + (5*. 1) = 35 +18 +15 +. 5 = 68. 5

In this way value can be compared with other vendors.

Merits:

a) Any number of evaluation factors can be included and they can be assigned relating weight as required by the organization.
b) Subjective evaluation is minimized.

3) Cost Ratio Methods:

it involves a good systems of determining the actual costs incurred in purchasing, follow-up, transaction, packaging, receiving, etc. and determining the unit cost incurred by the buyer on the material when actually received.

The higher the cost, the lower the supplier’s comparative rating.

Example: Cost related to quality—

Vendor’s Factory visit – Rs. 3000/-

Approval of sample - Rs. 1000/-

Inspection cost - Rs. 2000/-

Cost of rejected materials- Rs. 800/

Cost of losses in production- RS. 100/-

Total Rs. 6900/-

The value of purchase = Rs. 6, 90,000/-

Hence, quality –cost ratio = 6900:6, 90,000

= 1:100

= .01 =1%

Cost related to Delivery –

Cost of telegram, fax, telephone, etc = Rs. 900/-

Visit to plant for expediting = Rs. 6000/-

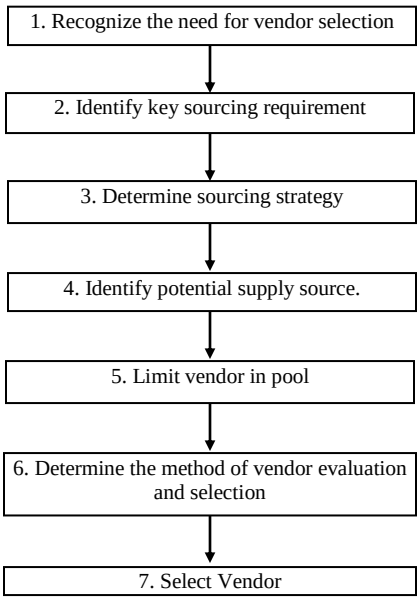
Total = Rs. 6900/-

Delivery –cost ratio = 6900:6,90000

= 1:100 =.01 = 1%

The three methods described above are intended to enable buyer to exercise better judgment over selecting his vendor.

Vendor evaluation and Selection process



1. Recognise the need for vendor selection:

Recognizing that a requirement exists to evaluate and select a supplier for an item or service is the first step of the evaluation and selection process. Purchasing may participate in a product development team to get an early insight into new product development plans and thereby obtain some preliminary specifications on the type of materials, services or process required. This preliminary information may trigger initial evaluation of potential sources of supply in anticipation of future material need. The recognition of a purchase requirement occurs when a standard purchase requisition is received by purchase department. Also, a supplier selection decision may arise when existing supplier fail to perform satisfactorily. Changing supplier for existing items is termed as *supplier switching.*

2. Identify key sourcing requirement:

The requirement that are crucial for a purchase often differ widely from item to item, organization to organization or industry to industry – while different requirements exist for each evaluation area, the minimum evaluation criteria include supplier *quality, cost, delivery – performance, and technological capacity.*

3. Determine sourcing strategy:

There is no single sourcing strategy which can satisfy the requirements of all purchases. Because of this, the purchasing strategy adopted for a particular item or service will influence the approach adopted for supplier evaluation and selection process.

A commodity sourcing strategy provides direction on the overall objectives to be achieved for the commodity such as the number of supplier that will be used (multiple sourcing), the type of contract (long term or short term) and the type of supplier to be evaluated.

Some of the strategy options available are:

- Single versus multiple supply source.
- Short-term versus long-term purchase contracts.

- Choosing suppliers who provides product design support versus those who lack design support capacity.
- Developing a close working relationship versus traditional purchasing.

The strategy option selected has great influence on the supplier selection and evaluation process.

4. Identify potential Supply Sources:

Purchasers rely on various source of information when identifying potential sources of supply several variables determine the degree to which a buyer must search for information about supplier.

Strategy importance of purchaser/ technical complexity requirement

		High	Low
Capacity of existing supply base to satisfy low cost, delivery, technology and service requirements	High	Minor-moderate information search	Minor information search
	Low	Major –moderate information search	Major information search

A major source of information comes from existing suppliers. Buyers often expect existing suppliers to satisfy a new purchase requirement. The advantage is that doing business with an already familiar supplier thereby saving time and resources required to evaluate a new items is that the purchasing manager may never know whether better supplier are available without information about other sources of supply.

Selecting an existing supplier for a new purchase requirement may be recommended if a list of *preferred suppliers is* maintained. A preferred supplier status conveys immediate information about the supplier's overall performance and competency.

Other information sources are:

a) Sales representatives.
b) Information Data bases.
c) Experience of purchasing personnel by which they gain knowledge about potential supplier.
d) Trade journals and trade directories.

e) Industrial trade shows or exhibition.
f) Other supplier.
g) Purchasing personnel of other units.
h) Internet search.

5. Limit Supplier in a Pool:

Based on the information gathered, the purchaser may identify many potential supply sources from which to choose. Because the performance capabilities of suppliers vary widely and also because of limited resource available to the purchaser, an in-depth visit to all potential supplier or evaluation of all of them will not be possible. Hence, a preliminary evaluation of potential supplier is often used to narrow down the choice before conducting an in-depth formal evaluation.

The various criteria used to making the preliminary evaluation are:

i) Financial Risk analysis.
ii) Evaluation of current and previous supplier performance.
iii) Evaluation of information provided by suppliers.

6. Determine the method of supplier evaluation and selection:

Based on the preliminary evaluation, supplier who are not capable are eliminated. The buyer or purchase team must decide hoe to evaluate the remaining suppliers who may appear to be equally qualified.

7. Select Supplier:

The final step of evaluation and selection process is to select one supplier or more than one supplier. The selection methods vary widely depending on the purchase item under consideration.

Vendor Rating

Vendor rating system is an asset and valuable tools of making the purchase decision and also providing a feedback to suppliers with low rating to encourage improvement in their performance

After selecting the most eligible vendor, the buyer must “educate” them with regard to his various requirement.

It is important to evaluate vendor’s performance as apart of continuing process of vendor development.

Example: Relative weights given to different parameters.

Quality – 40

Delivery – 40

Price – 20

i) Quality points = (quality product accepted / total quantity supplied) *40.
j) Delivery points = (Quantity delivered in time / Quantity ordered) *40
k) Price point = (last normal or emergency purchase rate / ordered price) * 20

Delphi survey was conducted in certain organization to identify the factors and corresponding weightages, which need to be considered for designing an effective system to monitor the performance of supplier.

In order to ascertain the view of management as to the criteria to be adopted for selecting a supplier a list of 8 criteria was circulated to 25 managers and purchase officers. Blank spaces were left for individuals to add additional criteria.

The **criteria** chosen were:

a) Quality of product.
b) Price of product.
c) Maintaining delivery date.
d) After sales service.
e) Reputation in the market.
f) Ancillary unit supply.
g) Technological expertise / experience of supply.
h) Financial standing.

The twenty three completed performances were received back and ranking was as follows.

Criteria ***Position***

a)	Quality of product.	I
b)	Price of product.	III
c)	Maintaining delivery date.	II
l)	After sales service.	VII
m)	Reputation in the market.	VI
n)	Ancillary unit supply.	VIII
o)	Technological expertise / experience of supply.	IV
p)	Financial standing.	V

The following additional criteria were also received—

b) payment terms
c) Testing facilities available.
d) Proximity of supplier.
e) Installed capacity

Objectives of vendor rating:

I. To establish a procedure by which quality control, purchasing and user departments can fulfill the corporate objective of obtaining quality product at minimum cost.
II. To develop an effective tool for rectification of defects, improving the vendor's ability to serve more satisfactorily and as a basis for making future purchase decisions.
III. To periodically evaluate the written evaluation aspects relating to quality, quantity, price and service pertaining to the vendor.

Advantages of Vendor Rating:

i. An efficient vendor rating system obtains the best vendor for the buyer organization which saves both and money and also achieve the best value for money spent on purchases.
ii. Provide scientific vendor rating system, which improves the image of the purchasing executive.

iii. Rates the entire performance of the vendors and is not guided by a narrow vision.

iv. Provide the buyer additional information on the capabilities of the vendor with regard to know-how, testing, transport, contractual willingness etc.

v. Objectively compares performance of vendors and improves their performance.

vi. Decision to disqualify , blacklist or discriminate the existing vendors can be arrived at on an objective basis.

vii. Inculcate a competitive spirit among vendors.

viii. Vendors rating can be used as an important mechanism to allocate the share of business amongst a large number of supplier.

ix. Vendor rating not only enables the buyer organization to know the performance of their suppliers but also gives an idea as to how much they out-perform their competitors regarding price, quality, lead-time, split shipment, service , technical assistance, meeting in emergency, ad hoc requirement of buyers etc.

Question

1. Discuss the need for vendor development and vendor rating.
2. Discuss different levels of Vendor evaluation.
3. Explain the various steps involved in vendor evaluation and selection process.
4. Distinguish between “vendor evaluation” and “vendor rating”.
5. Discuss factors to be considered in vendor evaluation.
6. State the objective of vendor rating.
7. Mention Advantages of vendor rating.
8. Discuss the parameters of vendor rating
9. State the necessity of Vendor development.
10. State the importance of vendor Development.
11. Discuss different types of Vendor development and State its advantage.
12. Discuss different techniques and procedures of vendor development.

Legal aspects of purchasing. Reference to Contract Act, Sale of Goods Act, Drug Control Act in respect to purchase activities. Import substitution

Legal Aspects of Buying

Legal aspect of purchasing – rules and regulation regarding purchasing

The Indian "sale of goods" act,1930 and "Indian contract Act", 1872 cover some of the important legal aspects.

Materials manage and purchasing manager should know some of the legal aspects, such as 1) law of agency, 2) Law of contracts, 3) law pertaining to sales of goods and 4) Arbitration.

1) **Law of agency**: - in this era trans-nation business enables a business man to depend up on the service of a person is called Agent.

Agent: - An agent is person employed to act on behalf of another called principal, for the purpose of bringing the principal in to contractual relationship with third person.

Essential of relationship of Agency:

i) Agreement between principal and agency.

ii) Intention of agent to act on behalf of principal.

1) Agreement:

- Agency depends on agreement but, not on contract. There is no consideration is necessary.
- Principal has to agree to be represented by the Agent is sufficient determinant.
- This relationship can be created by the express agreement and by the implied agreement.

a) Expressed agreement: it is usual power of attorney within stamp paper.
b) Implied agreement:- Agency is not contractual in nature.

   A ⟶ B ⟶ C

2) law of contract:

The Indian contract Act, 1872 (Act IX of 1872) lays down certain general rules regarding contracts. Sec (6) of Indian Law of Contract provides that "***An agreement when enforceable by law***" is a contract.

A contract has been defined as "an agreement enforceable by law" or "a legal binding, agreement between two or, more persons b which rights are acquired by one or more to act or, bearance on the part of other or others.

- In the definition there are two things – agreement and enforceability by law.
- To form an agreement there must be n offer from one party and acceptance by another party.

  Agreement = offer+ acceptance.
- But, social agreement is not contract. Contracts are those agreement which are enforceable by law courts. So , *contract = Agreement + enforceable by law.*
- *Therefore, all contracts are agreement but all agreements are not contract.*

Essential Elements of Contract

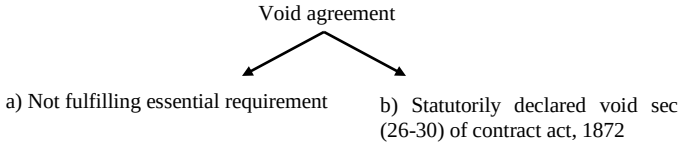
1) Offer and acceptance: There must be lawful offer by a party and lawful acceptance by another party.
2) Legal relationship: The agreement must create legal relationship between two parties.
3) Lawful consideration: The agreement to be enforceable by law must be supported by a consideration. The agreement is legally enforceable only when the parties give something and get something in return.

4) Capacity of parties -- competency: The parties to the agreement must be competent o enter in to the contract.

He or she is of –

- Sound mind
- Age of majority.
- Not disqualified from contracting by any law to which he is a subject.

5) Free and genuine consent:- Both the parties must be of the same min on all the material terms and condition.
6) Lawful object: the object must be lawful.
7) Agreement not declared void – the agreement must not have been declared void by any law of the country.
8) Certainty and possibility of performance: the terms and condition must be certain and not vague.
9) Legal formalities: the agreement may be oral or, in writing. Where it is in writing is must comply with the necessary legal formalities as to writing, registration and attestation.



a) Statutorily declared void:

Sec 26: Any agreement restraining the degree of freedom of choice of marriage in respect of adult is void.

Sec 27: Freedom of commerce, trade, business, unreasonable restriction on commerce, trade, business to any one is void.

Sec 28: Full restriction on legal proceeding is void.

Sec 29: If language of a contractual not meaningful and no capable to mean anything that is vague, then the agreement is void.

Sec 30: (wagering agreement or contract) – it is an agreement either parties stands to gain or loose depending upon the outcome of an uncertain event over which both the parties have no control , is void.

3) SALE OF GOODS ACT, 1930

When an offer to sell or buy goods for a price is made , It is known as contract of sale. However, it is binding only when accepted by the buyer and his acceptance is communicated to the seller on the terms and conditions is equivalent to a rejection of the original offer.

Definition of sale

Sale:- sale is defined by Indian Sale of Goods Act, 1930 as:

" A contract of sale of goods is a contract whereby the seller transfer or agrees to transfer the property in goods to the buyer for a price" .

from the above definition of a sale, the following may be derived as essential features of a sale:

1) A sale is a bilateral contract.
2) Money consideration is a must for a sale of goods.
3) There is distinction between sale and contract of work and material. But, according to *Avtar singh* " the dividing line between the two is not very clear. The only conclusion that can be drawn from English authorities is that every case must be judged of by itself".
4) Goods must be a subject –matter of the contract for sale. The " goods" is defined by Section 2(7) of Indian Sale of Goods Act, 1930 as under:

   "*goods means every kind of movable property other than actionable claims and money; and includes stocks and shares, growing crops, grass, and things attached to or forming part of the land which are agreed to be severed before sale or under the contract of sale"*

Condition: - It is stipulation in a contract of sale of goods, which is so essential to the fulfillment of the contract.

Important Ingredients of a transaction of sale:

- There must be two competent parties. (buyer and seller)
- There must be definite goods in the contract of sale.
- Goods must be in deliverable form.

- Title of goods or effects transfer of goods.
- Must be concluded by delivery in one hand and payment of price (consideration in one hand)

Document for title of goods:-

- Dock warrant
- Bill of loading
- railway receipt, etc.

4) **Arbitration:** Arbitration is a judicial process under which one or more outsider render binding award in the merit of the dispute.

   Voluntary arbitration is only in a name, in reality it is same as adjudication.

Legal Aspect of Buying

Question 1

Discuss some of the legal aspects of buying. Explain the law of agency in detail.

Or

Discuss the legal duties and responsibilities of purchase manager regarding buying of materials.

Answer:

It is the duty of purchase manager to see the finance available before placing order for purchases. The purchase manager being a capable person is expected to know the basic laws regarding the law of agency, law of Sale of Goods Act, law relating to sales tax, law relating to transport, law relating to excise etc.

Law of Agency

In legal term such persons who provide services are called agents. This is a law pertaining to the relationship between the agent and the principal and is known as law of agency.

Section 182 of the Act defines an agent as "person employed to do any Act for others or to represent another in dealing with third persons. The person for whom such Act is done, or who is represented, is called the "principal". The contract of law may be in writing or oral; when their law is in writing, it is in the form of "Power of Attorney".
The above definition is wide enough.

Essentials of an Agency

There are two essentials for agency: (a) Agreement between the principal and his agent and (b) Intention of agent to act on behalf of the principal.

(a) ***Agreement:*** It is very essential that agency will be created when there is an agreement. It is also true that no consideration is needed for creation of an agency. The fact that the principal has agreed to be represented by the agent is sufficient to principal to support the contract of agency. It is true that the agent is remunerated by the principal. The relationship between agent and principal may be created by the express agreement and by implied agreement.

Agency by Express Agreement: A contract of agency may be created by an express agreement. Normally the authority given by the principal in his agent binds the principal by the acts done within the scope of his authority. The usual form of written contract of agency is the power of Attorney on stamped paper.

Agency by Implied Agreement: Agency is not necessarily contractual in nature. The existence of agency may in certain cases be implied when it is to be inferred from the circumstances of the case. For example, 'A' owns a shop in Delhi himself living in Kolkata. The shop is being looked after by 'B' and during the course of business, he buys ordinary goods from 'C' for the purpose of running the ship, and pays for them out of 'A's fund with his knowledge. Here, 'B' has implied authority from 'A' to order goods from 'C' and make payment in the name of 'A'.

The implied agency arises when the principal conducts himself towards the agent or the third parties, in such a manner as if the principal had conceded to the appointment of the

agent. Implied agency includes Agency by Estoppels, Agency by Holding out, and Agency by Necessity. A detailed study of such agencies is beyond the scope of this text.

Creation of Agency

The law of agency between agent and principal can be created in any of the following ways:

v) By appointment either by words or by writing;

vi) By conduct either the principal or the agent behaving as principal and agent respectively though the concerned parties might have not expressed in clear terms;

vii) By situation created by the parties though again there may not have been any expression in clear terms;

viii) By necessity of the case, that is, agency of necessity;

ix) By ratification of an act not authorized previously.

Duties of an Agent:

The following may be described as duties of an agent:

1. To avoid conflict of interest between him and his principal in particular.
2. To execute the mandate given to him by the principal.
3. To follow instructions or accepted customs. The agent is bound to go by the directions of his principal. The agent may be held liable for the loss, if he does not abide by the instructions or follow the accepted customs.
4. To take reasonable care and skill, which is expected to a man of average caution.
5. Not to delegate since agent himself is the creation of delegation by the principal.
6. Not to make secret of any profit which he has earned due to creation of the agency.
7. To remit money whatever has been received on behalf of the principal and which is due to him.

Rights of an Agent:

The following are the some of the important rights of an agent:

1. Right to remuneration
2. Right to retain the principal's money till his claims are settled.
3. Right of lien of goods, other property, etc., till his claims are settled.
4. Right to indemnity which is extended to all losses and expenses incurred by the agent during the course of the business which he does for his principal.
5. Right to compensation, if the principal exposes himself to unreasonable risks by giving directions to his agent to work for him and in the process, of course, working as per the principal's direction, the agent suffers a loss which is to be compensated by the principal.

Section 225 in this connection provides:

"The principal must make compensation to his agent in respect of injury caused to such agent by the principal's neglect or work of skill."

Contract of Sale or Law of Sale

Question 2

Explain the law relating to sale of Goods Act 1930. Discuss the main points of difference between sale of goods and agreement to sale.

Answer:

When an offer to sell or buy goods for a price is made, it is known as contract of sale. However, it is binding only when accepted by the buyer and his acceptance is communicated to the seller on the terms and conditions are equivalent to a rejection of the original offer.

Definition of Sale

Sale Defined: "Sale" is defined by Indian Sale of Goods Act 1930 as: "A contract of sale of goods is a contract whereby the seller transfer or agrees to transfer the property is goods to the buyer for a price."

From the above definition of a 'sale', the following may be derived as essential features of a sale:

1. A sale is a bilateral contract.
2. Money consideration is a must for a sale of goods.
3. There is distinction between sale and contract of work and material. "the dividing line between the two is not very clear. The only conclusion that can be drawn from English authorities is that every case must be judged of by itself."
4. Goods must be a subject-matter of the contract for a sale. The "Goods" is defined by Section 2(7) of the "Indian Sale of goods Act 1930 as:

"Goods means every kind of movable property other than actionable claims and money and includes stocks and shares, growing crops grass, and things attached to or forming part of the land which are agreed to be served before sale or under the contract of sale."

Difference between Sale and Agreement to Sell: A sale is different from an agreement to sell.

Section 4(3) of the Act gives the distinction between the two and states:

"Where under a contract of sale the property in the goods is transferred from the seller to the buyer, the contract is called a sale, but where the transfer of the property in the goods is to take place at a future time of subject to some condition thereafter to be fulfilled, the contract is called an agreement to sell."

Difference between Agreement to Sell and Hire-Purchase

The following are the distinguishing points between an agreement to sell and a hire-purchase.

Sl. No.	Criteria	Agreement to sell	Hire purchase
1	***Entitlement***	An agreement to sell does not entitle to the intending purchaser to the possession of the goods agreed to be sold on a future date or on the fulfillment of	But a hire purchaser is entitled to have the possession of goods he has agreed to purchase on hire-purchases basis.

		some conditions as may have been laid down.	
2	***Title of goods***	In an agreement to sell the question of transferring the goods to the purchaser does not arise.	In a hire-purchase case too the title of goods remains with the seller and thus it is not transferred to the purchase though the possession and permission to use the goods is given to the purchaser.
3	***payment***	In an agreement to sell such a situation never arises. Neither there arises a question of any payment in an agreement to sell.	The hire-purchaser is the hirer and he has to pay the installments regularly as per the settlement failing which he invites the forfeiture of the amount he has already paid besides the lien which the seller enjoys over the goods.
4	***Law***	Agreement to sell is governed by the Sale of Goods Act.	The hire-purchase is governed by the Hire-Purchase Act
5	***Sales-tax***	Sales tax is not levied till they become a sale according to respective laws.	Sales tax is not levied till they become a sale according to respective laws. (Same as Agreement to sell)

Formalities of the Contract of Sale: The formalities of the Contract of Sale is clearly spelt out by Section 5 of the Act which lays down as given below:

(1) "A contract of sale is made by an offer to buy or sell goods for a price. The contract may provide for the immediate delivery of the goods or immediate

payment of the price or both, or for the delivery or payment by installments, or that the delivery or payment or both shall be postponed."

(2) "Subject to the provision of any lay for the time being in force, a contract of sale may be made in writing or by work of mouth, or partly in writing and partly by word of mouth or may be implied from conduct of the parties".

The Subject-mater of the Contract of Sale: The subject-matter of the Contract of Sale can only be 'goods' as defined by Section 2(7) of the Act.

The Price in a Contract Sale: Section 9 of the Act defines the price in a Contract of Sale. In fact this Section has not attempted a definition. Instead a through Section 9(1) and Section 9(2) enunciates *two principles* for the determination of the price under a Contract of Sale. The two sub-sections read as follows:

Section 9(1):

"The price in a contract of sale may be fixed by the contract or may be left to be fixed in a manner thereby agreed or may be determined by course of dealing between the parties."

Section 9(2):

"Where the price is not determined in accordance with the above principle. The buyer shall pay the seller a reasonable price. What is reasonable is a question of fact dependent on the circumstances of each particular case."

'Conditions' and 'Warranties': Section 12(2) of the Act defines the **conditions** as reproduced below:

"A condition is a stipulation essential to the main purpose of the contract, the breach of which gives rise to a right to treat the contract as repudiated."

Section 12(3) of the Act defines the warranties as given below:

"A **warranty** is a stipulation collateral to the main purpose of the contract, the breach of which gives rise to a claim for damages but not to a right to reject the goods and treat the contract as repudiated."

Section 12(4) of the Act concludes that the stipulation in a contract of sale *depends on the construction of the contract.* It lays down:

"Whether a stipulation in a contract of sale is a condition or a warranty depends in each case on the construction of the contract. A stipulation may be a condition though called a warranty in the contract."

The Implied Conditions in a Contract of Sale: The following are the implied conditions in a contract of sale:

1. ***Condition to the title:*** Section 14 provides that, "In a contract of sale, unless the circumstances of the contract are such as to show a different intention there is an implied condition on the part of the seller that in case of a sale, he has a right to sell the goods and that, in the case of an agreement to sell, he will have a right to sell the goods at the time when the property is to pass."
2. ***Sale by description:*** Section 15 provides that "where there is a contract of sale of goods by description, there is an implied condition that the goods shall correspond with the description."
3. ***Sale by description and also by sample:*** Section 15 further lays down that "….if the sale is by sample as well as by description, it is not sufficient that the bulk of the goods corresponds with the sample, if the goods do not also corresponds with the description. "Thus, the goods must correspond with the sample as well as with description, if the sale contract is affected taking both the stipulations into active consideration of the purchaser as well as of the seller.
4. ***Buyer beware:*** **The lesser is guarded against by exceptions provided in the Act to the Principle of Caveat Emptor** when the Act makes a provision under Section 16 which reads as follows:

   "Subject to the provisions of this Act and any other law for the time being in force, there is no implied warranty or condition as to the quality or fitness for any particular purpose of goods supplied under a contract of sale."
5. ***Goods of merchantable quality:*** Section 16(2) provides "where the goods are bought by description from a seller who deals in goods of that description (whether he is the manufacturer or producer or not) there is an implied condition that the goods shall be of merchantable quality."

6. ***Goods should be fit for the buyer's purpose:*** Section 16(1) of the Act provides that "where the buyer, expressly or by implication, makes known to the seller the particular purpose for which goods are required, so as to show that the buyer relies on the seller's skill and judgment and the goods are of a description which it is in the course of the seller's business to supply (whether he is the manufacturer or producer or not), there is an implied condition that the goods shall be reasonably fit for such purpose."

Sale by Sample: Section 17(1) says that "a contract of sale is a contract for sale by sample where there is a term in the contract, express or implied, to that effect."
Section 17(2) further explains and lays down three conditions which are implied in every contract of sale by sample. Section 17(2) (*a*), (*b*) and (*c*) read as under:

12(2) (*a*) "That the bulk shall correspond with the sample in quality."

(*b*) "That the buyer shall have reasonable opportunity of comparing the bulk with the sample."

(*c*) "That the goods shall be free from any defect rendering them un merchantable which would be apparent on reasonable examination of the sample."

Implied Warranties: According to the Act the following warranties are implied in every contract of sales of goods:

1. ***Quite possession of the goods:*** Section 14(*b*) of the Act provide that if there is no agreement to the contrary there is an implied warranty that "the buyer shall have and enjoy quite possession of the goods."
2. ***The goods should be free from any charge or encumbrance:*** Section 14(*c*) of the Act provides that "the goods shall be free from any charge or encumbrance in favour of any third party not declared or known to the buyer before or at the time when the contract is made."

Circumstances Reducing Conditions to Warranty: There are certain circumstances in which a condition can be reduced to warranty. When this happens:

1. the buyer loses his right to reject the goods supplied to him by the seller and

2. The buyer in such cases is left with no alternative but to be content with the remedy to claim damages for the breach of the condition.

The following courses are open to buyer when a condition is reduced to warranty under certain circumstances:

1. The buyer may waive off the condition.
2. The buyer may opt for the alternative of treating the breach of condition as the breach of warranty.
3. The buyer cannot reject the goods hence he is required to accept them as supplied by the seller particularly when the contract of sale is not severable and the buyer has either accepted the goods in full or any part thereof.

Exclusion of Liability for Implied Terms: Section 62 of the Act provides "where any right, duty or liability would arise under a contract of sale by implication of law, it may be varied by express agreement or by the course of dealing between the parties, or by usage, if the usage is such as to bind both parties to the contract."

There are *three* modes by which the liability for implied terms may be varied as envisaged by this Section of the Act:

1. By Express Contract;
2. By Course of Dealing, and
3. By Usage.

Stipulations as to Time of Payment: Section 11 of the Act lays down that "unless a different intention appears from the terms of the contract, stipulations as to time of payment are not deemed to be of the essence of a contract of a sale."

Passing of Goods sold on Approval: Goods sold on approval should either be accepted or rejected within the time specified or within reasonable time. If without rejecting the goods the buyer retains the goods up to the specified period or does not return the same within reasonable time then the goods would be deemed to have been purchased by the buyer and buyer becomes liable for the payment.

Section 24 of the Act lays down rules the essence of which is that the goods sold posses to the buyer either by acceptance or by failure to return within the stipulated time or within reasonable time.

Sale of Unascertained Goods: Section 18 of the Act provides that "where there is a contract for the sale of unascertained goods, no property in the goods is transferred to the buyer unless and until the goods are ascertained."

In this connection Section 23 also makes the following provision:

"Where there is a contract for the sale of unascertained or future goods by description and goods of that description and in a deliverable state are unconditionally appropriated to the contract either by the seller with the assent of the buyer or by the buyer with the assent of the seller, the property in the goods thereupon passes to the buyer."

Section 23(2) of the Act recognises one mode of appropriation in explicit terms when it lays down:

"When, in pursuance of the contract, the seller delivers the goods to buyer or to a carrier for the purpose of transmission to the buyer, and does not reserve the right of disposal, he is deemed to have unconditionally appropriated the goods to the contract."

Section 26 under section **"Risk *prima facie* passes with property"** deals with transfer of risk in a contract of sale of goods where risk and property go together. It reads as given below:

"Unless otherwise agreed, the goods remain at the seller's risk until the property therein is transferred to the buyer, but when the property therein is transferred to the buyer, the goods are at the buyer's risk whether delivery has been made or not."

Question 3.

Explain contract of sale? Discuss the law of breach of contract.

Answer:

When a person offers to sell or buy goods for a price, it is known as contract of sale. In general, orders for purchases are placed through telephones; in order to make these binding a formal purchase order should be forwarded.

The proposal cannot be revoked once the acceptance is put in a course of transmission to the proposer. In a similar manner the acceptance cannot be revoked once the communication of the acceptance comes to the knowledge of the proposer. Suppose, Ram proposes to sell an item to Rahim by a letter sent by post. Rahim accepts the proposal by a letter of post. Ram can revoke this proposal at any time before Rahim posts his letter of acceptance but not afterwards. Rahim can revoke his acceptance at any time before his letter teaches Ram but not afterwards. However, the party who give time to another to accept or reject an offer is not bound to wait till the time expires.

Special Contracts: The materials managers place contracts where a rate is agreed upon and the quantities to be supplied from time to time are left to be intimated by periodical releases. Here, the buyer does not commit himself to lift any minimum quantity. Such a contract is deemed to be only a standing offer and its acceptance only means that the buyer recognises this standing offer for the stated period. However, once a release order is issued by the buyer the contract becomes binding for that quantity. The supplier may feel free to withdraw his standing offer with reference to future order. This is not possible if the tender and its acceptance were intended by the buyer and the seller to be a contract involving an obligation on both of them during that time period.

Frequently contracts are entered into, stipulating a specific quantity to be lifted during a stated period, with clauses providing some variations in the quantity or the period during which it is to be lifted. This is a binding contract and enforceable for the supply of the entire quantity of goods contemplated in the contract. There is another type of contract in which, though the buyer is not bound to any specific quantity, he commits to buy and pay for all the goods needed by him or goods that are produced by the seller in a stipulated period. This again is a binding contract. a purchase contract becomes null and void when the price to be paid is not stipulated in the order or at least the manner in which the price is to be fixed is not indicated.

However, according to the Indian Sale of Goods Act, stipulations regarding time of payment are not condition of the contract. In other words, they are not of essence. This is because a delay in payment does not make the contract without consideration. But, if the contract is for the delivery of goods by installments and each installment is to be paid for

on delivery, then failure to pay for any one installment is adequate for the seller to repudiate the contract. When the time of payment for each installment is of the essence. In determining whether the time was of the essence of the contract the court will consider the intention of the buyer and the seller as expressed in the contract as well as in the course of dealing between the parties or usage, surrounding circumstances, equitable considerations, etc. Ordinarily, however, when a specific time is fixed, it is treated to be of essence and the ones of proving that it is not so lies with the one who asserts so. To quote Indian Sales of Goods Act, "Unless a different intension appears from the terms of the contract, stipulations as to the time of payment are not deemed to be of the essence of a contract of sale. Whether any other stipulation as to time is of the essence of the contract or not depends on the terms of the contract."

Condition and Warranty: A condition is a stipulation in contract of sale pertaining to goods which are the subject of contract and this stipulation is essential to the main purpose of the contract, the breach of which will make the contract liable to be rejected. A warranty is also a stipulation in contract of sale pertaining to goods which are the subject of contract *but* this stipulation is only of a collateral nature to the main purpose of the contract, the breach of which gives rise to a claim for damages but does not make the contract liable to be repudiated and hence the goods cannot be rejected.

Sale by Description: It is not uncommon to find orders which have only a description of the item in question. By implication, it follows that items must follow the description. It pays in the long run to make certain that such purchase orders are complete and clear in their description. This could be done by mentioning the end use, furnishing drawings, tolerances, finish etc., referring to the national or international specifications, and the tests which the item has to pass.

There are cases when the supplier produces the item under the full knowledge as to how exactly it will be used. In such a case there is an implied warranty in the sense that the buyer can claim damages when the item later proves to be unfit. There is a mistaken notion that there is an implied condition in the case of a contract for the sale of a specific item under its patent. There is actually no implied condition as to the fitness of the item for any specific use. Similarly, statements regarding the quality of the product and

performance appearing in commercial advertisements are not to be construed as an express or implied warranty.

Aspects in Contracts Based on Samples: Goods sold on the basis of samples give the buyer many advantages. The buyer must be able to establish the existence of a sample as well as a clear understanding to the effect that the sample is of the contract standard. For this purpose the sample needs to be preserved for future inspection.

Works Contract. This type of contract pertain to the supply and fixing of materials of buildings and such immovable properties. The important factor in works contract is that it is indivisible, *i.e.,* it cannot be looked upon as a contract of sale of goods plus a contract for labor. From the materials manager's point of view a good amount of saving can result by way of reduction in sales tam through resorting to works contract.

We will now see the difference between a guarantee and warranty. A warranty is a part of the contract through only collateral to the main object of the contract. it is a statement of something which a party undertakes to do. A guarantee is something which the promiser undertakes, to be answerable to the promise for the debt, default, etc., of another person whose primary liability to the promise himself exists or is contemplated. Therefore, it is an additional and conditional contract.

We will discuss the legal aspect involving the payment of price. It has three dimensions, viz., time of payment, place of payment and mode of payment. As regards the time of payment, the amount should be paid when it is due. The buyer has no right to delay it till a demand is made. As regards the place, it should be the place previously appointed. If unspecified it should be at the seller's place. The mode of payment is very important. In the purchase order, this should be clearly spelt out. Factors like sales tax, octroi, customs duty, excise duty, escalation classes, discounts, credit terms, dispatch terms, such as FOB, CIF, etc., must be clearly laid down. If the buyer complies with the mode of payment suggested in the contract, he is safe even if the money is stolen or does not reach the seller.

Risks and Rights: It is necessary for the buyer to recognize that the risk of deterioration of goods during transit rests with him even though the seller might have agreed to deliver

the goods at his own risk at a place other than where they are when sold. The buyer, has his rights. After reasonable verification within stipulated time buyer can accept the goods and unless it has been done it can not be called as "accepted". If the buyer fails to inspect the goods within a reasonable time and determine whether they conform to contractual terms, then he is deemed to have waived his right. If by mistake the buyer has failed to discover defects then subsequently he has a right to claim damages.

Acceptance of Goods: Acceptance by the buyer is complete only when he has intimated the seller of his acceptance, or when he fails to intimate the seller even after a reasonable time, during which he had opportunities to inspect the goods, or he does not act in relation to them which is inconsistent with the ownership of the seller. The buyer has another important right. Unless agreed upon, the buyer is under no obligation whatsoever to return the goods which he has rejected as they do not conform to contractual requirements.

Breach of Contract

Discharge of contract by breach is also provided in the Act. A breach of contract takes place when:

i) any of the contracting party renounces his liability under contract,
ii) any of the contracting party acts in such a manner as to make it impossible to discharge his liability under the contract, and
iii) Any of the contracting party totally or partially fails to perform and fulfill his obligations under the contract.

The breach may be *anticipatory* or *present*. An anticipatory breach occurs when there is prior repudiation of the contract by any of the party. Two legal effects emerge from this. First, the party concerned may be excused from performance or from further performance. Secondly, the injured party, however, is entitled for the following:

1. For compensatory damages, which is not penal.
2. For compensation for mental pain and suffering.
3. For mitigation of damages.
4. For liquidated damages and penalty.

The party which sustains a loss due to breach of contract will be so compensated with respect to damages that it will be placed in a position as if the contract has been performed. In commercial transactions the aggrieved party can claim only the actual loss caused to him and not to others.

Damages arising out of a breach of contract in the usual course are **recoverable**. Damages which arise owing to circumstances peculiar to the case and which would not have arisen during the normal course are **not recoverable**. Unless the circumstances are made known to the party which breaks the contract and such damages are contemplated by both the parties at the time of entering into the contract. The amount of compensation by way of replacing the damages will be, as far as possible, the amount which will place the hurt party in the same position as it would have been, had it not suffered the breach of contract which led to such damages. Very often the parties wish to reduce damages to a certainty. This, they achieve by fixing a rate or sum at which damages are to be assessed. If such an amount is a genuine pre-estimate of the damages, then it is known as liquidated damages. However, if the amount fixed is not at per with the greatest loss that could have followed form the breach, it would be considered a penalty. As per the Indian Contract Act, it is for the court to ascertain the actual loss and award the same. A similar provision is available for security deposit and earnest money. If the amount is small and in reasonable proportion to the price then the forfeiture of the deposit or earnest money paid by the defaulting party is not considered as penalty. If the amount is a large part of the price, then forfeiture of the amount will be in the nature of penalty. In such cases the Court will not permit forfeiture but will only award reasonable compensation.

Patent Rights: The buyer can protect his organisation by adding special patent infringement clauses to his purchase orders. However, when the buyer orders to his own specification, this does not apply. It is often the case that big buyers give financial and technical assistance to the supplier for developing new products and materials. It is advantageous, in such cases, for the buyer to include a clause in the purchase orders, covering the patent rights in inventions discovered by a supplier in the course of executing his purchase contract.

Settlement of Disputes: Arbitration is used for settling disputes between buyers and sellers. It is less time consuming when compared with litigation. For this purpose the buyer must include an arbitration clause and the supplier has to agree to it. Once such a clause is stipulated, the court will not entertain any suit unless the arbitration process is exhausted. Arbitrators have similar powers the courts and their award is binding and enforceable on the parties.

Transfer of Title

Question 4

What are the rules regarding transfer of title in the contract of sale?

Answer:

Section 27 of the Indian Sale if Goods Act lays down the principle of *nemo dat quod non habet.* Explaining this ancient maxim it provides as follows:

"Subject to the provisions of this Act and of any other law for the time being in force, where goods are sold by a person who is not the owner thereof and who does not sell them under the authority or with the consent of the owner, the buyer acquires no better title to the goods than the seller had."

Justice Deamin has clarified the position of modern law in the following words:

"In the development of our law, two principles have given for mastery. The first is the protection of property: no one can give a better title than he himself posses. The second is the protection of commercial transaction: the person who takes in good faith and for value without notice should get a good title. The first principle held away for a long time but it has been modified by the common law itself and by statute so as to meet the needs of our times."

Rules as to Delivery of Goods

1. ***Place of delivery:*** In the absence of any contract to the contrary, goods should be delivered at which the respective parties are having their business at the time of sale.

2. ***Time of delivery:*** In the absence of any contract to the contrary, goods should be delivered within a reasonable time.
3. ***Delivery by allotment:*** In the absence of any contract to the contrary, if the goods are in possession of any third person there cannot be any delivery unless the third person acknowledges to the purchaser that he is in possession at the goods on his behalf. Once he acknowledges, it amounts to a delivery to the purchaser.
4. ***Time for tender of delivery*** is said to be a reasonable hour when the buyer can demand the delivery from the seller.
5. ***Delivery of wrong quality: Short delivery:*** If the goods are diverted in short quantity as per the provisions of Section 37(1) the buyer has a right to reject them. The buyer, however, has every right to accept the goods in short supply but he has to pay only for the goods delivered to him and not for the ordered goods.
6. ***Delivery in wrong quantity: Excess delivery:*** If the goods is delivered in excess quantity as per the provision of Section 37(2), the buyer has a right to accept the contracted goods. The rest he may reject and may make payment for the goods actually accepted by him. However, the buyer has every legal right to reject the whole consignment, if it is in excess.
7. ***Delivery of mixed goods:*** Here also the buyer has a legal right to either accept the goods contracted and reject those goods which do not conform to the description of the contract or he may reject the whole lot.
8. ***Delivery in instalments:*** If otherwise not provided in contract for sale of goods according to Section 38(1) of the Act, "the buyer of goods is not bound to accept delivery thereof by instalments."

Deteriorating during Transit: Regarding deterioration during transit, Section 40 of the Act lays down as follows:

"Where the seller of goods agrees to deliver them at his own risk at a place other than that where they are when sold, the buyer shall, nevertheless, unless otherwise agreed, take any risk of deterioration in the goods necessarily incident to the course of transit."

No Rejection after Acceptance: Section 42 of the Act provides:

(a) If the buyer has once accepted the goods and has already intimated the fact to the seller he has no right to reject them.

(b) If the goods have already been delivered to the buyer and he does any act in relation to them which is inconsistent with the ownership of the seller, he will have no right to reject them.

(c) It the goods have already been delivered to the buyer and he has retained them without intimating to the seller within stipulated or reasonable time that he has rejected the goods, he will have no right to reject them.

An Unpaid Seller: According to the provisions of Section 45 of the Act:

"The seller of goods is deemed to be an 'unpaid seller' within the meaning of this Act—

(a) when the whole of the price has not been paid or tendered;

(b) when a bill of exchange or other negotiable instrument has been received as conditional payment, and the condition on which it was received has not been fulfilled by reason of the dishonour of the instrument or otherwise."

Right of an Unpaid Seller: The law provides that:

(a) An unpaid seller has a lien on the goods for the price if he is in possession of the said goods;

(b) An unpaid seller has a right to stop the goods in transit after he has parted with the possession of the sold goods and he comes to know about the insolvency of the buyer.

(c) An unpaid seller has a right to re-sell the goods.

Seller's Remedies against Buyer for Breach of Contract: The seller has a right:

i) To sue the buyer for the price, if he has not been paid and the goods have passed to the buyer.

ii) To claim damages for non-acceptance and move the court if he so wishes. Section 56 notes: Where the buyer wrongfully neglects or refuses to pay for the goods, the seller may sue him for non-acceptance.

Buyer's Remedies against Seller for Breach of Contract: The buyer has a right:

(a) To claim damages for non-delivery. Section 57 provides: where the seller wrongfully neglects or refuses to deliver the goods to the buyer, the buyer may sue the seller for damages for non-delivery.

(b) To reject the goods if there is breach of warranty or where the buyer elects or is compelled to treat the breach of condition as a breach of warranty.

Law of Contract

Question 4
What is contract of sale? What are the essentials of a valid contract?

Answer:

From the materials manager's point of view, the Law of Contract is an important branch of Mercantile or Commercial Law, because all commercial transactions start from an agreement between two or more people. The object of the Law of Contract is to introduce definiteness in commercial and other transactions. It may be said that the purpose of the Law of Contract is to ensure the realisation of reasonable expectation of the parties who enter into a contract. The Law of Contract may be defined as that branch of law which determines the circumstances in which a promise shall be legally binding on persons making it. The Indian Contract Act of 1872 (Act IX of 1872) lays down certain general rules regarding contracts.

Definition of Contract: "Contract is an agreement enforceable by law." — *Section 2*(*b*) *of Indian Contract Act, 1872*

Thus, the definition points out two basic features of a contract. To make contract there must be:

1. an agreement, and
2. The agreement entered into should be such which is capable of being enforced by law, *i.e.*, an enforceable agreement can only be classed as a contract in the eyes of law.

"Agreement", defines Section 2(*e*), is "every promise and every set of promises forming the consideration for each other." And a promise is an accepted proposal. Section 2(6) of the Contract Act defines the promise as a proposal when accepted, becomes a promise.

Therefore, in simple words it can be said that an agreement is an accepted proposal. Two points emerge from this:

1. Every agreement is made of a proposal, and
2. the proposal is accepted.

Hence, there should be two parties to make an agreement—the proposer and the acceptor. The accepted proposal, called agreement, should be enforceable by law.

From enforceability clause emerges that every contract is an agreement but every agreement cannot be a contract under the law. The agreement to become a contract has to satisfy the following conditions:

(i) There should be some consideration for the agreement.

(ii) The parties entering into agreement should be legally competent to contract.

(iii) The contracting parties should necessarily give free consent. There should be no coercion or pressure from any side.

(iv) The object of the agreement and of the parties should be lawful. The object should not be contrary to legal provisions.

To conclude, Contract = Agreement + Enforceability at law. All contracts, therefore, are agreements, but all agreements are not contracts.

An agreement may be made in writing, by word of mouth, by reference from the conduct and circumstances of the case or by any combination of the above methods.

Essential Elements of a Valid Contract: An agreement is a wider term than a contract, because an agreement may or may not be enforceable by law. It is extremely important to know what agreements are contracts in law. In respect of an agreement which is not a contract, no legal remedy is available to either party, if the other party fails to carry out the agreement.

A contract, in order to be enforceable by law, must have the following essential elements:

1. ***Offer and Acceptance:*** There must be a lawful proposal or offer by one party and a lawful acceptance of that proposal by the other party thus resulting in an agreement. The terms of offer and acceptance must be definite. The acceptance of the offer must be according to the mode prescribed and must be communicated to the proposer. A proposal when accepted becomes a promise.

2. ***Legal Relationship:*** The agreement must create legal relationship between the parties. If there is no intention to create legal relationship, there can be no contract between the parties.
3. ***Lawful Consideration:*** The agreement to be enforceable by law, must be supported by a consideration. The term consideration in simple words means 'something in return'. The agreement is legally enforceable only when both the parties give something and get something in return. A promise to do something, getting nothing in return, is usually not enforceable by law.
4. ***Capacity of parties—Competency:*** The parties to the agreement must be capable of entering into a valid contract. Every person in competent to enter into a contract if he is of sound mind, age of majority, and not disqualified from contracting by any law to which he is a subject. Flaws in capacity to contract may arise from minority, lunacy, idiocy, drunkenness, and status.
5. ***Free and Genuine Consent:*** It is essential to the creation of every contract that there must be free and genuine consent of the parties to the agreement. The parties must agree upon the same thing in the same sense, *i.e.*, they must be of the same mind on all the material terms of the contract.
6. ***Lawful Object:*** The object of the agreement must be lawful. It is lawful unless it is forbidden by law or is of such a nature that if permitted, would defeat any provisions of law.
7. ***Agreement not Declared Void:*** The agreement, though it might posses all the essential elements discussed above, must not have been expressly declared void by any law in force in the country.
8. ***Certainty and Possibility of Performance:*** The terms of the agreement must be certain and not vague or ambiguous. If the agreement is vague and if it is not possible to ascertain the meaning of agreement, it cannot be enforced. The terms of agreement must enable performance. An agreement to do an act impossible in itself cannot be enforced.
9. ***Legal Formalities:*** The agreement may be oral or in writing. Where it is in writing it must comply with the necessary legal formalities as to writing,

registration and attestation. If the agreement does not comply with these legal formalities it cannot be enforced by law.

Responsibilities of Purchase and Materials Managers

Question 5

What are the general rules relating to insurance especially for materials purchase manager.

Answer:

The responsibilities of the purchase manager as regards to insurance starts from the time he places an order with his supplier till the materials reach the hands of the ultimate consumer. He arranges:

(a) to ensure the materials in transit,

(b) to insure the materials in store,

(c) to analyse the insurance needs,

(d) to see that adequate insurance cover is provided to (*a*) incoming materials, (*b*) materials in storage, (*c*) outgoing materials, (*d*) work-in-progress and finished goods till they are in the charge of the materials department.

(e) to see that ocean voyage (in case of imports) is fully insured against all odds according to the terms, conditions and policy of the organisation.

Goods on Ocean Voyage

All *marine ocean voyage policies* lay down certain conditions which must be borne in mind while dealing with insurance of goods being imported. The following should be sincerely thought of and actions taken accordingly.

1. On delivery of goods either to the consignee or to the warehouse, etc. on the advice of either the consignee or the consignor insurance cover ceases to be effective.
2. The insurance cover ceases after the expiry of 60 days after the completion of discharge of the goods form the overseas vessel at the final port of discharge.

3. The insurance cover ceases as soon the goods start for the next destination after the discharge from the overseas vessel.
4. The insurance cover remains in operation if there is a change of the voyage or termination of the voyage before the goods reach the final destination provided a prompt notice is given and the insurer pays additional premium.

Goods in Inland Transit

The following points should be borne in mind for getting the goods insured while in inland transit:

(i) 'All risks' should conveniently be covered.

(ii) Commencement of policy should be clearly specified.

(iii) Termination clause should be clearly specified.

Usually the policy commences at the time of loading the goods and terminates (ceases to be in operation) seven days after the arrival of the inland carried at the destination town or on delivery whichever is earlier.

(iv) Period of storage at the destination should be clearly specified during which the risk will be covered. Also it should be clearly specified as to what type of insurance risk — theft, fire, pilferage etc., — will be covered during storage period. Usually 'fire risks' are covered in the absence of any specific terms.

Insurance of Goods in Storage: During storage goods are usually covered against the following risks and insurance policy is taken for either all of them (comprehensive) or any one or more of them:

1. Fire	5. Flood
2. Explosion	6. Earthquake
3. Riot	7. Theft and burglary
4. Strike	8. Pilferage etc.

Claims Management

Since claims settlement takes time and involves considerable effort, an Insurance or Claims Officer would be well advised to take the following points into considerations:

1. He should remember the termination date of the policy his organisation has taken.

2. He should take prompt action for the extension of period of insurance cover if the termination date is near or about to cease.
3. He should get the insurance cover extended well before the materials reach the ultimate destination.
4. He should intimate the insurer about the change, if it is to take place, in the ordinary course of transit, well before it actually takes place.
5. He should not give a clean receipt where he has any doubt regarding the condition of the goods received. Of course, he may receive goods in doubtful condition under written protest.
6. He must give written notice to the carrier or his representative within three days of the receipt of delivery if there is (*i*) any loss to the goods, (*ii*) any damages not apparent at the time of delivery but subsequently detected on opening the packing.
7. He must submit a prompt written claims statement for the loss, damage, etc.
8. He should take care that the claim so submitted should bear documentary proof for quick claims settlement the important documents are as listed below:

   i) The original policy,

   ii) Bill of Lading or Railway Receipt or Transport Agency Receipt,

   iii) Invoice.

   iv) Survey Report,

   v) Claim, Bill, and

   vi) Letter of Subrogation
9. He should make every effort to minimize the loss and risk by specifying the nature and quality of packing.

Minimising Insurance Cost and Self-Insurance: There are only two ways of minimising expenditure on insurance: First, by reducing the element or risk involved in materials movements, and secondly, by introducing self-insurance to take care of small and fairly predictable losses and seeking the insurer's assistance only for protection against catastrophic loss.

Risk Improvement and Loss Minimisation: The natural mode of transport and the extent of care exercised in materials management have a bearing on the probability of losses

occurring and on the quantum of such losses. Whenever there is an improvement in any of these, the insurers should be persuaded to consider a reduction in premium rates. It is sometimes profitable to discuss risk and loss minimization measures with the insurers themselves, who may come out with suggestions based on their experience over a wider range of clients.

It appears a sound policy, from the point of view of cost reduction, to consider self-insurance for certain risks, especially where the size of operations is large. The cost of processing claims has to be taken into account while calculating the expenditure on insurance. Self-insurance could operate more profitably for the transit of finished goods than for incoming materials, because, in the case of finished goods, all the risk factors attendant on transit are within the organisation's control and the experience of losses over the past few years should be readily available to ascertain the pattern of transit losses. But to transfer the insurance responsibility to the carriers is an unwise move. Self-insurance could be introduced progressively by first excluding claims for less than a specified amount and working up to a stage where only the most restricted form of insurance to take care of catastrophic losses is taken. But these are decisions which have to be taken in each case depending on the circumstance, taking into account the accumulation of risk, the maximum possible loss arising out of an accident, the administrative expenses in dealing with insurers, suppliers and buyers, regarding claims and the tax angle.

A systematic and periodic, preferably, an annual review of the existing insurance arrangements of the organisation, the cost and benefits, should be undertaken. The insurance needs of a large and complex organisation are ever changing and the insurance cover designed with care today may not remain a perfect fit two years from now.

Import substitution

Import Policy or International Purchase

For under developed and developing country like us the dependency is now not possible. So, we have to depend on other country in some manner i.e., we have to import some of the material from foreign countries. It is one of the most important economic functions.

To facilitate the easy flow of goods , the govt. announces its import policy usually in the beginning of new financial year i.e., on 1st April. The Import Trade control (ITC) ,policy popularly known as Red book is issued in two volume. These policies are periodically updated by amendments depending upon socio-economic and political factors.

Volume 1: deals with the policy of government with regard to users' import of materials.
Volume 2: deals with government policy with regard to registered exporters.

For spare parts:

- Import f spare parts, the procedures for which have been released, is usually on a repeat basis i.e., for same value as in earlier year.
- In case of non –priority industries, a consolidated license for spare parts and raw material is issued.
- Small scale industries can have a special status when they apply through their respective sponsoring authority.
- Application for import of capital goods must go through the sponsoring authorities. There is a special capital goods committee, which clears these applications on the basis of various factors, such as availability from indigenous sources, essentiality of import, availability of foreign exchange, etc.

- For emergency requirement of spare parts the procedures are simplified and there are special forms.
- Policy is more liberal for registered exporters.
- They are given preferential treatment in the allocation of foreign exchange.

They have a duty concession, like refund or exemption from payment from central excise duty, customs,/excise duty drawback on raw material used for the exported items.

Import policy: new import policy 1984-1985 by the then commerce minister V.P.Singh, announced new import-export policy. Then Narasimha Rao Govt. in 1992-1997 announced import – export policy which goes far beyond mere balancing of import and export or a favorite balance of trade. Trade policy should be spear head for better technology, greater investment and for efficient performance.

As the goal of a self reliant and dynamic economy has not been sacrificed, one has discarded certain restrictive odd assumptions. India is able to shed export pessimism and have finally realized that export and imports, taken together, could account for as much as 20% of gross Domestic Product and today aiming at the attainment of this level of trade.

2.11.1 Highlights of Import Policy, 1992-97

General guide lines:

3. Trade is free, subject only to a negative list of imports and a negative list of export which may pruned now and then with an aim to make the trade as free as the country's economic and trade environment warrants.
4. Stable policy for five years 1992-97 subject to review every quarter.
5. Simplicity and transparency is the sole aim of the new policy.

Import Guidelines:

1. Negative list of import is the smallest one.
2. Consumer goods will continue to remain under restraint.
3. Import of 3 items banned, 68 items restricted, and 8 items canalized.
4. Special import facilities for i) hotels and tourism industry, and ii) sports bodies.

5. Negative list to be administered by general schemes. Case –by-case licensing is minimized.
6. Actual user condition is eliminated except in a few special cases.
7. Import of capital goods liberalized. Capital goods have been removed from negative list of imports.
8. Import of second-hand capital goods allowed subject to certain sector wise restriction. In some sectors it is allowed with out license and in others under license.
9. Export promotion Capital Goods (EPCG) scheme is liberalized and two windows have been opened for concessional duty imports.

A) Rate of concessional customs duty	i) 25%CIF value
	ii) 15%CIF value.
B) Export obligation	i) 3 times CIF value
	ii) 4 times CIF value
C) Period of fulfillment of export	i) 4 years
	ii) 5 years.

10. EPCG scheme extended to components of capital goods with concessional customs duty of 15% .
11. Import of raw material liberalized barring few items.
12. All licenses under duty exemption schemes transferable.
13. Permission is accorded to install machinery on lease.
14. Special import licensing schemes makes the following three categories wherein special import licenses may be issued.
15. National campaign for quality awareness, along with up-gradation and accreditation of laboratories/ testing houses, to be launched.

    Procedures are to be made simple, transparent and easy to administer

IMPORT SUBSTITUTION – WHAT AND WHY IN INDIA

Substitution means to replacement of one material by another for performing the same function. Import substitution refers to indigenous source development so as to:

1. Get the materials indigenously.

2. Conserve valuable foreign exchange.
3. Reduce the cost of production.
4. Encourage and develop our industry and trade.

India is a developing economy. The policy of import substitution should enshrined as a major national objective. As far as possible the process of import substitution is the resultant effect of many national instruments aimed at achieving self-reliance as well as a favourable balance of payment position. The programmes of the various government departments like the planning commission, dept. of science and technology, director General of technical Development(DGTD), Chief controller of Import and Export(CCIE), Metals and minerals trading corporation(MMTC), State trading corporation(STC), etc should be proper coordinated to achieve the landable objective of self reliance. Thus import substitution implies substituting what is imported by indigenous items as far as practicable.

Strategy for Import Substitution

A new strategy for import substitution has to be based on: (a) Enlightened corporate policy of source development, (b) standardization, (c) National technology policy.

Many organizations like B.H.E.L., T.E.L.C.O, H.M.T., L.&T., T.V.S. group etc. have established company policies for source development. Financial and technical assistance are provided by some of these companies for developing indigenous suppliers. Many organizations pay only lip service to source development and they strongly feel that tangible results will be obtained only if the developments of suppliers are interwoven with the long term corporate plans of the buying organizations.

Source Development: In developing indigenous sources the Companies have to consider initial problems relating, to high cost, poor workman ship, need for technical help, availability of raw materials etc. These problems will arise only when the supplier is building up his competence. Hence the companies should take a long-term view and should, go out of their way to help the supplier technically and financially. A few companies ever go to the extent of allowing the sophisticated imported machines to be opened up and the drawings scrutinized by the suppliers. Some manufacturing organizations encourage their own craftsmen and engineers to become entrepreneur and suppliers. The concept of the industrial estate is a consequence of this logic. However, some of the industrial-estates are not flourishing in the highly industrialized parts of the country like Bangalore, Poona, etc. because they are asked to concentrate only on minor jobs and sufficiently challenging orders are denied to them.

Standardization: Standardization has been accepted as a basic necessity for mass production. It is a continuous and time consuming process. In the case of industrially developing countries, a systematic approach for establishing priorities, and a total view of industrial sectors is called for. Due consideration has to be given, for the rapid advances taking place in the technological fields in the world. The Indian Standards Institution has developed over 7000 standards.

Role of Materials Managers: Materials managers who are in change of source

development and import substitution can play a significant role in this direction. They should encourage the engineers to adopt value analysis techniques which is a very fruitful area for import substitution. In this context the country has made a significant break through by replacing imported copper through indigenous aluminium.

3. Quality Control & quality management-Principles & methods.

Quality Definition

Joseph Juran ~ Quality is fitness for use.

Juran's definition, though very simple, is very much generic in nature. It covers also all the basic elements of all other definitions. A best quality walking shoe does not serve any good purpose of a 'sports-shoe'. Each product or service has some specific purpose(s) to serve. That product or service which serves its intended purpose better, is taken to be of better quality.

The driving force which determines the level of quality is the customer. As their needs and expectations change, so should the level of quality be changed. This fact led **Feignbaum (Total Quality Control)** to say that "Quality is a customer determination, not an Engineer's determination, or a general Management determination. It is based upon this customer's actual experience with the Product or service, measured against his/her requirements – stated or unstated, conscious or merely sensed, technically operational or entirely subjective – and always operating a moving target is a competitive Market."

Dimensions for Tangible Product:

i) **Performance** ~ Colour and picture quality as well as audio-reproduction (Operation characteristic)

ii) **Features** ~ Remote and other available controls (What the product provides)

iii) **Reliability** ~ How frequently the repairing is needed. (Product survivability)

iv) **Maintainability/Serviceability** ~ How difficult/expensive to repair (Ability for repair)

v) **Durability** ~ How long it will last (i.e., life span)i.e., Economic length of use

vi) **Conformance** ~ How well it meets the design specifications (Physical characteristic)

vii) **Aesthetics** ~ How does it look.

viii) **Perceived Quality** ~ Brand image, Feeling confident about quality from past experience.(Subjective Assessment)

Introduction

Total Quality Management (TQM), a buzzword phrase of the 1980's, has been killed and resurrected on a number of occasions. The concept and principles, though simple seem to be creeping back into existence by "bits and pieces" through the evolution of the ISO9001 Management Quality System standard.

Companies who have implemented TQM include Ford Motor Company, Phillips Semiconductor, SGL Carbon, Motorola and Toyota Motor Company.

The latest changes coming up for the ISO 9001:2000 standard's "Process Model" seem to complete the embodiment. TQM is the concept that quality can be managed and that it is a process. The following information is provided to give an understanding of the key elements of this process.

Total Quality Management (TQM)

Total = Quality involves everyone and all activities in the company.

Quality = Conformance to Requirements (Meeting Customer Requirements).

Management = Quality can and must be managed.

TQM = A process for managing quality; it must be a continuous way of life; a philosophy of perpetual improvement in everything we do.

TQM Compared to ISO 9001

ISO 9000 is a Quality System Management Standard. TQM is a philosophy of perpetual improvement. The ISO Quality Standard sets in place a system to deploy policy and verifiable objectives. An ISO implementation is a basis for a Total Quality Management implementation. Where there is an ISO system, about 75 percent of the steps are in place for TQM. The requirements for TQM can be considered ISO plus. Another aspect relating to the ISO Standard is that the proposed changes for the next revision (1999) will contain customer satisfaction and measurement requirements. In short, implementing TQM is being proactive concerning quality rather than reactive.

TQM as a Foundation

TQM is the foundation for activities which include;

- Meeting Customer Requirements
- Reducing Development Cycle Times
- Just In Time/Demand Flow Manufacturing
- Improvement Teams
- Reducing Product and Service Costs
- Improving Administrative Systems Training

Ten Steps to Total Quality Management (TQM)

The Ten Steps to TQM are as follows:

1. Pursue New Strategic Thinking
2. Know your Customers
3. Set True Customer Requirements
4. Concentrate on Prevention, Not Correction
5. Reduce Chronic Waste
6. Pursue a Continuous Improvement Strategy
7. Use Structured Methodology for Process Improvement
8. Reduce Variation
9. Use a Balanced Approach
10. Apply to All Functions

Principles of TQM

The Principles of TQM are as follows:

1. Quality can and must be managed.
2. Everyone has a customer and is a supplier.
3. Processes, not people are the problem.
4. Every employee is responsible for quality.
5. Problems must be prevented, not just fixed.
6. Quality must be measured.
7. Quality improvements must be continuous.
8. The quality standard is defect free.
9. Goals are based on requirements, not negotiated.
10. Life cycle costs, not front end costs.
11. Management must be involved and lead.
12. Plan and organize for quality improvement.

Processes must be Managed and Improved

Processes must be managed and improved! This involves:

- Defining the process
- Measuring process performance (metrics)
- Reviewing process performance
- Identifying process shortcomings
- Analyzing process problems
- Making a process change
- Measuring the effects of the process change
- Communicating both ways between supervisor and user

Key to Quality

The key to improving quality is to improve processes that define, produce and support our products.

All people work in processes.

People

- Get processes "in control"
- Work with other employees and managers to identify process problems and eliminate them

Managers and/or Supervisors Work on Processes

- Provide training and tool resources
- Measure and review process performance (metrics)
- Improve process performance with the help of those who use the process

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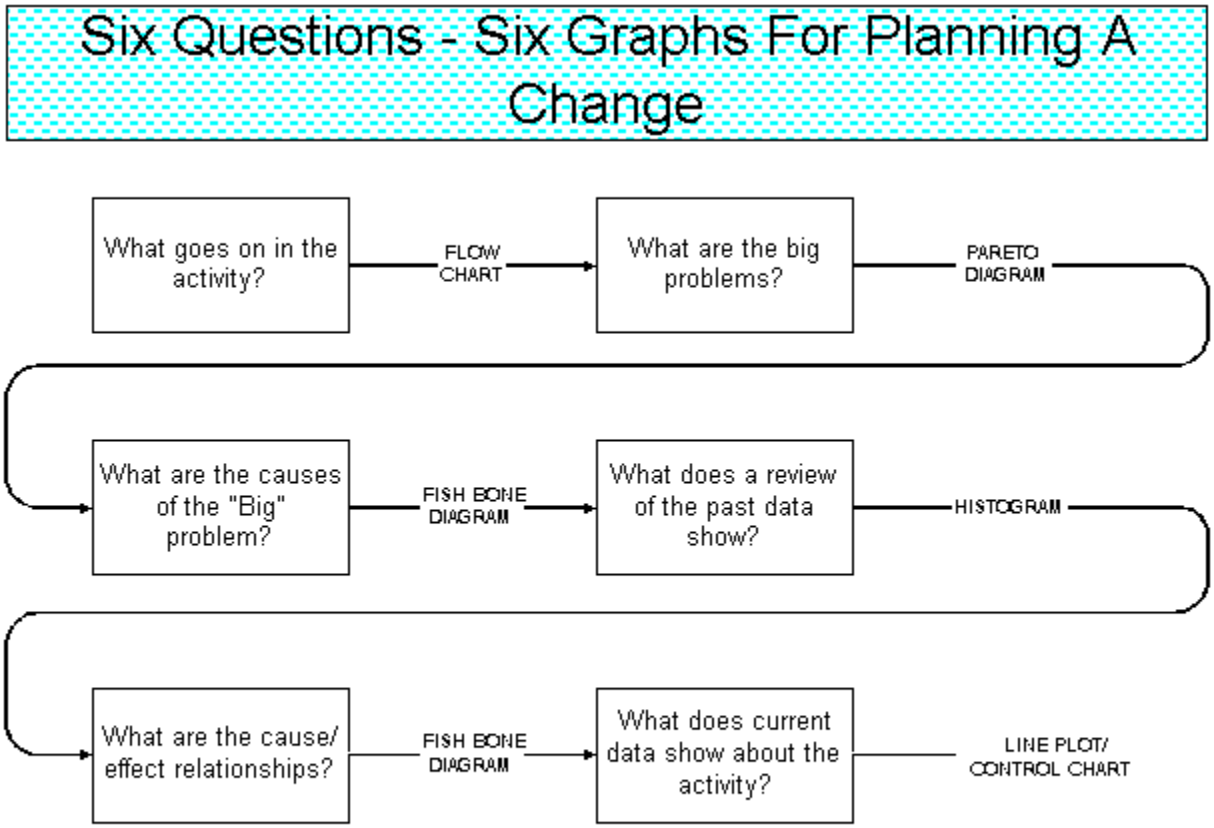
Planning a Change



TQM Process Improvement and Problem Solving Sequence					
PLAN (PLAN A CHANGE)			**DO** (IMPLEMENT THE CHANGE)	**CHECK** (OBSERVE THE EFFECTS)	**ACTION** (EMBED THE FIX INTO THE PROCESS FOR GOOD)
DEFINE THE PROBLEM	**IDENTIFY POSSIBLE CAUSES**	**EVALUATE POSSIBLE CAUSES**	**MAKE A CHANGE**	**TEST THE CHANGE**	**TAKE PERMANENT ACTION**
1. Recognize that what you are doing is a "PROCESS" 2. Identify the commodity being processed. - Process Inference 3. Define some measurable characteristics of value to the commodity. 4. Describe the "PROCESS" o Process Flow Analysis's o Flow charts o List of steps 5. Identify the "Big" problem	6. "BRAINSTORM" what is causing the problem. 7. Determine what past data shows. o Frequency distribution o Pareto charts o Control charts - sampling	8. Determine the relationship between cause and effect o Scatter diagrams o Regression analysis 9. Determine what the process is doing now o Control charts - sampling	10. Determine what change would help • Your knowledge of the process • Scatter diagrams • Control Charts - sampling • Pareto analysis ****Then make the change.	11. Determine what change worked (confirmation). • Histograms • Control charts - sampling • Scatter diagrams	12. Ensure the fix is embedded in the process and that the resulting process is used. Continue to monitor the process to ensure: A. The problem is fixed for good. and B. The process is good enough o Control charts - sampling ****To ensure continuous improvement, return

o Brainstorming o Checklists o Pareto analysis					to step 5.

Dimensions for Tangible Product / Concept of QUALITY IN SERVICE

- TIME
- TIMELINESS
- COMPLETENESS
- COURTESY
- CONSISTENCY
- ACCESSIBILITY
- ACCURACY
- RESPONSIVENESS

Determinants of Quality

A key issue is how to achieve 'Quality'. There are three distinct but very much interrelated aspects of Quality. These are **(i) Quality of design**, (ii) **Quality of conformance (iii) Quality of performance.**

Quality of design is the degree of achievement of purpose by the design itself. The more sophisticated a design, the more it cost.

Quality of conformance is assessed by the faithfulness with which the actual product or service conforms to the design-specification. It is obvious that both the quality of design and the quality of conformance must tally to make a good quality product.

Quality performance is another name for Reliability, which is determined by his probability that an item will perform its intended performance without failure is a given period of time under the specified environmental conditions.

A product or service most meet all the three requirements so that it is deemed to be of good quality.

OBJECTIVES

- Quality Management is the scientific search for the most effective way to deliver the best care to the patients; is about saving lives and money
- Quality can continuously improved over time.
- When standards of care are met, the standards are raised, and the improvement continues, and costs continue to decrease.

- Quality involves: ***appropriateness, availability, continuity, effectiveness, efficacy, efficiency, respect and caring, safety and timeliness.*** (Tabish, OUP, CH 76 p599)

Characteristics of Quality:

- QUALITY IS:
  - Defined By Customer
  - A Measure Of Achievement Of Customer Satisfaction
  - Fulfilling The Customers' Needs / Requirements
  - Keeping One's Word
  - Ensuring No Defects
  - Ensuring Fitness Of Use
  - Image Of The Company And Customer Confidence In The Organisation
  - A Precise And Measurable Variable
  - Utility To Society

Quality management principles

- To lead and operate an organization successfully, it is necessary to direct and control it in a systematic and transparent manner. Success can result from implementing and maintaining a management system that is designed

- to continually improve performance while addressing the needs of all interested parties. Managing an organization encompasses quality management amongst other management disciplines.

- Eight quality management principles have been identified that can be used by top management in order to lead the organization towards improved performance.

a) **Customer focus**
Organizations depend on their customers and therefore should understand current and future customer needs, should meet customer requirements and strive to exceed customer expectations.

b) **Leadership**
Leaders establish unity of purpose and direction of the organization. They should create and maintain the internal environment in which people can become fully involved in achieving the organization's objectives.

c) **Involvement of people**

People at all levels are the essence of an organization and their full involvement enables their abilities to be used for the organization's benefit.

d) **Process approach**
A desired result is achieved more efficiently when activities and related resources are managed as a process.

e) **System approach to management**
Identifying, understanding and managing interrelated processes as a system contributes to the organization's effectiveness and efficiency in achieving its objectives.

f) **Continual improvement**
Continual improvement of the organization's overall performance should be a permanent objective of the organization.

g) **Factual approach to decision making**
Effective decisions are based on the analysis of data and information.

h) **Mutually beneficial supplier relationships**
An organization and its suppliers are interdependent and a mutually beneficial relationship enhances the ability of both to create value. These eight quality management principles form the basis for the quality management system standards within the ISO 9000 family.

INTERNATIONAL STANDARD ISO 9000:2000

Quality management systems — Fundamentals and vocabulary

1. Scope:

This International Standard describes fundamentals of quality management systems, which form the subject of the ISO 9000 family, and defines related terms.
This International Standard is applicable to the following:

a) organizations seeking advantage through the implementation of a quality management system;

b) organizations seeking confidence from their suppliers that their product requirements will be satisfied;

c) users of the products;

d) those concerned with a mutual understanding of the terminology used in quality management (e.g. suppliers, customers, regulators);

e) those internal or external to the organization who assess the quality management system or audit it for conformity with the requirements of ISO 9001 (e.g. auditors, regulators, certification/registration bodies);

f) those internal or external to the organization who give advice or training on the quality management system appropriate to that organization;

g) developers of related standards.

2. Fundamentals of quality management systems

2.1 Rationale for quality management systems

— Quality management systems can assist organizations in enhancing customer satisfaction.

— Customers require products with characteristics that satisfy their needs and expectations. These needs and expectations are expressed in product specifications and collectively referred to as customer requirements.

— Customer requirements may be specified contractually by the customer or may be determined by the organization itself. In either case, the customer ultimately determines the acceptability of the product. Because customer needs and expectations are changing, and because of competitive pressures and technical advances, organizations are
— driven to improve continually their products and processes.

— The quality management system approach encourages organizations to analyse customer requirements, define the processes that contribute to the achievement of a product which is acceptable to the customer, and keep these processes under control. A quality management system can provide the framework for continual improvement to increase the probability of enhancing customer satisfaction and the satisfaction of other interested parties. It provides confidence to the organization and its customers that it is able to provide products that consistently fulfil requirements.

2.2 Requirements for quality management systems and requirements for products

— The ISO 9000 family distinguishes between requirements for quality management systems and requirements for products.

— Requirements for quality management systems are specified in ISO 9001. Requirements for quality management systems are generic and applicable to organizations in any industry or economic sector regardless of the offered product category. ISO 9001 itself does not establish requirements for products.

— Requirements for products can be specified by customers or by the organization in anticipation of customer requirements, or by regulation. The requirements for products and in some cases associated processes can be contained in, for example, technical specifications, product standards, process standards, contractual agreements and regulatory requirements.

2.3 Approaches to measurement of Quality:

- QUALITY MANAGEMENT IS THE:
  - Scientific way to deliver the best care to patients
  - About saving lives and money
  - Continuous improvement over time
  - Standards of care are met, the standards are raised, the improvement process continues, the costs continue to decrease

- Application of quality management to clinical processes can reduce post operative infection and eliminate many adverse drug reactions, reducing costs to hospital and patients.

- QUALITY INVOLVES:
  - APPROPRIATENESS
  - EFFICACY
  - EFFICIENCY
  - RESPECT AND CARING
  - SAFETY
  - TIMELINESS.

- CONTINUOUS QUALITY IMPROVEMENTS:

Customer-driven quality

TQM has a customer-first orientation. The customer, not internal activities and constraints, comes first. Customer satisfaction is seen as the company's highest priority. The company believes it will only be successful if customers are satisfied. The TQM company is sensitive to customer requirements and responds rapidly to them. In the TQM context, `being sensitive to customer requirements' goes beyond defect and error reduction, and merely meeting specifications or reducing customer complaints. The concept of requirements is expanded to take in not only product and service attributes that meet basic requirements, but also those that enhance and differentiate them for competitive advantage.

Each part of the company is involved in Total Quality, operating as a customer to some functions and as a supplier to others. The Engineering Department is a supplier to downstream functions such as Manufacturing and Field Service, and has to treat these internal customers with the same sensitivity and responsiveness as it would external customers.

TQM leadership from top management
TQM is a way of life for a company. It has to be introduced and led by top management. This is a key point. Attempts to implement TQM often fail because top management doesn't lead and get committed - instead it delegates and pays lip service. Commitment and personal involvement is required from top management in creating and deploying clear quality values and goals consistent with the objectives of the company, and in creating and deploying well defined systems, methods and performance measures for achieving those goals. These systems and methods guide all quality activities and encourage participation by all employees. The development and use of performance indicators is linked, directly or indirectly, to customer requirements and satisfaction, and to management and employee remuneration.

Continuous improvement
Continuous improvement of all operations and activities is at the heart of TQM. Once it is recognized that customer satisfaction can only be obtained by providing a high-quality product, continuous improvement of the quality of the product is seen as the only way to maintain a high level of customer satisfaction. As well as recognizing the link between product quality and customer satisfaction, TQM also recognizes that product quality is the result of process quality. As a result, there is a focus on continuous improvement of the company's processes. This will lead to an improvement in process quality. In turn this will lead to an improvement in product quality, and to an increase in customer satisfaction. Improvement cycles are encouraged for all the company's activities such as product development, use of EDM/PDM, and the way customer relationships are managed. This implies that all activities include measurement and monitoring of cycle time and responsiveness as a basis for seeking opportunities for improvement.

Elimination of waste is a major component of the continuous improvement approach. There is also a strong emphasis on prevention rather than detection, and an emphasis on quality at the design stage. The customer-driven approach helps to prevent errors and achieve defect-free production. When problems do occur within the product development process, they are generally discovered and resolved before they can get to the next internal customer.

Fast response
To achieve customer satisfaction, the company has to respond rapidly to customer needs. This implies short product and service introduction cycles. These can be achieved with customer-driven and process-oriented product development because the resulting simplicity and efficiency greatly reduce the time involved. Simplicity is gained through concurrent product and process development. Efficiencies are realized from the elimination of non-value-adding effort such as re-design. The result is a dramatic improvement in the elapsed time from product concept to first shipment.

Actions based on facts
The statistical analysis of engineering and manufacturing facts is an important part of TQM. Facts and analysis provide the basis for planning, review and performance

tracking, improvement of operations, and comparison of performance with competitors. The TQM approach is based on the use of objective data, and provides a rational rather than an emotional basis for decision making. The statistical approach to process management in both engineering and manufacturing recognizes that most problems are system-related, and are not caused by particular employees. In practice, data is collected and put in the hands of the people who are in the best position to analyze it and then take the appropriate action to reduce costs and prevent non-conformance. Usually these people are not managers but workers in the process. If the right information is not available, then the analysis, whether it be of shop floor data, or engineering test results, can't take place, errors can't be identified, and so errors can't be corrected.

Employee participation

A successful TQM environment requires a committed and well-trained work force that participates fully in quality improvement activities. Such participation is reinforced by reward and recognition systems which emphasize the achievement of quality objectives. On-going education and training of all employees supports the drive for quality. Employees are encouraged to take more responsibility, communicate more effectively, act creatively, and innovate. As people behave the way they are measured and remunerated, TQM links remuneration to customer satisfaction metrics.

- TOTAL QUALITY MANAGEMENT:
  - IS A STRUCTURED SYSTEMATIC PROCESS FOR CREATING ORGANISATION WIDE PARTICIPATION IN PLANNING AND IMPLEMENTING CONTINEOUS IMPROVEMENT IN QUALITY

Quality control /or statistical quality control

By inspection, quality of the product is conformed with standards and specification during or at the end of the production process. It is imperative to have certain disturbances with cause the product deviate slightly from the desired standard. This types of variability is inherent in the process and is known as **variability due to chance causes.** causes of deviations like conditions of production process , nature of raw materials, the behavior of operations , etc. such causes are **assignable causes.** Due to these causes , defective item are produced which lower the quality of the product.

Inspection has rather a limited use. It does not provide the extent to which certain products requiring high degree of precision conform to the strict standards set in this regard. Similarly, it does not provide all the information on the basis of which substandard finished products can be scrapped in the case of mass production industries, producing standardized product. These imperfections of inspection in quality control makes the use of statistical quality control indispensable in this cases.

Statistical Quality Control (SQC) is the application of statistical technique to determine how far the product conforms to the standards of quality and precision and to what extent its quality deviates from the standard quality. The purpose of statistical quality control is to discover and correct only those forces which are responsible for variations outside the stable pattern. The standard quality is predetermined through careful research and investigation.

Techniques of SQC:

The techniques of statistical quality control can be divided in to two major parts:-

1. Control chart and
2. Acceptance sampling

1. ***Control chart:*** Control chart is most important quality control technique. It is a chart and depicts three lines on the chart. One line is the central line showing the average size. The other two lines one below the central line and other above the central line, indicate the limits of the tolerances, within which deviation from standards permissible. The actual measurement values which fall outside the tolerance limits are considered to be out-of – control points and assignable cause may be said to exist. This will enable the manufacturers to know the causes of variation or causes of trouble which he could amend.
2. ***Acceptance Sampling:*** It can be described as the post – mortem of the quality of the product that has already been produced. Under this technique, a sample is selected at random to examine whether it conforms to the standards laid down. It can be assumed that a certain percentage of goods will not conform to the standards. So, certain percentage of defective products in a ot may be specified. This technique has all the limitations of sampling technique. There are two limiting levels of quality in an acceptance sampling plan.

i. **The Acceptable Quality Level (AQL)** that represents the lowest percentage of defectives which buyer is expected to accept and seller is expected to supply.
ii. **The Lot Tolerance Percentage Defective (LTPD)** that represents a limit at which the buyer wants to be quite certain that the lot will not be passed. the

acceptance sampling inspection may be of the following types: i) inspection by attributes, ii) inspection by variable, iii) inspection by number of defects per unit.

Importance or Benefits of Statistical Quality Control:

1. ***It saves on rejection:*** In absence of statistical quality control technique, many products may be found defective and worthless at the end of the manufacturing process and may be thrown away as scrap. SQC avoids such a situation and saves the cost of labor and material involved in production of defective items. It measures the extent of defect and defective products may be improved by re-working to the level of acceptable standard. It also saves loss which will arise out of re-working on the items rejected – out rightly or which cannot be brought to the acceptable standard.
2. ***It maintains high standard of quality:*** Statistical quality control ensures the maintenance of high standard of quality because lower standard of quality products are not marketed. They are improved, if possible, or rejected out rightly but in no case, substandard goods sold in the market. Thus the concern gains in goodwill.
3. ***Reduced expenses of inspection:*** It reduces the expenses of inspection to a great extent and enables the product to be manufactured at lower cost.
4. ***Ensures standard price:*** If certain products are not up to the desired standard of quality, and cannot be improved without much expense, they can be down graded and sold cheaper. SQC maintains the standard of the product, the producer is able to secure the standard price for all standard products, thus it increases the profitability of the concern.
5. ***Feeling of responsibility among workers:*** Among workers, a feeling of responsibility develops because they begin to under stand that their work is being inspected very minutely, hence, they work carefully. It helps increasing their moral.

4. Principles of storage & stores accounting-types of storage care & preservation of materials & equipments in inventory control.

Store keeping is a service function which deals with the physical storage of goods under the custodianship of a person called storekeeper or stock controller. Goods stored may be either "stores" or "stocks". Materials which are not used in work or raw materials are usually referred to as "stores" and the place where they are kept is known as "store room".

"Store function concerns receiving, movement, storage and issue of items – raw materials, bought out parts, tools, spares, consumable, etc.—required for production, maintenance, and operation of the plant and finished goods until its dispatched to customers."

"Stores , therefore is the custodian of all goods that are received in the company until they are consumed or sold and naturally it assumes the responsibility of receiving, storage, preservation, issue and accounting function".

Responsibilities of store management are:

- *to receive materials, to protect them while in storage from damage or authorized removal,*
- *To issue materials in the right quantities at the right time, to the right place, and to provide these services at the least cost.*

Objective of store management / objective of store

2) THE MOST IMPORTANT OBJECTIVE is to provide uninterrupted supply of the raw materials, equipments, tools, components, to the user department.
3) Receive the materials (raw materials, components, tools and equipments) and check the materials in respect to bills of materials.
4) Receive, inspect, and issue the materials.
5) Receive and issue finished goods
6) Accept, store, arrange scrap for disposal.
7) To provide adequate and proper storage & preservation so that obsolescence and damage of materials will be minimized.

8) To provide proper safety and security to materials & personnel.
9) To meet the demand of the consuming departments by proper issues and account for the consumption.
10) To minimize obsolescence, surplus, and scrap through proper codification, preservation and handling.
11) To ensure good house keeping, so that material handling, material preservation, stocking, receipt and issue can be done adequately.
12) To assist in verification and provide supportive information for effective purchase action.

Functions of Scientific Store Management

The working of store need to be organized to perform the following functions:

i. Requisitioning from purchasing department an economical quantity of material for delivery at the most appropriate time.
ii. Exercising control on quantity of material received.
iii. Storing and protecting materials against hazardous condition, weather, deterioration and pilferage.
iv. Issuing materials against properly authorized material requisitions.
v. Maintaining exact records of all receipt, issues, and balance to facilitate ordering of required materials.
vi. Maintaining adequate records of records of receipt and expenditures.
vii. Maintaining adequate stock of materials to serve production needs.
viii. Keeping inventory investment within desired limits.

In another Way We can Determine the Functions of Store Management

1) **Identification:-**it is the process of codifying and describing all items required to be stocked.
2) **Receipt: -** it is the process of accepting after inspection all materials required for use and (or) accepting of end product.
3) **Inspection:** It is the quality checking pf all incoming materials.
4) **Storage:** It is the process of storing the materials in the warehouse.

5) **Stores accounting:** It is charging each issue voucher in a manner so that the money spent is properly allocated.
6) **Stock control:** It is the process of provisioning , which mean continuously arranging, receipt and issuing of stocks so as to ensure required service consistent with economy.
7) **Stock record:** It has to be maintained day to day position of receipt issue.
8) **Stock taking:** It is the process of verifying physically the quantity of materials.

Corresponding with the above responsibilities and functions every store should have the following sections.

- Identification
- Receipt and inspection
- Stocking wards that look after storage issue.
- Dispatch
- Ledger, which looks after stock records.
- Stock verification.

9) **Issue and dispatch:** It is process of receiving demands from consumers and issuing the demanded materials without loss of time.

Functions of Store can be divided into Two Heads

1) Primary function and 2) Secondary function

1) Primary function:

a) To make available a balanced flow of raw materials, components, tools, equipment and other stores required for operation.

b) To provide maintenance materials spare parts and general stores as required.

c) To received and issue materials after physical inspection and proper identification.

d) Storage and preservation of materials.

e) Safety and security of materials.

f) To arrange for collection, acceptance of scrap & other discarded materials for disposal.

2) **Secondary function:**

- **a)** Quantity inspection and acceptance.
- **b)** Stores accounting.
- **c)** Stock control.
- **d)** Feedback information to material control section.
- **e)** Help in standardization and variety reduction.
- **f)** Service information such as:
  - i. Demand for materials and its specification quantity, etc.
  - ii. Notification when stocks are running low.
  - iii. Details of deliveries rejected on inspection.
  - iv. Certification of invoices for quantity and quality.
  - v. Particulars of anticipated change in consumption.
  - vi. Warning of items urgently required for breakdown.
  - vii. Listing of obsolete, surplus and scrap materials for disposal.

Benefits of Scientific Store Keeping / Advantages of Store Management

i) Scientific stock control reduces losses due to accumulation of inventories.

ii) Efficient stores issues reduce down time in production and increases profit.

i) Periodic review detects obsolete and non- moving items and helps the firm to get rid of unproductive inventory.

ii) Follow up with purchase, helps to avoid stock outs and the production losses.

iii) Proper record keeping, provides exact picture of inventory in store to higher level of management.

3.7 Types of Stores

According to function (functionally) stores are of five types.

1. Receiving stores.
2. Main store.
3. finished product store (warehouse)
4. Special store.
5. Scrap yard.

According geographical area stores are of four (4) types:

1. Central store
2. Regional store.
3. Divisional store.
4. Sub store.

Methods of Storing

Preservation of Stores

Storage means that as long as materials belong to the custody of stores department, they should free from damage and deterioration and maintain their required properties. preservation, technically is the protection of stores from heat, moisture, dust corrosion rust fire, etc so as to maintain materials in their original form.

Preservation measures serve three objectives:

a. To keep materials safe from all kinds of damage.

ii) To maintain materials to their original value and quality.

iii) To reduce storage losses and consequently to reduce production cost

Deterioration is caused by

I. climate and environment

II. physical and chemical agents

III. Biological agents.

General rules of preservation:

i) Materials should not be kept in direct contact with the floor; they should be kept on raised platforms. This prevents moisture from ground, dust, insect,

ii) All items those are perishable – should be issued on "first in first out" (FIFO) principle.

iii) Expiry date should be informed for replacement

iv) The store should be kept free from rodent menace for which the following measures are essential:

a) The floors should be in concrete or bricks.

b) Windows should be kept at least one meter above the floor level.

c) The edges of the windows should lined by tin plates. Should be sealed properly

d) Drains pipes should be kept provided with wire mesh inside.

e) The rat holes should be fumigated.

v) Pesticides and fungicides should be sprayed in order to remove termites, ants and fungi in the store.

Precautions and specific method of preservation

Preservation methods for a particular class of material depend on its properties, characteristics and special problems. The following are the preservation methods of some common items.

Leather goods:

i) They should be protected from exposure to sunlight and high temperature.

ii) They should be periodically turned to changes their location.

iii) Yellow soap powder and should be applied to make the goods soft and long lasting.

Wood and wooden product:

i) They should store above the ground so that the moisture does not attack them.

ii) They should be treated with creosote oil which makes the wood unattractive to insect to eat. Pesticides should be applied periodically to remove termites.

iii) They should be kept away from heat.

Textile and cloths:

i) Small textile goods should be stored in airtight container and boxes.

ii) The goods should be occasionally exposed to sun.

iii) Naphthalene balls should be placed layer and packed.

iv) DDT should be sprayed.

Rubber and rubber goods:

i) Rubber parts should be stored in dry and cool place.
ii) Rubber tubing and rubber sheet should not be folded.
iii) Oils and greases should not be allowed to contact with rubber items.
iv) Specially designed racks should be used for storage of rubber parts.
v) They should be periodically rotated to change their position so that they are not on the same end for long.
vi) Rubber goods being perishable should be issued on first in first out basis.

Metals (iron and steel):

i) The material should be stored at least 15 to 20 cm above the ground.
ii) These should not be stored in open space since sun rays are harmful particularly for bright finished steel.
iii) Metals should be painted by anti-corrosive paints or covered by a film of oil, grease or mineral jelly.
iv) Black bituminous paint should be applied to cast iron surface exposed to sun and rain.

Chemicals:-

i) It should be stored in cool and dry places.
ii) Air –conditioned room or places having refrigeration facilities are ideal location for storage of chemicals.
iii) All containers – drums, carboys, etc must be properly corked.

Paints:

i) These are inflammable and must be protected from fire.
ii) It should be kept in cool and dry place protected from heat and water.
iii) Paint drums should be periodically rolled to prevent sedimentation.
iv) Issue of paint should be first in first out basis.
v) It should not be exposed to atmosphere.

Machinery:

i) It should be kept in packed condition or in running condition.
ii) Sophisticated machines should be kept in air- conditioned room.

Security of Stores

Security of stores include measures against i) theft against outsider, ii) pilferage by employees, iii) mal practices by store staff, iv) precaution against fire protection , and v) pest control measures to control rodents and termite menace.

The measure should include:

- Daily closing of stores and custody of keys.
- Marking of stores.
- Storage of expensive and scarce materials.
- Movement control of incoming/ outgoing vehicle. Number should be taken in the time of
- Indemnity bonds or bank security from stores employees.
- Enquiry and disciplinary action against defaulter.
- Fire fighting equipment.
- Pest control measures.
- Theft by outsiders.
- Pilferage by employees.
- Malpractices by stores staff.
- Prevention of fire.
- Menace of rodents and termites.

1. Daily closing of stores :

1. Daily closing of stores should be properly supervised.
2. Stores keys must be numbered and registered. Written instructions should be issued nominating the persons responsible for them.

   During off duty hours, keys should be kept in a locked key safe under the supervision of a responsible person of watch and ward.

3. “Key movement” register should be maintained. Any one collection and depositing a key, particularly after the factory hours, should be required to sign the register.
4. All duplicate keys should be kept under the custody of the senior officer.
5. Loss or misplacement of a key on each occasion should be thoroughly investigated.

2. Theft by outsiders

To check theft by outsiders, the following measures should be taken:

1. The entire factory should have a high compound wall or high barbed wire fencing.
2. The number of windows, glass shutters and open ventilators should be minimum possible.
3. All windows and skylights should be capable of being securely fastened. They must also be fitted with bars and iron-mesh.
4. Security guards should be put on duty at strategic points.
5. Outsiders should not be permitted to enter inside the store beyond the serving counters.
6. During closed hours a watchman should be provided to look after the stores building or alternatively to have visits made at short intervals by the patrolling watchman.
7. Expensive and scarce items should be kept locked in steel almirahs.
8. Stores should be insured against burglary.

3. Pilferage by employees

Pilferage is gradual removal of material in small quantities by the employees of the company. The following steps should be taken to check pilferage by the employees.

1. Only authorized personnel should be allowed to enter the store.
2. No personal property should be allowed to be kept inside the store.
3. Items liable for pilferage should be monogrammed with company’s name or marked for some identification details before taking them into stock. Specific points to be observed are:
    - Heavy items generally do not require identification marks.

- Articles of domestic use such as hand tools, soaps towels, clothing, lamps etc. should be marked.
- Cutting tools should also be marked.
- Measuring instruments such as micrometers, verniers scales etc. particularly need marking.
- Costly office stationery liable for pilferage should also be marked.

4. Fresh materials may be used against return of old ones.
5. Surprise checks of a section of store items should be done now and then.
6. Store staff should be searched before they are allowed to leave the main gate after day's work.
7. Immediate enquiry should be conducted as soon as any case of malpractice is brought to light.
8. Prompt disciplinary action should be taken against defaulters (i.e., employees caught while pilfering).
9. The punishment for pilferage should be given adequate publicity.

4. Malpractices by stores staff

Malpractices are the manipulations by store personnel with active help of outsiders. The following steps help in checking malpractices by store staff:

1. Outgoing trucks, trolleys, parcels etc. should be thoroughly checked.
2. Gate passes should be made to take material out of the company.
3. Indemnity bonds or bank security must be taken from stores employees at the time of employment.

5. Prevention of fire

Fire is an accident caused by chemical reaction between combustible and oxygen. Possible causes of fire are:

1. Careless handling, storage and disposal of inflammable materials
2. Open electrical circuits causing short circuits.
3. Smoking by staff and clients in the store.
4. Spontaneous ignition due to slow oxidation with inadequate ventilation to remove heat.

5. Lightning.
6. Spark from shunting engine.
7. Auto-combustion

The following precautions should be taken towards fire prevention in stores:

1. Doors and staircases should be made of fire resistant materials
2. Smoking should be prohibited in and around the store houses.
3. Adequate earthing should be provided to the external and internal wiring of store buildings.
4. Materials should be properly stored. Leakage of inflammable oils, greases and fluids should be prevented.
5. Combustible materials should be stored in covered metal bins and if they are in large quantities, they should be stored in small lots at more than one location with enough gap to limit spread of fire.
6. Defective wiring should be immediately replaced to eliminate possibility of fire due to short circuit.
7. Since inflammable materials can undergo spontaneous ignition on account of slow oxidation coupled with inadequate ventilation to remove resulting heat, they should be stored appropriately.
8. Those parts of buildings where inflammable materials are kept should be well insulated to reduce risk of fire spread.
9. Exits should be provided with panic bolts and should open outwards.
10. Telephone should be provided at strategic location with ready access to telephone numbers of fire brigade.
11. Material inside the stores should be stocked keeping fire hazard in mind:
    a) Enough clearance (around half a meter) between wall and the storage rack should be provided.
    b) A minimum of meter of aisle space should be maintained between the storage racks.
    c) A meter of clearance is recommended between the stacked items and tube light fittings.

12. a) Appropriate fire fighting equipment (e.g. fire extinguishers, sand and water-buckets, water sprinklers etc.) should be provided at the appropriate places.

    b) Fire extinguishers should be located as near as possible to exits or stairs landing.

    c) Fire extinguishers should be visible and easily accessible (the bottom of the extinguishers should be 75 cm above the ground) with a completely unobstructed passage leading to them.

    d) A framed plan showing the location of the fire fighting equipment and other useful information should be displayed at the suitable places.

    e) Fire fighting equipment should be checked periodically and kept ready to be used during any eventuality.

13. a) In a large organization, a separate fire fighting department manned by trained manpower should be created.

    c) In small companies, 2-3 persons from each work area and all personnel from security and store should be given training on 2 to 3 occasions in a year.

    d) Smoke detectors should be installed in the storehouses to provide early warning.

    e) Fire alarms should be provided at appropriate place to sound emergency communication to all.

    f) Store buildings, stocks, machinery and other assets should be covered by fire insurance.

6. Rodents and termites

Measures against rodents and termite should be like the following:

1. If rodents is observed, the stores should be fumigated and rat poison traps kept.
2. For termites, white ants and fungi, suitable pesticides and fungicides should be sprayed.

Stores Receipts

Receiving concerns *control on quantity and quality of materials from the time they are received until they are accepted and taken into stock.*

Receiving is though a clerical operation but is the most important function of stores management. Its importance can be gauged from the following:

- Errors in purchase transaction can be detected more easily at the time of receiving of material rather than afterwards.
- Correctly performed receiving function can prevent malpractices.
- Receiving department can assist purchase department in improving effectiveness of their vendors.

Receiving department therefore, should not be looked upon as merely a clerical operation but an important function designed to throw up error automatically.

Responsibilities of Receiving Stores

The receiving store is charged with the following responsibilities.

i) Verification of correctness of paperwork and appropriateness of supply before accepting the goods.
ii) Verification of quantities.
iii) Unloading of materials.
iv) In warding of the consignment.
v) Informing purchase/ indenter/PPC regarding receipt of goods.
vi) Preparing necessary documents
vii) Arranging inspection of materials.
viii) Returning all rejected goods and all chargeable empties back to suppliers.
ix) Forwarding accepted materials to appropriate stores for storage.

Nature of Materials Receipt

Material in the receiving store are received from following sources:

i) Purchased materials received from supplier.
ii) Materials received from customer for processing.
iii) Materials returned by customer as defective.
iv) Materials returned by the customer on account of excess supply.
v) Defective materials returned by customers during the warranty period.

vi) Tooling , gauges, etc. received from customers “on loan” to be returned .

vii) Semi-finished or finished jobs received from vendors.

viii) “On loan materials” returned by the vendors.

ix) Delivery of materials by the couriers.

x) Materials received from forwarding agents against imports.

xi) Petty cash purchases by the purchase personnel.

Documents Used in Receiving

The following documents are received/ raised for receipt transaction:

1. ***Delivery challan:***
   - Also called dispatch memo, delivery note or delivery advice.
   - It is sent by the supplier with the material.
   - It lists the item details and the number of packages sent by them.
   - Two copies of the delivery note are sent by the supplier to the buyer.
   - Buyer retains the original and returns the second copy duly signed asa proof of safe delivery.
2. ***Rail way receipt (RR)/ lorry receipt (LR):*** Issued by the railways/ transporter is the document which acts as an authorization slip for getting the delivery of goods by the purchaser from the railways/ transporter.
3. ***Bill of entry:*** It is received from the clearing agent against imports.
4. ***Cash memo:*** when cash purchased is made then it is used as a receipt for the money which buyer has paid to the seller.

Receiving Procedure

1. ***In warding at the security gate:***
   - It guards against the malpractices by store personnel.
2. ***Verification correctness of paper work and appropriate ness of supply:***
   - Right kinds of goods are only received.
   - Goods should be received in the right time, not very early and not late.

- Right quantity goods should be received.

3. ***In-warding in the receiving stores***
    - Verifying number of packets (quantity checking).
    - Observing apparent damages (damage and shortage checking). If any damage or shortage found then discrepancy note should be made and it should be duly signed by carrier.
    - Unload materials with the help of carrier and own people at the appropriate place.
    - Entering details of the consignment in to a goods received register.
    - The carrier copy is also stamped duly signed by receiving authority.
4. ***Verification of quantities:***
    - **Loss of packages*:*** the number of packages received may be less than the number that is supported to be received.
    - **Damage of packages***:* the packages on receipt may be found to be broken or damaged.
    - **Wrong items:** material different than what has been ordered may be found on opening of the package.
    - **Damage to the materials:** materials within the package should be checked that these are in the right condition. If any thing found damaged then company has to do according to the contract.
    - **Shortages in supply:** If the quantity will be found short, then discrepancy notes should be issued.
    - **Excess supply: If** the quantity will be found excess than mentioned in the packing slip or in the delivery challan/ delivery advice note, then again discrepancy note to be made.
5. ***Notifying indentor about receipt of materials:***
    - It notifies the indentor/ purchaser of the receipt of materials and thereby avoid unnecessary follow up on their part.
    - It provides a formal “communication channel” between receiving department and purchasing department as well as receiving and indenting department.
    - It enables the indentor to inspect “urgent materials”.

6. ***Preparation of goods received report:***

GRR - *Goods Receipt Report* – is prepared by the receiving store on completion of physical verification of quantities. It is also known as GIN (Goods Inward Note) , MRR (Materials Received Report), MIN(Materials Inward Note), RCIA (Receipt-cum-Inspection Advice), etc. is an important linking document between supplier, store, inspection, purchase and accounts department. It has different copies like following:

i. **Original copy**
ii. **Accounts Copy** should be provided to accounts department on quantity invoiced by the supplier. It is given to facilitate the movement of payment.
iii. **Indenter's copy** to intimate the receipt and inspection results of the materials received.
iv. **Main stores copy** to give custody of the accepted materials, post stock cards, etc.
v. **Purchaser's copy** to enable buyer to record receipt materials on follow up copy of the P.O.
vi. **Receipt cum inspection copy** to maintain records on GRRs.

7. ***Inspection of materials:***

1. After the materials have been verified for quantity, quality control department has to undertake quality certification process.
2. Most commonly used approach is to forward the complete set of GRR.
3. The task of inspection of material may require the receiving store to –

i) Send the material to inspection department.
ii) Indicate the location where material is kept in the receiving stores.
iii) Provide copy of drawing of the component to be inspected.

1. Inspection department after receiving GRR carry out the inspection function.

2. They have to issue another report after completion of inspection; this is known as *Inspection Note*.
3. Few companies use a document called "Rejection Note" to report inspection results.

8. Delivery of inspected materials to stores :

- Accepted / rejected materials after inspection are forwarded to their respective stores accompanied by a copy of goods receipt report (GRR).
- Accepted materials are moved to main store while non-conforming materials are sent to rejection store.

9. Return of defective materials back to suppliers:

- Defective materials are kept in the rejection store.
- A **rejection register** is maintained where each rejected material is recorded. **Local supplier**s are informed to collect rejected materials. **Outstation suppliers** are notified and defective materials are sent back to suppliers.

10. Returning all changeable empties back to supplier:

All empties are returned to the supplier as per terms of purchase order.

Issue Control

Issue can be divided into

i) ***Issues to consuming department:*** These issues are based on work order and bills of materials. Listing for each material. Quantity to be issued against each components requiring that material.

Sometimes materials are issued on loan basis.

ii) Issues to outside supplier for processing or conversion: this aspect of issuing materials depends on delegation of authority. Personnel of different hierarchy has the authority to issue different quantity of materials

Methods of Issue

1. ***Issue on request:*** This is most traditional and conservative way of issue where in the indenting dept. sends a man and collect materials from stores.

2. ***Issue per schedule:*** In a batch production unit sometime the requisition for issue of stores is sent before indicating time & date.
3. ***Imprest issue:*** In this system a list of certain items with quantities is approved for the processing to be in charge of sub store near the shop floor mainly for tools and components.
4. ***Replacement store:*** When a fresh issue has to be made to replace old machine.
5. ***Loan store:*** Supplying of stores to meet emergency need to repair some break downs.

Issue of Materials

One of the major activities of stores is to issue materials of the right quality , in the right quantities and at the right time. "*Efficiency of the stores infact is judged by the user departments how correctly and efficient their demands for materials are met".*

Materials, in general, are issued for the following purposes. Material is issued to –

i) *Production* for manufacture of goods against customers' demand to make supply equals demand.
ii) *Maintenance department* for plant repairs and maintenance of machinery.
iii) *Suppliers* to be returned.
iv) *Sister companies* from stock held.
v) *Employees / scrap contractors* (e.g. sale of scrap materials).
vi) *Customers* as sale of finished products. etc

Timing of Issue

Material should be issued only during specified hours of the day. This enables the storekeeper to attend to other duties outside these hours.

Issue Procedure

The following issue procedure may be followed:

i) **Logging the request for material:** Request for the issue material should be logged in the *requisition register* which maybe maintained date wise like a diary.

ii) **Scrutiny of request:** Material requisitions should be scrutinized to verify whether it is signed by an authorized signatory. For this, the storekeeper should be provided a list of authorized officials and their designations and specimen signatures.

iii) **Checking availability of materials:** Stores keeper should check the availability of indented material in full or part in any one of the following categories:

a) Second hand / repaired
b) Obsolete
c) Surplus
d) Material returned from shops.
e) New material

Issue should be made in the above order to ensure that accounts of second hand repaired and surplus are closed first.

If the material of the exact specifications is **not available**, the store should check the availability of **substitute** material. The indentor should be informed of the availability of substitute material and the same should be issued if it is acceptable to the indentor.

If the exact material or substitute material is not available, the requisition may be kept pending for issue later on receipt of material.

iv) **Logging the record in issue counter:** Before taking the material to the issue counter, the issue clerk should update the material requisition and make an entry in the bin card and in the stores ledger / ledger card.

v) **Preparation of gate pass:** Where the material is to be taken out by vendor, sub-contractors, customers, a gate pass prepared giving the following details:

- Description / code no. of the material
- Quantity
- Issue voucher no. and date
- Name of the person/ organisation

vi) **Disseminating the materials:** Material should be collected by the indenter's representative from the issue counter after signing the issue voucher.

Where the materials are bulky / heavy, they may be delivered by the store on door-delivery basis.

Essentials of Correct Issues

Essentials of correct issue are as follows:

- Material should be issued only against written requisitions.
- Material should be issued only against authorized requisitions.
- Material should be issued only in pre-fixed quantity especially for C-class of items.
- Material should be issued on the basis of FIFO principle (first-in-first-out), particularly for the items of low shelf life.
- Material should be issued in bags, cartons to avoid damage.
- Materials should be issued only during the specified hours of the day.
- Material requisitions should be received preferably a day in-advance.

Stock Taking

Stock taking, also called stock verification, is the process of ascertaining – by counting, weighing or measuring – whether the physical stock of materials tallies with the balances shown in the stock records i.e. bin cards or stock ledger.

Purpose:

Stock taking –

- Is required to correct discrepancies between physical stock and book balance and thereby ensure better material control.
- Is the primary requirement for the financial statements.
- Is an indicator of overall stores efficiency and management control.
- Helps early detection of obsolete and dormant stocks.
- Helps to audit store procedures.
- Acts as a form of performance check on stock record

Common Causes of Discrepancies

Some of the common causes of discrepancies between physical stock and book balance are:

i) Manual (clerical) error.
ii) Materials handling error.
iii) Incorrect stock taking.
iv) Incorrect location of parts.
v) Environmental factors.
vi) Preservation error.
vii) Poor storage conditions.
viii) Theft and malpractices.

Identification, Codification, Verification

Identification

- Any organization stores a large no. of items. There should be some means of identifying them.
- The normal way of identification of stores by simple description. But, an item can be described in many ways.

Example: An electrical firm found that a simple item of a screw with a width of 3/8" and length 6" had as many as 118 names, depending on types of usage and department using the screw.
A proper codification removes 117 stocking points (names).

Codification

Definition: Codification in an industry is the systematic concise representation of equipment, raw materials, tools, spares, supplies, etc. in an abbreviated form complying alphabets, numerals, colors, symbols etc.

Objective of codification:

a. to classify and codify the items on some logical basis to suit the objective of the organization.
b. to assist the process of standardization and variety reduction .
c. to facilitate proper functioning of the store house.
d. to ensure that each item is kept under one unique code.
e. to prepare a catalogue – nomenclature list to reduce ambiguity.
f. to make available the catalogue to all concern dept.

Benefits of codification:

1. Accurate and logical identification:

- Codification distinguishes one item from another item.
- helps in accurate identification which eliminates any confusion.

2. ***Avoid long description:***
   - codification reduces the chance of using long description of items which leads to incorrect and wrong typing of item name.
3. ***Prevention of duplication:***
   - Items in absence of codification require to be described by brand names, trade names, or, technological names and as such they often requisitioned by different personnel under different names.
   - This can result in stocking of an item at different places under different names.
   - Codification system prevent duplication.
4. ***Product simplification:***
   - A basic reason for codification is the simplification (i.e. reducing unnecessary variety) and standardization (i.e., regulating variety).

- **Objectives**

a) For proper codification, grouping of identical items is essential.

- The most commonly used method is to take up an item group. Within its main classification list down variety in use, identify variations among the variety and prepare the vocabulary.

b) Vocabulary (i.e., manual on codification) lists down all the items in stock according to the basic characteristics which provides a quick reference for the personnel from design and planning to know what variety already exists.

5. ***Smoothening the purchase activities:***

   Codification makes the purchase activity efficient due to the following :

   i) It helps to avoid long clerical activities.
   ii) Codes avoid the ambiguities.
   iii) Buying instructions to the suppliers become easy and quick if there is proper understanding of codification by the supplier.
   iv) Codification classifies the items in to groups which in a large sized firm enable section wise organization of purchase department based on group codes.
   v) Purchase records too can be maintained according to the group codes and different documents can also be filed accordingly.

6. ***Efficient storekeeping:***

Codification facilitates locating and indexing of the materials in the main stores, sub stores, finished parts stores and warehouses.

7. Accurate and reliable recording and accounting:

Codification leads to effective store control, efficient recording and result-oriented accounting.

8. Others:

Codification –

a. Helps to look for alternative at a time when the stock of a particular item is nil.
b. Ensures proper quality description of the item which assists in efficient inspection.
c. Simplifies costing and pricing since, due to codes cost can be calculated job wise for which different cost heading can be provided.
d. Assist production since manufacturing runs can be planned on the concept of group technology.
e. Assists management in their efforts to earn a good return on its investment. This being possible due to the various benefits of codification enumerated above.

Three Basic Stages of Scientific Codification

Three basic stages of scientific codification are: **Identification, Classification, and Codification.**

a) Identification: it is the process of assigning correct nomenclature to the items. The nomenclature description must take into consideration the specification and technical name of the item. Identification system should be uniform for all items.

b) Classification: it is the arrangement of items in to groups according to the common features of the items. The grouping of stores items which may be done on the basis of "description", "supplier", "origin", "value" or "function" of the item.

i. **Description classification**: it refers to the grouping of items of similar description, irrespective of the end use, into one class.
ii. **Nature of materials**: items under this method are classified according to their nature such as raw materials, machinery, machinery and equipments, consumable items, chemicals, etc.
iii. **Source of purchase**: Under this the items are classified according to sources of procurement.
iv. **Source of origin**: This classification is rarely used except in cases of spares of machinery.

c) Codification: "*it is the allocation of specific codes to the items within the broad frame work of the selected items".*

Characteristics:

i) **Simplicity**: The system must be simple in its form. Personnel without any external assistance and without much of training can handle this.
ii) **Small:** The codes should use minimum possible number of characters/ digits/ symbols.
iii) **Flexibility:** The system should be flexible enough to comply with different situation.

iv) **Uniqueness**: The system should ensure one and only code for each item. That is each item should have a unique identification code.

v) **Proper choice of coding symbol:** The selection of the symbols especially in alpha – numerical system, should be such that there is no possibility of confusion. O, s, z, l, v should not be used because, they are similar to 0,5,2,1,u.

vii. **Self decoding**: The codes should be capable of being easily decodified.

viii. **Convenient for accounting**: the system should be convenient for accounting.

Basic System of Codification

Coding systems adopted by various organization may be classified under—

2) Alphabetical.
3) Numerical
4) Alpha-numerical
5) Color coding system

1. Alphabetical:

In this method letters or alphabets are used to describe an item. Sometimes combination of alphabets are designed to give a mnemonic meaning.

Methods:

a) Each material group is assigned alphabet(s) as basics. Usually first one or two alphabets in the title of its material group is taken as the starting point of the code.

b) Sub-alphabets, based on the characteristics of the items, are used for their identification.

Example:

Particulars	**Sub-alphabet**	**vocabulary**
I Acids (AC)		
i. sulfuric acid	SU	AC-SU
ii. Nitric Acid	NI	AC-NI
iii. Hydrochloric Acid	HC	AC-HC

Advantages:-

i. Simple to understand and easy to use

ii. do not require any formal training in code allotment or in its implementation.

Disadvantages:-

a) Codes are difficult to remember.

b) Chances of duplication.

c) Repetition of alphabets often leads to confusion.

d) Selection of proper alphabet is difficult.

2. Alpha- numerical system:

Alpha-numerical system in the combination of alphabetical and numerical systems and is in improvement over the alphabetical method. The system operates as under:

i. Each item is identified and is given a proper description.

ii. Identical items are classified into groups and each group is provided a suitable title.

iii. Codes are assigned next to the group, which are usually alphabets.

iv. Individual items within each group are assigned numbers based on block system or decimal system.

Item Group	Items	codes main	sub	vocabulary
I	**Acids:**			
	i. Sulfuric acid	**AC**	**51**	**AC-51**
	ii. Nitric acid	**-**	**52**	**AC-52**
	iii Hydrochloric acid	**-**	**53**	**AC-53**

3. Numerical system:

The numerical system is based on use of numbers. Three sub-systems within this category are:

a) Sequential code:

- The system is suitable for small organisation where the number of items is limited.
- Each item is assigned a number and the new item is allotted the next higher number.
- After each allocation few numbers are reserved for future. These reserved numbers are allotted to new items in future when expansion takes place.

Example: manufacturer of water pumps coded following items as under.

Sr. No.	Item description	code
1.	Housing	1001
2.	Impeller	1002
3.	Cover	1003
4.	Inner gasket	1004

b) Block code:

- Under a block code system, items of similar nature are grouped together and a block of numbers is assigned to each group.

Example:

1001 to 2000	for raw materials
2001 to 3000	for electrical items
3001 to 4000	for mechanical item.

The block numbers may be further subdivided according to the need.

c) Decimal code:

- In this system a group of numbers separated from each other by a decimal point, identifying the general (or main) category of items, sub –category and individual items.
- The **first** category called *general category* divides the items in to main classification such as raw materials, electrical items, mechanical items, packaging and packing materials, etc.
- The **second** category called *sub-category* divides the main classification into sub-classification according to their nature, use or characteristics. Electrical items group may be sub-divided in to wires, fuses, plugs, shades, cutouts, etc.
- The **third** category called *item category* divides the sub-category into individual items within the sub-category.

Main class : tools (40)

Sub-classes : drills (01), tapes (04), etc.

Process of Codification

1. ***Coding by end-use:*** groups or sub groups may be evolved keeping in view the purpose for which the items are eventually employed.
2. Coding by source of supply: sometimes codification of an item done according to its source of supply.
3. Coding by nature of items: groups may be based on inherent characteristics.
4. Coding customer wise: groups may be based on end user.

Codification systems:

There are two universal systems of classification and codification:-

1. Brisch system: Brisch system of classification was conceived by Mr. E.G. Brisch and was evolved by jointly with Mr. J. Gombinski. It is one of the most popular systems used in the industrial applications. The system is primarily design oriented and it divides all facets of an organization into number of main categories (ten or less) according to the nature of the business.

A 10 – digit typical class breakdown in an engineering company is given below.

0 – Organization and operations.

1 – preliminary materials.

2 – bought out commodities.

3 – components (single piece parts) to user's own design.

4 – sub assemblies and assemblies to user's own design.

5 – tools and portable equipment.

6 – plant and machinery.

7 – building , services, and utilities.

8 – Scrap and waste.

9 – Reserved.

The first digit of the code , depending upon class of the item, is picked up from the above series and all other digits are tailor-made to the client organization.

How to use this system:

The Brisch system basically consists of blocks separated by decimal point :

a) The first block represents major classification. The materials to be coded are grouped together based on one of the classification systems (e.g. raw materials, packing materials, fasteners, stationary, tools, finished products, etc.). Usually, 2 digits (xx) are adequate for most of organization.

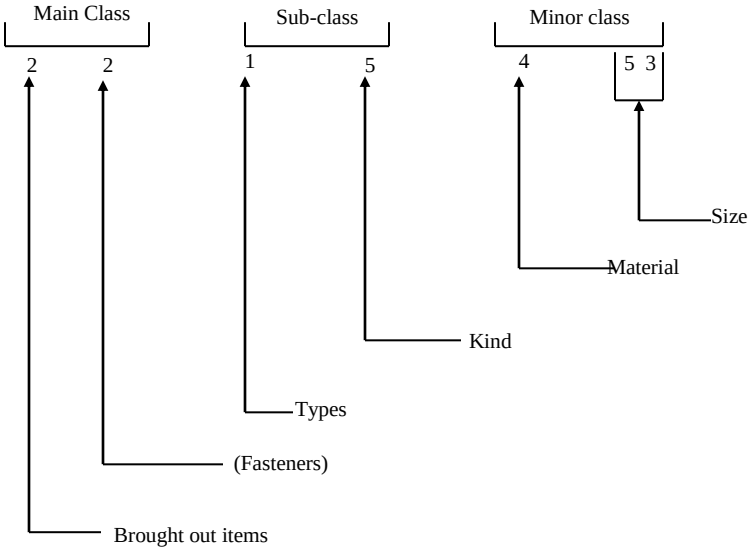
b) The second block represents the next level classification (next to primary classification). The items are divided and sub-divided based on their types and kinds. Usually, two digits are enough for secondary classification.

c) The third block represents the lowest level classification (usually dimensions, appearance, other distinguishing features)

As mentioned above, the codes are assigned in three blocks separated by decimal point. Total number of digits can vary as per needs of the user but the common figure is seven.

2. The Kodak system:

this system was originated by the Eastman Kodak Company of New York, USA. The system is primarily based on numerical system of codification and it combines all the good points of other systems. The system employs ten –digit-numerical-code. The composition of code is in three groups of digits that are separated by hyphens. The hyphens serve to break the code into easy to handle group of digits.



Salient characteristics:

1. Materials are divided in to few broad classifications, called basic classifications.
   i. The number of basic classification is restricted to 100 and first two digits, between 00 to 99, are employed to indicate the basic classification of the item.
   ii. The classification of materials is primarily based on purchase category. This system does not consider nature, use or characteristics for basic classification.
   iii. The items are grouped according to purchase categorization.

Basic class (preliminary class): First two digits (00-99) Main class code	Materials based on purchase categorization:
00-20	Raw materials
21-35	Machines and Mechanical Equipment.
36-40	Mechanical product and loose tools.
41-49	Electrical products and Electrical equip.
50-52	laboratory equip.
53-68	chemical, chemical equip. and misc. chemical product. office equip. and other misc. items.
69-78	office equip. and other misc. items.
79-83	furniture and fixture.
84-87	Semi- finished and finished product.
88-99	Miscellaneous.

2. Each preliminary class is further divided in to sub-classes (representative group). To illustrate the concept, we list below the subclasses of one of the basic classifications (say 40 covering cutting tools).

Sub –division of basic classification

Main class	Sub-class (code)	Description (kind of cutting tool)
Cutting tools (40)	0	Drills, reamers, counter bores, broaches
	1	Cutters and hobs.
	2	single point tools
	3	Taps and dies.

To form sub-groups, same factors and reasoning which apply to classes themselves are made applicable to sub-classes. For example,

a) Drills, reamers, counter bores, etc. which are generally made by the same manufacturers and are used on the same machine are grouped together and are placed in one class.

b) Tools required to cut threads – taps, dies, chasers, etc. are pt together.

c) Bar stocks are placed in one group while casting and forging are placed in different group.

3. Each sub-class is next amplified/ divided in to kinds of items. The kinds are arranged alphabetically within the sub-class.
4. Each kind of item is further split up into "types". Identification by alphabetization is continued further and each kind is divided in to "types". Next two digits – sixth and seventh digit of the two digits are assigned to the various types of particular kind.
5. Each type of item is next arranged according to its size and next two digits – eighth and ninth digits – are utilized for the purpose.
6. the last digit – tenth digit – is assigned to minor variables such as :

a) Types of steel.

b) Surface finished desired

c) Special tolerance.

d) Source of origin.

e) Usage value classification.

7. Exception to the pattern discussed above is made in order to fit special kind of materials

3. Color Coding:

Color codes find their use in identification of raw materials, lubricants, pipelines, etc.

i) color code for identification of pipelines

The Bureau of Indian Standards (BIS), vide there is : 2379 of 1963 has suggested the color identification as under:

Sr. No.	content	Ground color	First color band	Second color band
1	Cooling water/ chilled water	Sea green	French blue	-
2	Drinking water	Sea green	French blue	Signal red
3	treated soft water	Sea green	Light orange.	-

ii) Color code for raw materials:

No standard system exists for color code for identification of raw materials. The practice varies from company to company.

iii) Color codes for lubricants:

Color codes find their application in the maintenance departments as well. Different colors may be specified for different types of lubricant. Color codes can also be applied to identify the lubricant to be applied to a particular point.

Stock Verification

It is the process of physically counting, measuring or weighing the entire range of items in the stores and recording the result in a systematic manner.

Stock verification is usually carried out by the material audit department, reporting to either the material manager or the internal audit.

Purpose of stock verification:

a) To support the value of stock shown in the balance sheet by physical verification.
b) To verify the accuracy.
c) To disclose fraud, theft or loss of or, deterioration.
d) To reveal the weakness of the system in regard to safe custody.

Various methods of stock verification:

i. **Periodic stock verification.**
ii. **Perpetual / continuous verification.**
iii. **Blind stock verification.**

i) Periodic stock verification: Under this method of stock verification the whole stock should be verified in periodical manner in one financial manner.

ii) Perpetual / continuous stock verification: Under this method, verification is done throughout the year as per a pre- determined plan of action. 'A' item may be verified thrice a year, 'B' item twice a year and 'C' items once a year.

iii) Blind stock verification: In this system, the stock verifiers are given the location, but not details about code numbers, description and stock record. After verification, the report of verified compared with the actual record.

Process of verification:

1) Stores personnel first locate the stock item.
2) Verifier verifies stock with the help of sales personnel.
3) If any discrepancy found, reports it in discrepancy voucher.
4) Reason of discrepancy.
5) Re-verification, if necessary.

Store Accounting / Valuation of Stock

In relation to the estimation of the cost of the product for pricing decision, store accounting assumes a key role.

Material costing is very important in terms of the valuation of the cost of materials consumed by the production department as well as in terms of the estimation of the value of the materials held in stock.

Costing of the issues to production:

1. **First In First Out (FIFO)**
2. **Last In First Out (LIFO)**
3. **Weightage average cost**
4. **Simple average cost.**
5. **Base stock method**

1. First In First Out (FIFO)

The assumption made here is that the oldest stock is used first. Therefore at the time of issue, the rate of pertaining to that will be applied.

It is logical for the products which deteriorate with time

Date	receipt			Issues			Stock on hand		
	Qty	Rate (Rs.)	Value (Rs.)	Qty	Rate (Rs.)	Value (Rs.)	Qty	Rate (Rs.)	Value (Rs.)
1.2.2007	1000	1.00	1000	-	-	-	1000	1.00	1000
1.3.2007	2000	1.10	2200	-	-	-	1000 2000	1.00 1.10	1000 2200 3200
14.3.2007	-	-	-	500	1.00	500	500 2000	1.00 1.10	500 2200 2700
28.3.2007	-	-	-	1500 ↓ 500 1000	 1.00 1.10	 500 1100	 - 1000	 - 1.10	 - 1100

Advantages

1. Simple to understand and easy to operate.
2. Material cost represents actual cost which should be charged to a product.
3. Closing stock will be as closely representative of current prices as possible.

Disadvantages

1. In case of fluctuating price this method leads to clerical error.
2. In case of price change, comparison of one job with another can not serve any useful purpose.
3. When price rise, it will give low charges to product. It leads to many problem related to price of product.

2. Last In First Out (LIFO)

The basic assumption here is that the most recent receipt are issued first. We can apply LIFO system to the example discussed earlier.

It is logical for the products which deteriorate with time

Date	receipt			Issues			Stock on hand		
	Qty	Rate (Rs.)	Value (Rs.)	Qty	Rate (Rs.)	Value (Rs.)	Qty	Rate (Rs.)	Value (Rs.)
1.2.2007	1000	1.00	1000	-	-	-	1000	1.00	1000
1.3.2007	2000	1.10	2200	-	-	-	1000 2000	1.00 1.10	1000 2200 3200
14.3.2007	-	-	-	500	1.10	550	1000 1500	1.00 1.10	1000 1650 2650
28.3.2007	-	-	-	1500	1.10	1650	1000	1.00	1000

Advantages

1. Material cost will represent current price as far as possible.
2. It minimizes losses in inventory.
3. When there is wide fluctuation in price level LIFO tends to minimize unrealized gain

Disadvantages

1. In case of fluctuating price this method leads to clerical error.
2. In case of price change, comparison of one job with another can not serve any useful purpose.
3. When price are falling it will lead to low charge to production.

3. Average cost method or Weightage average method:

In this method, the issues to the production department are spit in to equal batches from each shipment at stock.

It is logical for the products which deteriorate with time

Date	receipt			Issues			Stock on hand		
	Qty	Rate (Rs.)	Value (Rs.)	Qty	Rate (Rs.)	Value (Rs.)	Qty	Rate (Rs.)	Value (Rs.)
1.2.2007	1000 (Q 1)	1.00	1000 (V1)	-	-	-	1000	1.00	1000
1.3.2007	2000 (Q2)	1.10	2200 (V2)	-	-	-	3000	1.067 *	3200
14.3.2007	—	—	—	500	1.067	533	2500	1.067	2667
28.3.2007	—	—	—	1500	1.067	1600	1000	1.067	1067

***** Average price = (V1+V2) / (Q1+Q2) = (1000+2200) / (1000 +2000) = 3200/3000 = Rs. 1.067.
Weightage average Price = Value of material stock / Quantities in stock.

Advantages

1. In case of fluctuating price, it will smooth out the fluctuation.
2. New issue price need not be calculated each time issues.

Disadvantages

1. Tedious calculation.
2. Material cost does not represent actual cost price.

4. Simple average cost method:

In this method, the issues to the production department are spit in to equal batches from each shipment at stock.

It is logical for the products which deteriorate with time

Date	receipt			Issues			Stock on hand		
	Qty	Rate (Rs.)	Value (Rs.)	Qty	Rate (Rs.)	Value (Rs.)	Qty	Rate (Rs.)	Value (Rs.)
1.2.2007	1000	1.00	1000	-	-	-	1000	1.00	1000
1.3.2007	2000	1.10	2200	-	-	-	3000	1.05 *	3150
14.3.2007	—	—	—	500	1.05	525	2500	1.05	2625
28.3.2007	—	—	—	1500	1.05	1575	1000	1.05	1050

***** Average cost = (1.00+1.10) / 2 = 2.10 /2 = Rs. 1.05.

It can be used in following situations:-

1. Materials are received in uniform lot quantities.
2. Difficult to identify each issue of materials.
3. Purchase price does not fluctuate.

Advantages

1. Simplicity.

Disadvantages

1. Material cost does not represent actual cost price.
2. In case of fluctuating price it gives very incorrect result.

Comparison of FIFO and LIFO in profit center point of view:

Raw materials:-

Opening stock =400 units @ Rs. 10/-

Purchases = 1000 units @ Rs. 15/-

Closing stock = 500 units

Wages ,etc. = Rs. 2000/-

Sales = Rs. 20,000/-show the profit if raw materials issued are priced at a) FIFO & b) LIFO.

Raw material consumed

	a) FIFO	b) LIFO
Opening stock	400 units @ Rs. 10/-=Rs. 4000/- + 1000 units @ Rs. 15/-=Rs. 15000/-	400 units @ Rs. 10/-=Rs. 4000/- + 1000 units @ Rs. 15/-=Rs. 15000/-
	Rs.20000	Rs.20000
Less closing stock	- Rs. 6000 (400 units @ Rs. 15/-)*	- Rs. 4000 (400 units of Rs. 10/-) **
	Rs. 14000	Rs. 16000
Add wages etc	Rs. 2000	Rs. 2000
Cost of production	Rs. 16000	Rs. 18000
Sales	Rs. 20,000	Rs. 20,000
Profit	Rs. 4,000	Rs. 2,000

* in case of FIFO the stock in hand (closing stock) will be charged according to last receipt price.

** in case of LIFO the stock in hand (closing stock) will be valued according to first receipt price.

What is Material Handling? Why mechanization is needed? What are the various factors which influence material handling? Explain briefly the commonly used material handling equipment in modern organisation.

Answer:

- Materials handling can be defined as the function dealing with the preparation, placing and positioning of materials to facilitate their movement or storage.
- Material handling may be defined as the art and science of movement, handling and storage of materials during different stages of manufacturing considered as material flow in to , through and away from the plant. It is in fact, the technique of getting the right goods safely to the right place at the right time and at the right cost.
- Goods material handling practices require systematic recording, critical review and improvement of all material handling activities to eliminate as many movements as possible, and mechanization / simplification/ modification of remaining movements to reduce cost and improve efficiency.

Mechanization of material handling in a country like India is ridiculous where huge labour force is available. In most of the warehouses, godowns and transport organizations involved in bulk material handling mainly depend on contract or casual labour. The reason of slow growth of Indian economy is the improper handling of materials and other extra unwanted cost incurred.

The present technology can boast of equipment designed on the basis of movement requirements, types of materials, safety and economy of operations. A variety of equipments are available to move materials up and down.

Economies of using equipment are as:

- Low cost of operation.
- Work speed goes up.
- Quality of work will improve.
- Low wastage and break down, some works etc.

Choice or Factors which influences equipment:

There are various equipments which are manufactured in India and abroad and are used in material handling.

The choice of the equipment is influenced by a variety of reasons; the following is the list of such factors:

(i) The location of plant or warehouse where equipment will be needed.

(ii) Product lay out.

(iii) Process lay out.

(iv) Integration or raw materials manufacturing plants and finished goods warehouse.

(v) Floor structure.

(vi) Transportation logistics.

(vii) Packaging.

(viii) Unit load.

(ix) Industrial relations in the organisation.

(x) Industrial inspections.

(xi) Location of vendors.

(xii) Location of warehouses.

(xiii) Capital and operating cost of equipment.

(xiv) Desired movement of materials and availability of process flow chart.

(xv) Material parameters or size, density, viscosity, flow ability, hazardous nature, safety, security raggedness, fragility, temperature, angle of repose etc.

(xvi) Distance to be traveled.

(xvii) Rate per hour / frequency.

(xviii) Loading and unloading levels.

(xix) Slope, inclination and aisle width.

(xx) Cross-traffic sequence.

(xxi) Technological obsolescence and standardization.

(xxii) After sales service availability.

(xxiii) Location of high tension points.

Cost Benefit Analysis:

The best method of handling material whose cost is lowest thus is that a detail cost analysis has to be carried out before deciding on the choice of handling equipment. The cost is divided into:

(a) **Capital Cost.**

(b) **Operating and maintaining costs**. These costs have adverse relationship. Increased capital costs are usually associated with lower operating costs and vice versa in a given environment. Suitable remedial measures are necessary if the cost exceeds the norms. The cost depends upon the distance moved, load carried, equipment conditions etc.

One should try to reduce cost; the following facts are kept in mind:

1. ***Demurrage Charges:*** The same may be compared with the standard demurrages.
2. Utilization of warehousing space.
3. Breakage in materials due to mishandling.
4. Inefficient use of materials.
5. Down time adequacy of handling equipment.
6. Excess maintenance.
7. Accumulation of raw material, work-in-progress and finished goods inventory.
8. Safety of operating personnel.
9. Depreciation of Equipment.
10. Rate of obsolescence.
11. Maintenance cost.
12. Spare parts consumption.
13. Cost of power and energy.
14. Cost of fuel and lubricants used in handling equipment.
15. Direct labor cost.
16. Departmental supervision.
17. Clerical records keeping.
18. Training cost.
19. Investment in equipment.
20. Salvage value of the handling equipment.
21. Estimated after sales service charges.

Different types of material handling equipments

One should make a rough estimate of benefits obtained by different equipment before deciding on the purchase of the equipment.

Hand operated equipment:

Material handling involves moving packing and storing articles in any form.

1. Hand Barrow:

- It comprises a frame which is provided with short shafts.
- It is used by two or more men.

2. Wheel Barrow:

- It is a shallow box with shafts and wheels.

- It is used for movement of small parts within the store.

3. Sack Truck or Two Wheel Truck:

- It is generally used for moving bags or container containing loose parts or granular materials.
- It consists of two wheels and a frame.

4. Handy Trolley:

- It can have three or four wheels.
- it is worked by a hand lever.

5. Sliding Wheel Trucks:

- Sliding wheel box trucks made with wood or steel.
- They carry loose parts.

6. Lift Truck:

- Hand operated lifting trucks are used for running beneath stillages on which parts are loaded.
- Hand operated trucks are also available.
- These are fitted with a platform which can be raised to a height of about 1.3cm. with the handle which is attached to the frame of the truck.

7. Pallet Trucks:

- These are very similar to stillage trucks.
- The only difference being that they are fitted with forks.

8. ***Stillage Trucks:***

- These consist of wheeled platforms fitted with a small hydraulic unit. It can raise or lower the platform.
- The trucks platform when lowered is run in between the legs of the stillage, the platform is raised and the truck complete with load is taken way.

9. Hand Stacker:

- It consists of a vertical iron frame carrying a platform which can be raised or lowered for loading and unloading.

10. Pulley Blocks:

- This is a geared pulley system fixed to an overhead beam and fitted with a lifting hook at the end of a chain.
- A separate endless chain is used to operate the mechanism.

11. Monorails:

- A monorail is a single ‘I’ section rail fixed overhead and bearing small wheeled trolleys which run on the rails.

- Monorails are often used in conjunction with pulley blocks or handling heavy loads out of or into vehicles in receipts or dispatch hays.

1. ***Roller Conveyors:***

- These are made of a metal frame work bearing horizontal rollers spaced at intervals, and the goods to be conveyed are pushed along the top of the rollers.
- Conveyors can be used for horizontal movement at or above floor level.
- Roller conveyors are generally used for fairly heavy packaged goods.

Power-Driven Equipment

All exhaustive treatment of all the available types of power-operated materials handling equipment is neither possible nor desirable here. For the purpose material handling is being considered in selection to store house and the stock yard work only.

1. Forklift Trucks:

- These are heavy duty machines with large wheels for stock yard work. Some trucks are suitable for operation both inside and outside, these have smaller wheels.
- The normal attachment is a fork consisting of two flattened Prongs which can be pushed under a load and then raised automatically to lift the load such pallet transporters are suitable for lifting transporting and laying down stores.
- Some trucks have detachable forks which can be replaced by crane attachments. They are not suitable for transporting goods for long distance and for heavy materials.
- Where handling by fork lift is done, gangways must be wide enough to allow the tracks.

These are of two types: **fixed platform** and **platform lift trucks**.

- The **lifting platform** has a platform instead of forks which can run under a suitable container or stillage and lift the load clear off the floor. Both type of can carry the materials for long distance.

3. Tractors:

- Tractors of various sizes are used in store houses and stock yards.
- The use of two or more trailers per tractor is more economical.
- Platform trucks most suitable for long distance.
- Some tractors can be fitted with various attachments for particular jobs.
- Tractors can also be useful for shunting railway wagon or putting narrow gauge rail trolleys.

4. ***Overhead Electric Cranes:***

- Five tone and ten tone overhead electric traveling cranes are most commonly used in stores.
- There are two main types—***floor controlled*** and ***cab controlled***.
- In case of ***floor controlled***, there is a control panel attached to a cable leading from the crane bridge overhead. The operator holds this panel in his hand to control the machine and walks up and down underneath it as it travels to and fro along its rails which are supported on the columns of the building.
- In case of ***cab controlled***, the crane is managed from above by an operator setting in a cab mounted in the crane bridge.

5. Goliath Cranes:

- It is a special version of the overhead traveling crane.
- The crane is mounted on legs fitted with wheels Instead of erecting an overhead gantry track all along the traveling distance.

6. Mobile Jib Crane:

- It is mounted on wide wheeled chassis and requires wide gang ways and the long jibs need high doorways for clearance.
- These are not suitable for inside work. But they are very useful in a stockyard.
- The commonest types in use for stores, can work from 2 to 6 tonnes with jibs up to 12 meters. It can be lifted or lowered to increase or decrease the reach of the crane.
- It can unload box wagons.

7. Locomotive Cranes:

- Self propelled jib cranes similar to the mobile road crane.
- They are suitable for use is stock yards served by railway lines.

8. Screw Conveyor:

- A screw conveyor or spiral conveyor consists of a revolving shaft provided with continuous or broken spill fighting.
- The screw is designed in such a way that it operates unidirectionally.
- An electric motor drives the screw at a suitable speed through a gear train.
- Screw conveyors are commonly employed for conveying non-coarse and fine materials.

8. Chain Conveyors:

- For continuous transport of goods, the overhead system of chain conveyors is more advantageous.
- It carries goods or components through a complete circuit bringing the parts to the points where the operation is to be carried out.

9. Skids:

- Skids are platforms usually constructed with metals or wooden platforms supported by four metal legs.
- They can carry more than the average size pallet.
- But they have the disadvantages of not being able to be tiered with damage.

10. Pallets:

- These are elevated platforms usually constructed of wood having a top platform and supported by two or three stringers.
- Unit loads are constructed of metal.
- They have the advantage of long life.

5. Distribution management (logistics Management)-distribution of materials to various departments & auxiliary services.

DISTRIBUTION

What is a Distribution Channel ?

- It is a set of *interdependent* organizations involved in the process of making a product or a service *available* for use or consumption by the consumer or the business user
- Web of capabilities embedded in extended enterprise.

Objective of channel intermediaries :

1) Transportation and storage
2) Replenishment
3) Market coverage
   i) Reduce time gap (temporal closure)
   ii) Reduce space gap (spatial closure)
4) Anticipation of demand.
5) Minimize the cost.
6) Provide market information
7) Distribution of resource.
8) Good relationship with manufacturer and consumer
9) Promotion or self place promotion.
10) Control over channel members
11) Issue of adaptability – with the change in market channel has to be changed.
12) Service out-put level
   i) lot size
   ii) waiting time
   iii) spatial convenience
   iv) product variety
   v) other services.

Why are marketing channels used ?

- Marketing channels are used for their greater efficiency in making goods available to target markets because of their *contacts, experience, specialization and scale of operations.* They can offer the producer more than it can achieve on its own
- Channel intermediaries can provide economies. The use of channels implies *fewer* relationships to manage & hence intermediaries *reduce* the amount of work that must be done by producers & consumers
- Marketing intermediaries transform the assortment of products made by producers into assortments wanted by consumers
- Producers produce *narrow assortments* of products in large quantities, but consumers want *broad assortment* of products in small quantities
- The channel intermediaries buy moderately large quantities from many producers & break them down into smaller quantities & broader assortments which are bought by consumers. Thus, channels play an important role in matching demand & supply

Role and function of Distribution system :-

1) storage
2) logistics
3) replenishment
4) promotion
5) After sales service.
6) Financial service.
7) Provide market information about product , customer, price stability and competitors.
8) Risk taking
9) Good relationship.

Functions of distribution channel members

- *Information* : Gathering & distributing info about producers' offers to consumers, collecting market intelligence from consumers & providing them to producers for their planning & control
- *Promotion* : Persuading target consumers to buy products & services produced by the producers
- *Contact* : Locating & communicating with prospective buyers
- *Matching* : Shaping the offer to suit the buyers' requirements
- *Negotiating* : Reaching a negotiated price & other terms of the offer so that ownership can be transferred
- *Physical storage & distribution*
- *Financing* : Funding the cost of channel work
- *Risk taking* : Bearing & sharing the risk of carrying out channel functions

The different channel functions create different *flows* through the marketing channels :

- ❖ Physical Flow (Forward Flow)
- ❖ Title Flow (Forward Flow)
- ❖ Payment Flow (Backward Flow)
- ❖ Information Flow (Forward & Backward Flow)
- ❖ Promotion Flow (Forward Flow)

Channel Levels

A) *Consumer Marketing Channels*

Zero – level (Direct marketing channels) : Manufacturer → Consumer
One – level : Manufacturer → Retailer → Consumer
Two – level : Manufacturer → Wholesaler → Retailer → ConsumeR
Three-level : Manufacturer → Wholesaler → Jobber →Retailer → Consumer

B) *Industrial Marketing Channels*

Zero – level : Manufacturer → Industrial Consumer
One – level : Manufacturer → Industrial Distributor → Consumer
Two – level : Manufacturer → Manufacturer's Rep → Industrial Distributor → Consumer

Different intermediaries perform different functions :

- Wholesalers & Retailers *buy, take title to & resell* merchandise. They are *Merchandisers*
- Manufacturer's representatives, brokers, sales agents *search* for customers & may *negotiate* on behalf of the manufacturer, but *do not take title* to the goods. They are called *Agents*
- Transportation companies, independent warehouses, banks, ad agencies *assist in* the *distribution* process but *do not take title* to the goods & *do not negotiate* purchase or sale. They are called *Facilitators*

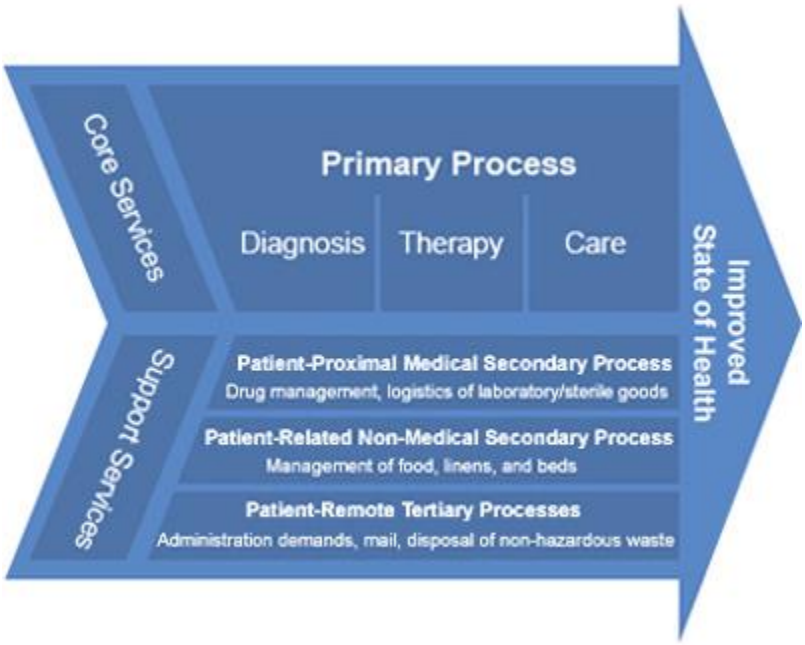
Hospital Logistics Processes

Healthcare Solutions > Healthcare Resources > Hospital Logistics
Share:|

The main task of all hospitals is the improvement of the patient's state of health. The provision of the medical treatment and patient care – core processes of the hospital, create demand for patient-related support services. These secondary processes can be of medical or non-medical nature.

Additional services summarized in tertiary processes are not directly linked to patients, but are necessary for proper operation of the healthcare facility.

Hospital Processes



Tasks of Hospital Logistics

Hospital logistics is coordinated cross-departmental with the flow of goods and information as well as a part of patient care. Examples of logistics tasks can be found in secondary and tertiary processes:

Logistics tasks in patient-related medical secondary processes

• Patient logistics • Drug management • Laboratory logistics • Management of medical goods	• Logistics of sterile goods • Information and documents • Disposal of hazardous waste

Logistics tasks in patient-related non-medical secondary processes

• Food management • Management of linens	• Management of beds

Logistics tasks in patient remote tertiary processes

• Management of administration demands • Mail service	• Disposal of non-hazardous waste

Automated Hospital Material Transport

In many processes of hospital logistics, material transport plays a decisive role. According to requirements, hospital material transport is either scheduled (planned) or on-demand (un-scheduled).

Compared to industrial applications, the quality of material transport in healthcare facilities must be of the highest quality. Wrong or inaccurate deliveries could have fatal implications for patients, hospital employees, and visitors. .

Automated Drug Management

The conventional drug management in hospitals is typically a manual process. The preparation of the individual patient daily medication is managed by nurses on the wards. Therefore, hospitals operate many drug storages in several departments. In addition, the preparation processes in the hospital pharmacy are characterized by manual handling. Automated drug management in the hospital pharmacy (pharmacy automation) and on hospital wards supports and increases patient safety.

Health care absorbs such a large proportion of public finances that it's not surprising the health system hasn't escaped cutbacks. For instance, the health sector accounts for about a third of the Quebec government's budget and about 7.5% of the province's GDP.

There's a pressing need to review health care practices to improve hospital operations and bolster their efficiency and effectiveness. Improved operations should provide better cost control, while maintaining the quality of care delivered to the public. Support processes are excellent targets because they don't necessarily have a direct impact on the quality of care provided. Hospital logistics is one such process, the goal of which is to efficiently deliver medical supplies and pharmaceutical products to the final consumer, the patient.

Comparative analysis, or benchmarking, is an improvement process that compares practices used by several organizations to identify best practices. Organizations use this management tool to identify both the problem areas in their business processes and opportunities for improvement.

A benchmarking process targeting hospital logistics was recently initiated in five Quebec hospitals. The aim of the study was to develop and test the validity of an analytical model and data collection tool tailored to hospital logistics. With these tools, the costs of current hospital practices and their impact on the quality of care could be evaluated. Eventually this could offer a clear set of benchmarked best practices for large hospitals.

Hospital logistics

A poorly understood and often unappreciated process, logistics accounts for a sizeable portion of a hospital's operating budget. Studies have shown that from 30% to 46% of hospital expenses are invested in various logistical activities, and that almost half of the costs associated with supply chain processes could be eliminated through the use of best practices.

In hospitals, logistics cover not just support services such as purchasing, stores and the pharmacy, but also health care services such as patient care units and operating rooms. Many activities that could be carried out by support personnel are often on the list of duties performed by health care personnel. The result is that the internal supply chain within a hospital is often highly fragmented.

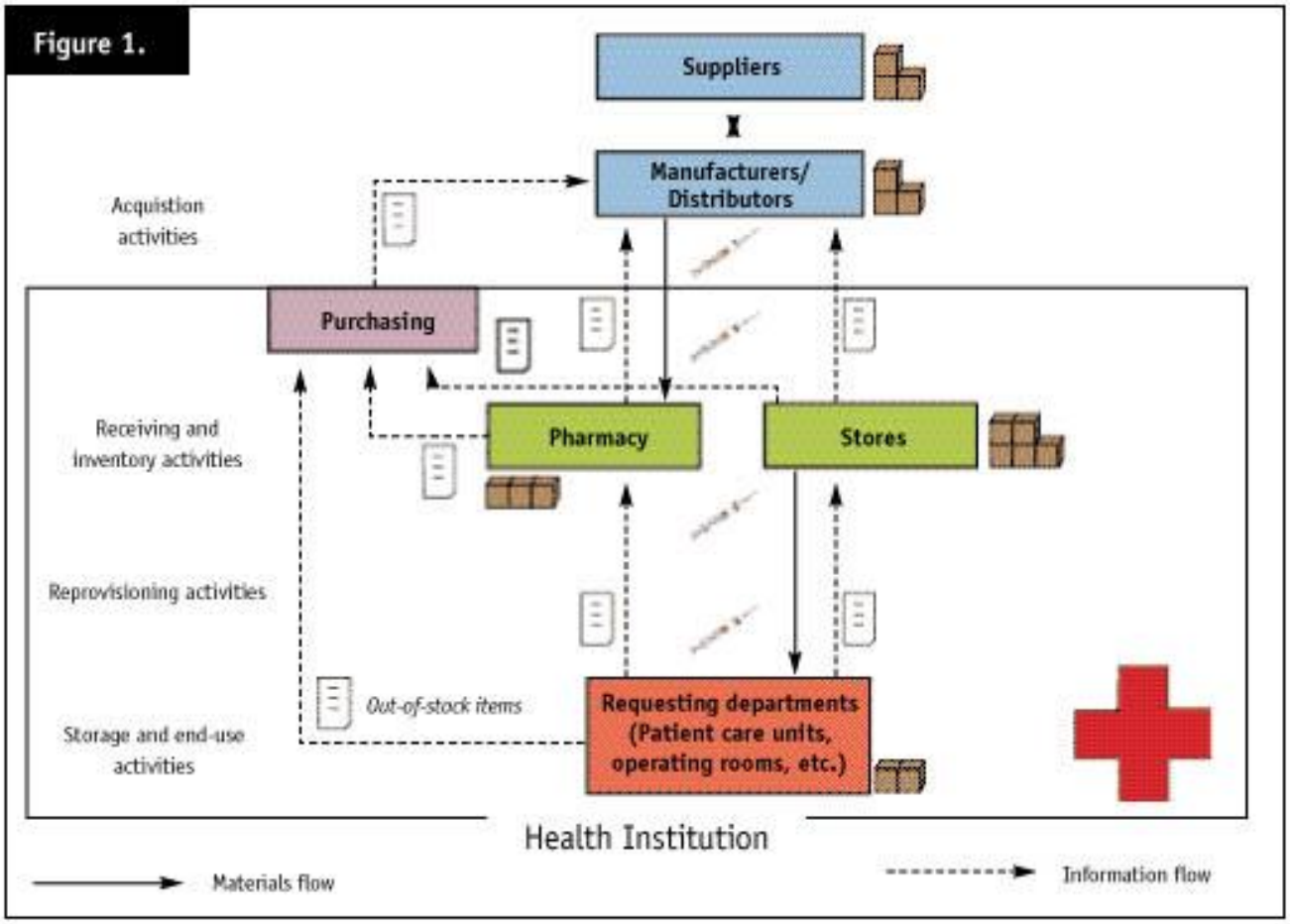
Logistics is a complex process. The people involved vary with the type of products in question: for example, stores manage medical and office supplies; the pharmacy looks after pharmaceutical products; and food services manages the procurement and processing of food products.

Two major management methods are applied by hospitals. Certain products are managed and stored in the hospital's stores (or pharmacy) before being distributed to specific departments: these are called inventory products. Other items are ordered directly by specific departments from the purchasing department, which oversees the purchases as needed and delivers them upon receipt to the departments: these are non-inventory or direct purchase products. The latter are generally not stored in the institution's stores.

Pharmaceutical products, meanwhile, are divided under two main headings: general products and prescription drugs. The inventory and non-inventory distinction doesn't apply to pharmaceuticals because all requisitions must go through the pharmacy.

Figure 1 summarizes the main players and their role in the replenishment of two critical items, medical supplies and pharmaceutical products. These relationships may differ at certain institutions. The pharmacy and stores order their products directly from suppliers in some cases. Yet, in other hospitals, ordering is taken care of by a purchasing department.

Figure 1 also shows the people who store medical supplies and pharmaceutical products, and indicates the two types of flows between them: the flow of materials and information. These two flows were studied in the project to benchmark hospital logistics processes.



Benchmarking

To benchmark hospital performance, performance measures for material and information flows had to be established. However, the information technology systems in place for support services in the health sector are very limited. In Quebec, few hospitals have an integrated system with extensive information retrieval capabilities. Moreover, this information is rarely standardized, which further complicates data collection and analysis. Given these limitations, a strictly results-oriented comparison was not feasible, so a process-based approach was taken.

Three steps were taken to provide structure to the benchmarking process. First, the hospital logistics process was divided into several components to facilitate the analysis. Second, an analytical model and related data collection tool was developed. The model identifies the performance measures used. And third, a data collection approach was selected to get the project up and going.

Breaking down the process

The hospital logistics process was divided into three main sub-processes, namely Ordering and Managing Supplies, Receiving Orders, and Replenishing User Departments. These sub-processes were also divided into various activities to obtain a sufficient amount of detail without unduly complicating data collection. The processes associated with inventory and non-inventory products are similar, with the exception of three activities not used in processing non-inventory items. Table 1 presents the breakdown of the logistical processes, with the activities not used in processing non-inventory items identified in italics.

Table 1: Breakdown of hospital logistics process

Hospital Logistics Process
Stock ordering and management
Identify needs
Create, process and track orders
Review stores and pharmacy management parameters
Order receipt
Take receipt of and check items
Transfer and stock items in stores or the pharmacy
Authorize payments to suppliers
Stock replenishment
Order items from stores or the pharmacy
Collect items from stores or the pharmacy
Transport items to patient care units
Store items in patient care units
Review patient care unit management parameters

Analytical model

The analysis hones in on two types of performance measures — efficiency and quality. Breaking down these processes into a number of specific activities offers a more detailed cost analysis. It is then feasible to assess a hospital's efficiency for each activity as well as the overall process. These costs become the primary point of comparison for identifying hospitals with the best practices. Since labour accounts for about 60% of a hospital's costs, the primary element affecting cost calculations is the time spent by personnel in carrying out various logistical activities. A second element, technology maintenance expenses, was also considered to build a more thorough picture of the situation.

The quality of the logistics process was also considered in the measurements of hospital performance. The health sector stands out specifically because of the critical nature of the service levels associated with logistical activities: the lack of an item or a drug in the operating room could, for example, greatly compromise the work performed by health care workers and threaten the health of a patient. The quality of support services must be maintained. Table 2 lists the service quality performance measures used in this study.

Table 2: Service quality performance measures

Sub-processes/activities	Performance Measures
Ordering and inventory management	
Identify needs	• Number of requisition lines for out-of-stock items at stores / total number of requisitions for stores/pharmacy • Number of urgent orders to suppliers / number of orders to suppliers
Create, process and track orders	• Number of order lines for items out-of-stock at supplier / number of order lines for orders to supplier
Receiving and storage	
Transfer and store items in stores or pharmacy	• Inventory turnover • Accuracy of inventory on the books • Storage area compactness (inventory value / area)
Replenishing	
Order items from stores or pharmacy	• Percentage of emergency orders for products stocked per unit • Percentage of urgent requisitions • Percentage of order lines for out-of-stock items

The study then examined various contextual elements such as hospital budgets, number of beds and type of specialties, as well as process-related elements (the current administrative practices in the hospital).

The analytical model helps identify best practice hospitals while shedding light on why the institutions perform well. Analyzing costs and quality through the practices employed by hospitals also leads to an understanding of the effectiveness and efficiency of various hospital-specific practices. It should be noted that the methodology identifies optimal practices for large hospitals, not residential and extended care centres.

Data collection

The data collection combined case studies of five hospitals, which provided an in-depth study of a limited number of sites, and questionnaires, which delivered the advantage of reaching a large number of respondents at a reasonable cost, but didn't allow for an in-depth analysis of these hospitals. This hybrid method thus identifies the practices deployed by institutions and determines their performance level.

The method doesn't, however, explain the nuances among hospitals with the same practices, because such hospitals are grouped together in the analysis. A second data collection phase specific to the most promising practices will therefore be needed to identify the impacts of implementing these practices. This phase will be conducted on a much more limited scale, since only the most promising hospitals will be visited.

In the current context, where health institutions must carefully manage public funds, where public accountability is becoming more and more pressing, and where governments are trying to motivate hospitals by rewarding high performing ones and penalizing those that are not performing as well,

improving the logistics process through the implementation of best practices has become a must. Identifying best practices by benchmarking the hospital logistics process can help hospital managers find cures for the ailing health sector.

Logistics in hospitals

The concept of using logistics' principles within UK hospitals is not new and the wide spread use of small bore pneumatic tube systems to distribute pharmaceutical products and test tubes is evidence that the optimisation of material movement, including the potential costs savings these systems generate, has been considered. However, the continuously increasing requirements imposed on existing developments mean that the 'logistics' components of the operation are increasingly seen as a way to not only reduce costs but also improve the effectiveness of the day-to-day processes, including the use of IT, and the service provided.

A further driver for the development of logistics concepts within hospitals is the use of the PFI procurements route. Operators of these new developments now have a financial horizon of 25-30 years making longer pay-back periods that may not be suitable to NHS trusts, on new processes or technology acceptable. They also have strong incentives to ensure that the layout of any new development, however integrated to existing buildings, provides the most cost effective solution for the delivery of FM services, whether in terms of space take up, shortest delivery routes, maximization of lift usage, etc, and that the operational processes proposed follow corporate sustainability targets. This paper looks at some of the logistics concepts that can be applied to the development of new hospitals and technologies used to provide the service required. Topics include site planning and the relationship with internal logistics, how internal logistics can be reviewed as horizontal or vertical distribution concepts and the impact on building layout and movement segregation. Automation and its use are described; covering small and large bore hole pneumatic tube systems, automated guided vehicles, track vehicle units and automated storage and retrieval systems.

Site planning

The logistics strategies used to define delivery routes, separation of flows, use of automation, etc, must start with a suitable strategy for the overall site planning. This should ensure that vehicles, used for goods and people, drop off at the closest possible point of usage, whether the main entrance or FM storage or the A&E department, to reduce travelling distances within the building. Any site planning should therefore start with the rationalisation of site entrances, ensuring that road access leads towards these entrances and suitable car parking is provided close to each entrance.

Site access and car parking seem to be issues regularly raised by hospital staff; either because of the lack of car parking, complexity of road infrastructure and the impact on way finding, unsatisfactory blue light access or the mixture of staff access with FM deliveries, patients, visitors and blue light. Some recent PFI projects have provided the opportunity to re-assess site planning in terms of providing clear access and parking by creating visitor 'front of house' and staff and FM 'back of house' areas. If this separation can be matched by the provision of alternative access roads then this solution ensures that the front of the hospital is not seen as a pool of cars (as 70 to 80% of car parking is typically allocated to staff which would be at the back) and traffic can be controlled around the site with vehicle flows being separated before the entrance of the site. The suitable location of barriers also ensures that rat-runs around the site are avoided.

The rationalisation of entrances to the buildings and site access also impacts on the level of double handling that may exist. In this context double handling is used to describe the process of offloading from a delivery vehicle and re-loading the goods onto a smaller vehicle, to be offloaded again at another entrance, etc. By the rationalisation of entrances, it is intended that not only visitors be directed to a single point of entry, but also FM deliveries be taken to a single FM industrial zone.

This industrial zone could include a catering distribution point, pharmacy bulk storage and manufacturing, security hub, telephone operators, porters, waste rooms, gas storage, etc. From this central FM zone, ideally, all goods would be broken down and re-packed into containers ready for their end point of use, therefore avoiding any middle processing

or storage spaces. In some instances, the goods should be delivered in a format already suitable to be taken directly to other departments, such as linen cages.

Rationalisation of roads?

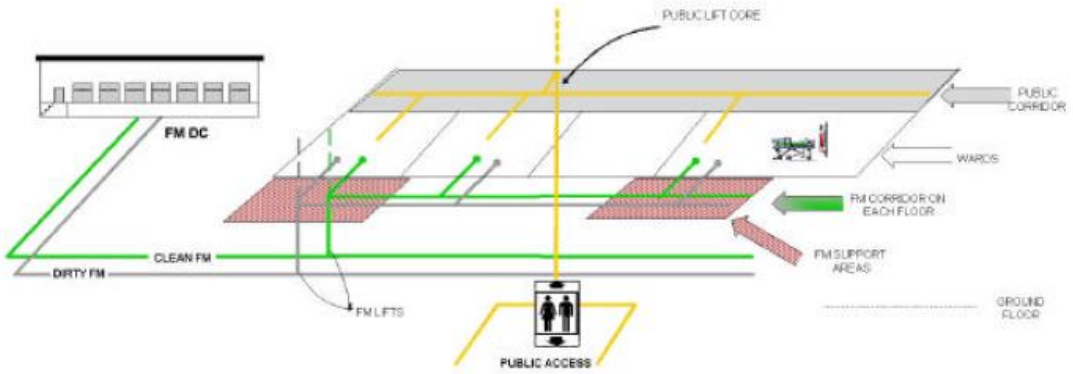
Internal logistics

The outcome of a logistics review of internal routes should be to ensure that goods and people can reach their final destination in the shortest and most direct possible route but still maintain any required segregation, such as between the public and patients in beds going to and from theatres. Visitors would therefore enter the main reception and then be directed towards main departments such as the day treatment centre, women and children's unit, wards and A&E. Each with clear signage and a clear identity to facilitate way finding.

Goods would be taken directly to the point of use from the central FM building through FM corridors and service lifts that provide direct access into the support spaces of each department without, for example, having to be taken through wards. It is obvious that in some instances it will not be possible to affect the layout of existing hospitals to such an extent that all double handling and cross-overs are removed and, in fact there may times when they are required as part of the normal operation, particularly if the central FM area is used to then distribute to a number of other buildings possibly on other sites.

Within the building, there are two principle methods of setting out the distribution routes and achieving the required movement segregation: horizontally and vertically. Horizontal segregation is based on providing separate corridors for the different types of activities; so for example, if the intention was to segregate between front of house (visitors and the general public) and back of house activities (FM, patients in beds and staff) then two corridors could be provided linking into each of the departments. This is illustrated on the sketch below.

HORIZONTAL FM / HORIZONTAL PUBLIC

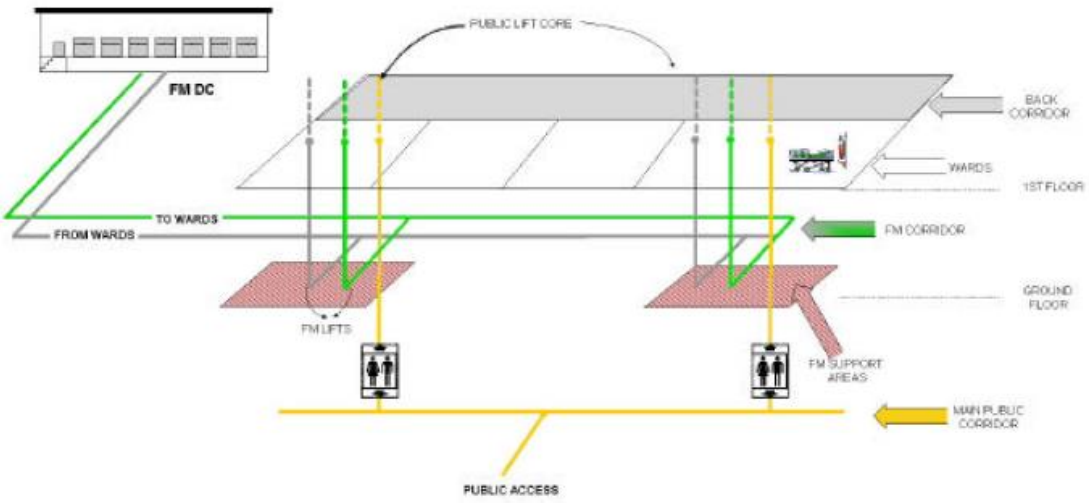


Horizontal segregation is in fact the hardest to achieve, not necessarily because it is difficult to ensure the corridors do not cross but, because the corridors would run the length of the building in several locations, it would be much harder to integrate this

strategy with the requirements for daylight. There is also a risk of simply creating too much circulation space.

Vertical segregation makes use of lifts to bring the different flows of goods and people into appropriate spaces on each floor. It is still necessary to provide the horizontal access routes to access these lifts, but the corridors are only duplicated on one floor and also possible to have them at different levels. This is by far the easiest to integrate with the requirement for daylight, but has a considerable impact on the number of lifts that need to be provided.
Vertical segregation is therefore not particularly suited to large sprawling developments with only two or three floors, but can easily be fitted when several departments such as wards are located above each other with identical access points. The use of vertical segregation is shown on the following sketch.

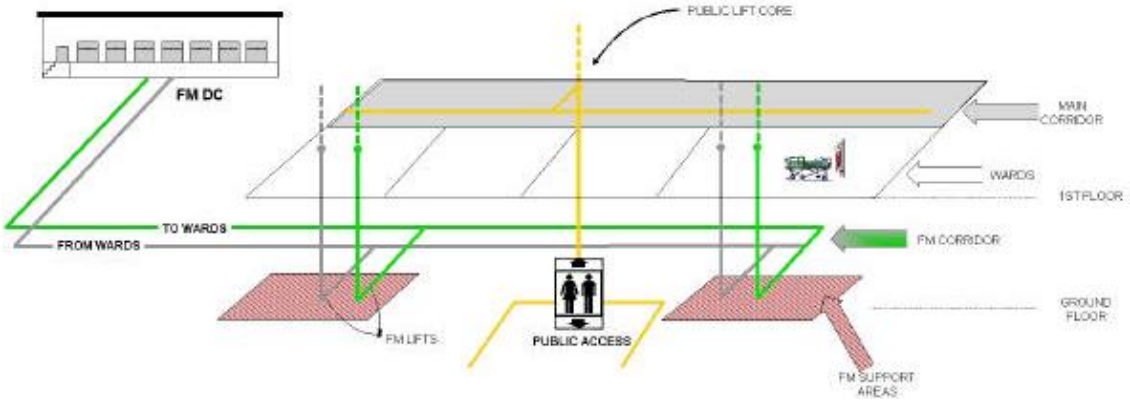
VERTICAL FM / VERTICAL PUBLIC



In practice the chosen solution is most likely to be a combination of both strategies to suite the site layout and access, and any existing buildings. In smaller hospitals, it may also be that there is in fact very little choice in terms of the solutions that can be put forward.

The sketch below makes use of vertical distribution for FM deliveries with a back of house corridor at ground floor, whereas the public is taken to a central core of lifts and distributed on each individual floor. The central lift core ensures that lift usage is maximised, which can reduce the number of lifts required compared to the vertical solution.

VERTICAL FM / HORIZONTAL PUBLIC



The choice of distribution strategy has a fundamental impact on where service spaces are located. For the horizontal distribution, FM spaces are located on each individual floor, as in the example given above, and link into each department on each floor across the corridor on that floor. In the vertical distribution, FM spaces are centralised on level A and are located remotely from the areas they serve.

This should not have an impact on the level of service provided and response times, but changes the interaction between ward/department staff and FM support staff. The vertical distribution option should also reduce the FM space required as functions are centralised. There are clearly a number of alternatives as to what spaces can be centralised and others that should be physically close to the various departments and each needs to be reviewed to ensure the most suitable solution is selected.

Automation

Various types of automation have been used in the UK healthcare industry for several years, but the

NHS has shied away from some types because of the payback periods involved and capital investment. The use of different procurement routes such as PFI, where a contractor has say a 25 year contract to run the building, bring new opportunities to look at automation and understand the benefits.

The descriptions below give a very broad-brush view of payback periods and need to be estimated for each specific project. However the other drivers for using automation should not be underestimated, these are:

- Improvement of service provision
- Better use of existing staff
- Reduction in the wear and tear of the building
- 24 hour availability

The improvements in service provision are typically centred on increased speed of response and flexibility. A single test tube can be transported several hundred meters within minutes and without a second thought to the need for manpower, with results coming straight back once ready. Similarly dirty linen and waste can be collected during the night when the hospital is quite and re-scheduled within minutes without having to re-organise staffing. The shift of goods portering functions from man based to automated

also means that porters' interface can be moved from dealing with deliveries to patients. This not only moves resources to dealing with patients and the public, but also creates a much more rewarding working environment.

A benefit of automation that is harder to quantify is the reduction in wear and tear. The movement of trolleys, cages, etc around corridors and in lifts creates considerable damage to finishes and transferring these movements to automated solutions can considerably reduce the impact of FM activities. Other benefits are the reduction in lift usage and subsequent reduction in wear and tear of the mechanical drives, car finishes, etc. Below is a short description of different types of automation that can be used for the distribution of goods.

Small bore hole pneumatic tube systems

These systems are commonly used in the UK and abroad. They basically consist of a PVC tube network that links a number of the hospital departments and into which carriers can be inserted for the distribution of pharmaceutical goods, specimens, Xrays, documents, etc. The movement of the carriers is achieved by one or more fans pressurising various runs of the tube network and either blowing or sucking air to move the carrier. The tubes are typically 110mm, 160mm, 200mm or 300mm in diameter depending on what needs to be distributed, however the smaller system is the most common in the UK.

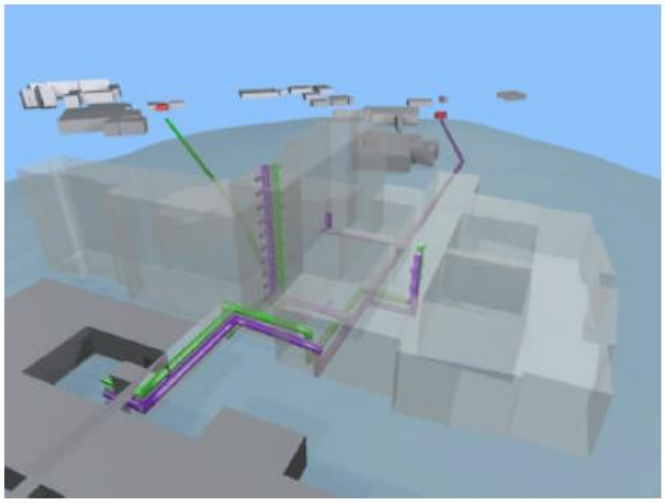
The design of pneumatic tubes is well understood, however hospitals sometimes complain about the waiting time of a carrier before it is actually sent. This is normally caused by bottlenecks in the system, usually at the bridge between two or more zones, or simply because there is insufficient spare capacity on the system. The capacity is limited by the number of fans (or zones) available, as one fan is able to push or pull one item at a time, two fans can move two items at a time and so on. Therefore if the fans are busy, any new carrier has to wait until the system has cleared any backlog and is free again. Most manufacturers now produce linear transfer stations, also called linear coupler servers, that can move carriers from one tube network to another and also store them when the network is very busy. As a rule of thumb, these should be considered if three or more zones are required. It is also normal to design a network to run at 50 to 70% of its capacity, which should ensure maximum waiting times of 5 to 10 minutes.

A simple system will cost approximately £3,750 per station. So for a hospital with say 16 stations (i.e. 10 wards, pharmacy, pathology unit, A&E, theatres, etc) on two zones, the cost will be in the region of £60,000. If it is used to replace three daily bulk deliveries from each station taking 15minutes each, the system is saving 12 man-hours a day, which means a possible pay back period of 2 years. This is extremely crude and the pneumatic tube system should hopefully provide a much better service than the equivalent of three bulk deliveries every day. Payback periods are however reasonably easy to calculate and short enough to justify the usage of these systems in most large NHS developments.

Intergartion of this system with an automated pharmacy dispensing unit & optimisation of the use of the dispensing system?

Large bore hole pneumatic tube systems

These systems are very similar to the ones described above but are, as the name suggests, larger and can therefore transport bulkier items. They are typically used to transport dirty linen and waste from various departments to central collection points that can be up to a mile away, and are therefore particularly suitable to either very large sites or high buildings. There is, as far as we are aware, not a single system in operation in a UK hospital despite this being reasonably old technology regularly used in airports and high-rise residential buildings. Arup are involved with two installations in Ireland and feedback from the users is so far very positive.



These systems are considerable more expensive than their smaller counterparts and, for example, a unit taking waste and linen from 16 areas (i.e. two networks running in parallel) to a central collection point with around a mile of pipe-work will be in the region of £450,000 to £500,000. If the FM team are expected to collect 3 waste containers and two linen cages from each area twice a day, then with a 15 minute travel time the system is saving approximately 40 man hours a day. The payback period is therefore much longer than the small diameter pneumatic systems, around 5 years in this example, which may be too long term for NHS trusts to consider their usage. Such payback periods would however be perfectly reasonable with longer service agreements, such as on PFI projects, and it is anticipated that new procurement routes will encourage their use within new developments.

Automated Guided Vehicles

AGVs are small robotic platforms approximately 400mm wide, 1500mm long and 300mm high that can lift and carry a variety of containers to pre-set destinations. They are programmed to carry out particular tasks at defined times, such as collecting waste containers and delivering catering boxes, and use the same corridors as staff, as they are fitted with anti-collision protection to ensure they do not collide with other moving objects or people. There are several types of guidance systems but the least obtrusive to install into existing buildings use laser technology that can locate walls and obstructions, which enables them to determine where they are and where they are going. AGVs will also call for a lift and request a floor ensuring that any point accessible through corridors and lifts can be reached. Doors also need to be fitted with actuators to ensure the unit can open them when approaching.

There are, as far as we are aware, no AGVs currently used in hospitals in the UK, although they are common in hospitals in the US, Germany, etc, and are used in many other industries such as manufacturing and aviation. Their main drawback is the payback period, which typically ranges between 5 to 7 years. Manufacturers do quote amortisation

periods as low as 3 years, but these estimates clearly need to be done project by project. As a guide, the CAPEX for a large system using 20 vehicles to distribute around an 800 to 900-bed hospital will be around £2 to £2.5m including containers. Another important factor is the ease with which they can be introduced to existing developments. As noted previously, they can be mixed with staff and are able to control lifts, but routes need to be clearly defined and they may not be ideal in developments where buildings are at different levels accessed via stairs and corridors are consistently blocked by doors. A further item that needs to be resolved early on is the type of container that will be moved. AGVs can usually deal with the weights involved in FM deliveries but having a wide variety of containers can be an issue. The AGV has a preset dimension and the choice of containers needs to be suitable to fit under the AGV so that it can be moved. Most AGVs can be modified to take other containers than those specified by the manufacturer but there is limited flexibility. The rationalisation of container sizes is probably a good thing in any logistics review, but the affects need to be understood throughout the supply chain (i.e. why have clean linen delivered in cages if it needs to be transferred to something else once at the hospital?).

Monorails and track vehicle systems

Track systems exist in many different formats ranging from the smaller units used for document handling and hospital distribution (typically around 30kg) to the systems used in manufacturing that can handle loads up to 500kg (some industrial units will move higher loads). The smaller track systems are slower than pneumatic tube systems and generally more cumbersome because of the space requirements. Their main advantage is their capacity to transport goods of different sizes in one container horizontally and vertically, making them suitable for constant flows of larger deliveries.

For individual projects, the choice will come down to either the monorail or the pneumatic tube systems and different diameters of pipe need to be reviewed to compare like for like. It may be that a mixed solution is the most suitable with the monorail used to deliver the larger containers between particular departments.

The larger monorail systems are marketed as being suitable to distribute FM containers for catering and linen. These systems need to be planned in at the early stages of design because they take up considerable space and do interfere with people movement if high ceilings are not available. As with the smaller systems they need to be reviewed against AGVs when looking at automation options, however their impact on the building infrastructure is a considerable disadvantage.

Pharmaceutical storage and dispensing systems

There are typically three different types of storage and retrieval systems: mini loaders, vertical carousels and dispensers. A mini loader is a small version of the Automated Storage and Retrieval Systems (ASRS) found in distribution centres and warehouses. Systems vary between manufacturers but typically products are stored and retrieved by a two-axis robot mounted on a rail system that operates between two rows of shelving. The tight shelving means that product density is maximized and use of the robot ensures fast

picking (450 to 900 products / hour). The automated storage software will also deal with channel allocation, stock management (including expiry date) and lot tracking.

Vertical carousels consist of vertical shelving units that rotate around a conveyor belt to ensure that the shelf with the product required is located at the picking face. They usually come with various module heights to suite the quantity of items than need to be stored. Pick-to-Light displays highlight the item that is requested so that the operator does not have to check all the labels available on one shelf. As with the mini loader, software is available for full stock control.

Dispensers use the same principle as the mini loader but items are stored vertically within a sealed container. One unit should be able to dispense approximately 300 units every hour, with a storage capacity dependent on the height available. They are usually smaller than the min loaders and therefore suite slightly different applications. Again they are able to provide full stock control.

The choice of storage and dispensing system needs to be based on the required storage, the number of dispensing points to be covered and the handling speed. Once these parameters have been defined, it should be easy to make a comparison between the various options.

Exception Management

Exceptions are special cases that deviate from the normal behaviour in a business process and need to be cared for exceptionally, normally by human intervention. Their cause might include: process deviation, malformed data, infrastructure or connectivity issues, poor quality business rules, etc. **Exception management** is the practice of investigating, resolving and handling such occurrences by using skilled staff and software tools. Good exception management can contribute to efficiency of business processes.

Need:- Logically, any company attempting this does so to alleviate perceived problems with recurring errors, things like rework, slow cycle times and information defects. The theory is that a formulaic approach will drive down process variability with benefits similar to how a standard tooling process will drive down defects and scrap in manufacturing. Unfortunately, this is not without its costs. Scalability is a critical component to most successful organizations. In many ways, the same could be said for individual consultants and their respective consulting firms. However, in order to effectively scale and drive revenue, a consultant needs to possess an excellent foundation in project management principles and techniques.

Exceptional project management skills help a consultant avoid being in a constant state of "catch up" and instead focus on managing an engagement, anticipating risk, resolving issues, and refining his/her ever-expanding library of best practices. It is these exact best practices that will help a consulting firm win business over a competitor during a competitive RFP process. While it is not always the case, a firm that is able to convey value, experience, and credibility through repeated success will end up well-positioned in good and not so good economic periods.

BLOOD BANK:-

Blood transfusion is an essential therapeutic intervention. We all may need blood in an emergency, and some of us need regular transfusions. The purpose of a transfusion is to provide the blood component(s) that will improve the physiological status of the patient.

Various blood components can be harvested from a single donation of whole blood. Most blood banks are able to separate red cells and plasma components. Others are able to prepare components such as platelet concentrates and cryoprecipitate. All these components, prepared by centrifugation, are often referred to as 'wet or labile products'. Other plasma products, generally referred to as plasma derivatives, can be harvested from plasma by a pharmaceutical process called plasma fractionation, which renders their properties stable. The collection of blood from donors may take place within the blood transfusion centre or hospital blood bank. It is also often collected from donors during mobile blood collection sessions. The blood is then taken to a laboratory for testing and processing into components and for storage and distribution as the need arises.

Safe storage of blood:-

Whole blood and red cells must always be stored at a temperature between +2 °C and +6 °C. The main reasons for giving a blood transfusion are to restore or help to maintain the body's oxygen-carrying capacity and the volume of blood circulating around the body. If blood is not stored at between +2 °C and +6 °C, its oxygen-carrying ability is greatly reduced. The anticoagulant/preservative solution in the blood bag contains nutrients for the blood during storage and stops the blood from clotting. The red cells can only carry and deliver oxygen if they remain viable: that is, if they retain the same properties as they have during their normal circulation in the body. The most important substances in maintaining the viability of red cells are glucose and adenosine triphosphate (ATP).

Packing and transportation of blood and blood components

An efficient system must be in place to ensure that all blood and blood components shipped by or received into a blood bank or blood transfusion service have been maintained within the correct temperature ranges. Red blood cell components must be kept at a temperature of +2 °C to +10 °C during transportation. All components routinely stored at +20 °C to +24 °C should be kept at these temperatures during shipment. All frozen components should be transported in a manner to maintain their frozen state. The transit time for blood and blood components should not normally exceed 24 hours Whole blood and packed red cells must always be stored at +2 °C to +6 °C and transported between +2 °C and +10 °C.

- Blood components and plasma derivatives should never be stored in unmonitored equipment.
- Red cells, platelets or whole blood must never be allowed to freeze.
- The optimal storage temperature for conditions for fresh frozen plasma and cryoprecipitate is –30 °C, and they must always be frozen solid. They can be stored at lower temperatures, but must never be warmer than –20 °C.
- Platelets must be stored at +20 °C to +24 °C with constant agitation and transported at temperatures within this range.
- During transportation, frozen components must be maintained at a temperature that ensures they will remain frozen.
- It is important to use a temperature monitor during transportation in order to check temperature ranges on receipt of the shipment.

- To assist the maintenance of temperatures for blood components, it is often useful for hospital wards to possess a refrigerator for short-term storage of issued blood from the blood bank.

A database is a collection of persistent data that is used by the application systems of a given enterprise. Traditional databases are organized by fields, records, and files. A field is a single piece of information; a record is one complete set of fields; and a file is a collection of records. To access information from a database, we need a database management system (DBMS). This is a collection of programs that enables the service providers and clients to enter, organize, and select data in a database. Presently in Bangladesh 113 Safe blood transfusion centers are providing safe blood to the clients. The effective management of blood transfusion program encompasses a good number of areas and one of the areas is development of data base for the safe blood transfusion centers and also for the donors. The development of a data base always needed good documentation.

Objectives of Data base

• To develop and maintain an appropriate integrated blood donor tracking database system for the efficient and effective recording and management of blood donor data and blood donor retention

• To significantly improve the quality of recording and management of information about blood donors to facilitate the effective tracking of repeat blood donors and the establishment of a reliable pool of regular repeat blood donors

• To significantly improve the accuracy, efficiency and effectiveness of tracking information on blood donations, and ensure blood safety through accurate labeling and identification of blood units at every stage

• To ensure sustainability through capacity building, staff skills training and the integration of plan and operations.

• Obtain the best available information on blood transfusion services in the country

• Assess the country situation on blood safety

• Monitor trends and progress

• Identify problems and needs in order to provide appropriate technical assistance

• Identify the areas and issues for providing support.

Data base software of Blood bank transfusion system

The Blood Bank Transfusion System consists of seven separate but interrelated application software modules:

• Blood Processing

• Patient Processing

• Inventory Management

• Recipient History

• Reports

• Purge Processing

• File Maintenance

Blood donation, also called blood banking, refers to the process of collecting, testing, preparing, and storing whole blood and blood components intended primarily for

transfusion. Blood registry refers to the collection and sharing of data about donated blood and donors. Donors who have been determined to be temporarily or permanently ineligible to donate blood are listed in a confidential national data base known as the Donor Deferral Register. A possible definition is that a database is a collection of records stored in a computer in a systematic way, so that a computer program can consult it to answer questions. For better retrieval and sorting, each record is usually organized as a set of data elements (facts). The items retrieved in answer to queries become information that can be used to make decisions. The computer program used to manage and query a database is known as a database management system (DBMS). The central concept of a database is that of a collection of records, or pieces of knowledge. Typically, for a given database, there is a structural description of the type of facts held in that database: this description is known as a schema. The schema describes the objects that are represented in the database, and the relationships among them. There are a number of different ways of organizing a schema, that is, of modeling the database structure: these are known as database models (or data models). Strictly speaking, the term database refers to the collection of related records, and the software should be referred to as the database management system or DBMS.

Blood Bank Management Software

Blood Bank Management Software, readily scalable and adaptable to meet the complex need of Blood Banks Who are Key Facilitator for the Healthcare Sector, it also supports all the functionalities of Blood Bank

Features of Blood Bank Management Software

• Generating reports on Stocks-Blood Group wise, Area wise and Expiry date wise.
• Donor Database-Blood Group wise and Area wise
• Maintain and update Unique Donor Identifications.
• Complete Key Consumables Inventory Management.
• Track and maintain all the Donor Types-Voluntary, Exchange and Directed.
• Improve the Effectiveness and efficiency of Blood Bank-Faster Response Time and Better Control
• Accurate database/Record Management.
• Blood Cross Match and Result Storage Facility.
• Digital Record archival backup and restoring facility-Better Housekeeping and Record Maintenance.
• Rejected Donor Database for Donor Control and Identification-Blood Transfusion related disease control and prevention.
• Searched Facility for Destroyed and Expired Blood.
• Comprehensive Donor database with Search Facility.
• Unique Donor ID and Patient record ID for managing future list.
• Improve Blood Bank processes by providing efficient and continuous software support

A comprehensive application software called **Blood Transfusion Management System** is required to implement by incorporating blood screening data, blood donor profile, valid documentation of laboratory testing , schedule for regular blood donation and motivational camp, schedule for training program, management of routine blood supply, monitoring of transfusion hazard, quality control of blood and its product, procurement and finance related activities. By establishing networking system between the centers will

enhance optimum use of information and data exchange to oversee, monitor and evaluate the quality of the services of the centers from a National Reference Center and dissemination of collective information worldwide through Website on Safe blood Transfusion Program of Bangladesh.

Software features

To fulfill the entire need of computer based operation, the following application modules to be developed and implemented in each computer c enter of the department :

• Donors Health Profile Management Module
• Recipient Health Profile Management Module
• Blood Screening Data Management Module
• Cross-Matching Data Management Module
• Medicine Inventory Control Management Module
• Blood Stock Inventory Management Module
• Check-List Management Module
• Investigation and Surgical Management Module
• And other module as required

VARIOUS TYPES OF LAUNDRY SERVICES

Laundry and Linen are one of the important supportive services in a hospital. The word Laundry is derived from Launderer/ Laundress which literally means washerman or washerwoman. Hospital linen means clothing made up of cotton, wool, synthetic fibers. Linen is comprised of basic fabrics which are spun into yarn and finally woven into cloth.

Hospital having its own laundry or in – plant system.

Justified and applicable in large hospitals, teaching hospitals as it is very expensive. In this system hospital has its own linen and laundry and all activities of laundry services like

- Collection
- Washing
- Pressing
- Distribution Marking
- Drying
- Folding
- Controlling Sorting
- Ironing
- Storing

Are done in hospital laundry and premises.

Rental Linen Supply System:

a) In this hospital hire laundered linen from the contractor
b) Contractor is responsible for everything i.e laundering of patient and staff linen as well as replacement.
c) Hospital does not have to spend on inventory of Linen.

Contractual System:

a) Hospitals own their linen but no means of laundering.
b) Washerman or Dhobi in contract takes the linen and after laundering brings back.

c) If hospital is providing the platform for washing area and water a subsidized contract type of system can be introduced.

Co-operative System:

a) When a group of small hospitals pool together and adopt a single laundry cooperative system.

b) Economical if adopted for small government hospitals also as they share services of highly qualified laundry manager/ supervisor, back up by modern machineries and automation.

c) Can evolve common policies of purchase, supplies, maintenance of standardization of linen after pooling their resources.

PURPOSES OF AMBULANCE SERVICES:-

Early detection

Members of the public, or another agency, find the incident and understand the problem

Early reporting

The first persons on scene make a call to the emergency medical services and provide details to enable a response to be mounted

Early response

The first professional (EMS) rescuers arrive on scene as quickly as possible, enabling care to begin

Good on-scene care

The emergency medical service provides appropriate and timely interventions to treat the patient at the scene of the incident

Care in transitthe emergency medical service load the patient in to suitable transport and continue to provide appropriate medical care throughout the journey

Transfer to definitive care

the patient is handed over to an appropriate care setting, such as the emergency department at a hospital, in to the care of physicians

Treatment of material losses, obsolete, surplus and scrap management

Surplus: Surplus stocks are those materials which are procured and stocked in excess of an organization's production and operation requirement. Surplus stock arises because of unplanned purchase, lack of coordination between production, maintenance and materials department or wasteful processes in production and unscientific inventory management.

Scrap, Obsolete And Waste

Scrap: Scraps are generated through production waste or as product of operations in the production shop floor through the production system.

Obsolete: Because of technological changes and advancement every organization may have to adopt the latest operational practice or production method. In such case any material, part, component, assembly, sub assembly required to be kept in the sore to meet up the demand of consumption for previous product is called ***obsolete***.

Waste: Waste is known as discarded substances having no value or very low value

Reasons of Waste

1) ***Changes in product design:*** These may lead to some items getting invalid so far as the final product is concerned. Hence, the entire stocks of such items become obsolete.

2) ***Rationalization:*** Sometimes raw materials rationalized to minimize variety and simplify procurement. The rationalization process renders some items as surplus or obsolete.
3) ***Cannibalization:*** When machine breakdown occurs, sometimes it is rectified using parts of an identical machine which is not functioning due to various reasons. This process is called cannibalization. It is common in many Industries.

4) **Faulty planning and forecasting:** the marketing department may have a projected a sales forecast which might be on higher side. It causes obsolete.

5) ***Faulty purchase practices:*** Decisions like buying in bulk to take care of discount and transportation economy without taking in to account factors such as shelf life, storage space requirement and technological changes once again lead to the accumulation of surplus and obsolete stock.
6) ***Other causes:*** Many items are held as insurable spares for many years without any consumption. **Faulty store-keeping**, **without adequate preservation**, lead to spoilage. Improper codification, poor manufacturing methods and inferior materials handling are also result in obsolete, surplus and scrap item.

Rules of Disposal of scrap:

Movement Analysis:

- The process of combining the stock records and movement analysis has been found very effective in locating such stocks in the total inventory.

Disposal of Scrap:

- Disposal of scrap when handled in an imaginative manner can result in handsome returns to the organization.
- Continuous market survey on the prices of various categories of scrap generated in the plant is necessary
- Disposal action follows when the scrap cannot be utilized within the organization.
- In practice, it is profitable to dispose the scrap directly to end-users rather than to middlemen.

Management of Waste, scraps, and obsolete:

Waste is known as discarded substances having no value or very low value. Waste may arise due to the inherent nature of materials, chemical reaction, evaporation drying, sublimation of goods etc.

Wastes are also be in the form of smoke, gas slag or dust which arises in the course are manufacturing process. The waste may be visible or non-visible.

The waste can be:

(a) Normal waste and (b) Abnormal waste

a) Normal Waste: It is due to natural causes, it cannot be checked only it can be minimized by exercising strict control. The effect of such waste is to reduce the quantity of output and to calculate the cost per unit of the output.

b) Abnormal Waste: Any loss caused by unexpected or abnormal conditions such as sub-standard materials, carelessness, accident etc. or loss in excess of the margin anticipated for normal process loss should be regarded as abnormal waste.

Scrap of materials and scrap at works (factory): Scrap is discarded material having some values. It represents fragments or remnants of material that are left from certain type of manufacture

There are three types of scrap, namely **(a) legitimate scrap, (b) administrative scrap**, **(c) defective scrap**.

a) *Legitimate scrap* arises due to the nature of operation like turning, boring, punching, etc. This type of scrap can be pre-determined and efforts should be made that it should not be more than the pre-determined quantity.
b) *Administrative scrap* arises due to administrative action, such as, a change in the method of production.

c) *Defective scrap* arises because of use of inferior quality of material or bad workmanship or defective machines. Such type of scrap is abnormal because it arises due to abnormal reasons.

Treatment of Scrap: The useful methods for the treatment of scrap are as follows:

1. When realizable value of normal scrap is insignificant (i.e., legitimating scrap administrative scrap) it may be credited to Profit and Loss Accounts like other income. This method is not suitable effective control over scrap because

The sale value of scrap may be deducted from the cost of material consumed or factory overhead

Defective output: Defective products or units are those which do not meet with dimensional or quality standards and are reworked for rectification of defects by application of material, labour and/or processing and salvaged to the point of either standard product or sub-standard product to be sold as seconds. So defectives are that portion which can be rectified at some extra cost of re-operation. Defectives may arise due to the following reasons:

a) Sub-standard materials.
b) Poor workmanship.
c) Poor maintenance of machines.
d) Wrong tool setting.
e) Faulty design of products.
f) Bad supervision.
g) Careless inspection.
h) Poor working conditions.
i) Lack of control, such as humidity, furnace temperature etc.
j) Excessive short runs.

Treatment of Cost of Rectification of Defectives: The following methods may be adopted for the treatment of this cost:

When the defective production is identified with a specific job or department, the cost of rectification is charged to that specific job or department.

Spoilage of Production: Spoilage refers to production that does not meet with dimensional or quality standards in such a way that it cannot be rectified economically and is junked and sold for a disposal value. So it occurs when goods are so damaged in course of manufacturing process as to become not rectifiable

Treatment of cost of Spoilage: The treatment of cost of spoilage depends upon the nature of spoilage. If the spoilage is normal, the cost is borne by good units of output. In case of abnormal spoilage, cost of spoilage is transferred to Costing Profit and Loss Acount

Control of Wastage, Scrap, Defectives and Spoilage: Every effort whould be made to reduce the cost of production by exercising control on wastage, scrap, defectives and spoilage. The following steps may be taken in this direction.

Biomedical Waste

Introduction: The waste produced in the hospital in the course of health care activities carries a higher potential for infection and injury than any other type of waste. Therefore, it is essential to have safe and reliable method for its handling. Inadequate and inappropriate handling of health care or bio-medical waste may have serious public health consequence and significant impact on the environment. Hospital generates 75-90 percent of non-hazardous general waste and rest 10-25 percent is the hazardous waste generated by hospital.

Definition: According to bio-medical waste (management and handling) rules, 1998 of India, Bio-medical waste means any waste, which is generated during the diagnosis, treatment or immunization of human beings or animals or research activities pertaining thereto or in the production or testing of biological, and including categories mentioned in schedule-I.

Categories of biomedical waste: Hazardous, toxic and bio-medical waste should be segregate into following categories for the purpose of its safe transport in a site and fir specific treatment or disposal. The categories are as follows:

1. **Human Anatomical Waste:** This contains human tissues, organs, body parts etc.
2. **Animal waste:** This contains animal tissues, organs, body parts, bleeding parts, fluid, blood and experimental animal used research. This waste generated in veterinary hospital and animal house.
3. **Microbiological and biotechnological waste:** This contains waste from laboratory culture, stocks and specimens of microorganism live or vaccines, human or animal cell culture used in research.
4. **Sharp waste:** This contains needles, syringes, scalpels, blades, glass, etc
5. **Discarded medicine and cytotoxic waste:** This contains waste comprising of out date, contaminated and discarded medicine.
6. **Soiled waste:** this contains item contaminated with blood and body fluids, including cotton, dressing, soiled plaster casts, linens, bedding and other material contaminated with blood.
7. **Solid waste:** This contains waste generated from disposable items, other than the waste sharps, such as tubing, catheters, intravenous sets, etc.
8. **Liquid waste:** this contains waste generated from laboratory and washing, cleaning, housekeeping and disinfecting agents.
9. **Incineration waste:** this contains ash from incineration of any biomedical waste.
10. **Chemical waste:** this contains chemicals used in production of biological, and chemical used in disinfection and insecticides, etc.

Collection, segregation, storage and transportation of biomedical waste:

A. **Segregation of waste:** Segregation of biomedical waste is very important part of hospital waste management department. The operation of segregation consist:

1. It should be done at the sources of generation of biomedical waste, e.g. the entire patient care active area; diagnostics services area Operation Theater, labor room, treatment room, etc.
2. Responsibility of Segregation should be with the generator of biomedical waste i.e. doctors, technicians etc.
3. Bio-medical waste shall be segregated into color coded containers / bags at the point of generation.
4. Whenever possible, biomedical waste must not be mixed with chemical, radioactive or other laboratory trash. This may be unavoidable (i.e. radioactive e carcasses) and in such instances special handling may be required.
5. The various types of biomedical waste should be segregated from each other.
6. Fluid waste should be contained separately from solid waste

B. **Collection of Biomedical Waste**:

Collection of biomedical waste varies for different services or departments depending upon waste generation practices, available resources or management approaches. Collection of biomedical waste should be done as per biomedical waste (management and handling) rules 1998. However, a separate container with color code shall be placed at every point of generation. The trolleys, which are used to collect hospital waste, should be designed in such a way that there should be no leakage or spillage of biomedical waste while transporting to designated site.

Type of container and color code for collection of biomedical waste

Sl	Category	Type of container	Colour coding	Treatment/ disposal
1	Human anatomical waste	Plastic bag	Yellow	incineration\ deep burial
2	Animal	Plastic bag	Yellow	incineration\ deep burial
3	Microbiological and biotechnology waste	Plastic bag	Yellow\ red	Local autoclaving, microwaving, incineration
4	Waste sharps	Plastic bag, puncture proof container	Blue\white\translucent	Disinfection(chemical treatment, autoclaving, microwaving, mutilation/shredding
5	Discarded medicines and cytotoxic waste	Plastic bag	black	Incineration, destruction & drugs disposal in secured land fills
6	Solid waste (soiled)	Plastic bag, puncture proof container	Yellow\ red	Incineration, autoclaving, microwaving
7	Solid waste (plastic)	Plastic bag	Blue\white\translucent	Disinfection by chemical treatment, autoclaving, microwaving, mutilation/shredding
8	Liquid waste	Plastic bag		Disinfection by chemical treatment and discharge into drains
9	Incineration ash	Plastic bag	black	Disposal in municipal land fill
10	Chemical waste(solid)	Plastic bag	black	Chemical treatment and discharge into drains for liquids and secured land fill for solids.

C. **Storage of waste:** Storage refers to the holding biomedical waste for a certain period of time. Although biomedical waste should be treated as promptly as possible it can be held temporarily. Treatable waste should not be allowed to accumulate. Waste that is to be disposed off-site should be stored in designated areas which are secure and access is limited to delegated individuals. Guide line for storage of biomedical waste are:

1. No untreated biomedical waste shall be kept or stored beyond a period of 48 hours.
2. The authorized person must take the permission of the prescribed authority, if for any reason; it becomes necessary to store the waste beyond 48 hours.
3. The authorized person should take measures to ensure that the waste does not adversely after human health and the environment, in case; it is kept beyond the prescribed limit.

D. **Transportation:** after collection, segregation and storage biomedical waste needs to transfer to disposal site. There are two type of transport, like: 1) transportation within the hospital and 2) outside the hospital. The parameters of transport are as follows:

1. Transportation within the hospital:

- Within hospital, waste routes must be designed to avoid the passage of waste through patient-care areas as far as possible.
- Separate time should be fixed for transportation of biomedical waste to reduce changes of its mixing up with general waste as far as possible.

- Dedicated wheeled containers, trolleys or cars should be used to transport the waste bins\ plastic bags to the site of storage/ treatment.

- Trolleys or carts should be thoroughly cleaned and disinfected in the event of any spillage.

- The wheeled containers should be designed in such a way that the waste can be easily loaded, remains secured during transportation, does not have any sharp edges and easy to clean and disinfect.

2. Transportation of clinical waste treatment or disposal site outside the hospital:

- Untreated biomedical waste shall be transported only in such vehicles as may be authorize for the purpose by the competent authorities as specified by government under the motor vehicle act 1988.

- The counters for transportation must be leveled as given in schedule-III and IV of BMW 1998.

Conclusion: The hospital waste, in addition to the posing risk to the patients and personnel who handle these wastes, is also a threat to the public health and environment. It is emerging as a health hazard to the community at large. Keeping in view, inappropriate management of biomedical wastes, the Ministry of Environment and Forests notified the "Bio Medical Waste (Management and Handling) Rules 1998." These rules are meant to protect the society, patients and health care workers. The most imperative component of the waste management plans is to develop a system and culture through education, training and persistent motivation of the health care staff.

Why is safe disposal of biomedical waste important?

Introduction: The problem of bio-medical waste disposal in the hospitals and other healthcare establishments has become an issue of increasing concern, prompting hospital administration to seek new ways of scientific, safe and cost effective management of the waste, and keeping their personnel informed about the advances in this area. The need of proper hospital waste management system is of prime importance and is an essential component of quality assurance. The safe disposal of biomedical waste important because exposure to hazardous health care waste can result in disease or injury due to one or more of the following characteristics:

a) It contains infectious agents, b) it contains toxic and hazardous chemicals or pharmaceuticals c) it contains sharps, d) it is genotoxic, e) it is radioactive.

Unscientific handling of bio-medical waste is the risk for the groups like:

1. **Medical staff:** doctors, nurses, sanitary staff and hospital maintenance personnel;
2. **In and out patients** receiving treatment in healthcare facilities.
3. **Visitors** of hospitals.
4. **Workers in support services** linked to healthcare facilities such as laundries, waste handling and transportation services.
5. **Workers in waste disposal facilities**, including scavengers.
6. **The general public** and more specifically the children playing with the items they can find in the waste outside the healthcare facilities when it is directly accessible to them.

Objectives of safe disposal of Bio Medical Waste Management are:

1. Inappropriate treatment and disposal of bio-medical waste contributes to environmental pollution. To prevent environmental pollution safe disposal of Bio Medical Waste important.
2. To prevent transmission of disease from patient to patient, from patient to health worker and vice versa.
3. To prevent injury to the health care worker and workers and workers in support services, while handling biomedical waste.
4. To prevent general exposure to the harmful effects of the cytotoxic, genotoxic and chemical biomedical waste.
5. The proper bio-medical waste management will help to control nosocomial diseases (hospital acquired infections)
6. Reduces HIV/AIDS, sepsis, and hepatitis transmission from dirty needles and other improperly cleaned / disposed medical items
7. Prevent illegal repackaging and resale of contaminated needles, cut cycles of infection and avoid negative long-term health effects like cancer.
8. To maintain healthy public health safe disposal of Bio Medical Waste needed.

Conclusion: From the above discussion it is found that improper management of biomedical waste has serious effect on environment, occupational and public health and to maintain the hygiene and health environment, occupational and public health safe disposal of biomedical waste is very important.

Disposal process: Answers is in K. Park or Madhuri Sharmas support and utility

1. All the items sent to incineration\ deep burial (categories 1,2,3 and6) should be placed in yellow coloured bags.
2. All the biomedical treatment should be placed in red coloured bags.
3. Any waste, which is sent to shredder after autoclaving/microwaving\chemical treatment, is to be packed in blue\white translucent bag.
4. Location of containers: all containers having different coloured plastic bags should be located at the point of generation of waste i.e. near OT tables, injection room, diagnostic service areas containers\plastic bags used for collection of segregated biomedical waste should be identifiable.
5. Labeling: all the bags\ containers must be labeled according to the rules (schedule-III of biomedical waste rules, 1998).
6. Bags: it should be ensured that waste bags are filled up to three-fourth capacity, tied securely and removed from the site of generation regularly and timely.
7. Certain categories of waste, which may need pre-treatment (decontamination\ disinfection) at the site of generation such as plastic and sharp materials. Etc. should be removed from the site of generation only after treatment.
8. The process of collection should be documented in a register. The coloured plastic bag should be replaced and garbage bin should be cleaned with disinfectant regularly.

Classification of hospital waste

1. **General waste:** Largely composed of domestic or house hold type waste. It is non-hazardous to human beings, e.g. kitchen waste, packaging material, paper, wrappers, and plastics.
2. **Pathological waste:** Consists of tissue, organ, body part, human foetuses, blood and body fluid. It is hazardous waste.
3. **Infectious waste:** The wastes which contain pathogens in sufficient concentration or quantity that could cause diseases. It is hazardous e.g. culture and stocks of infectious agents from laboratories, waste from surgery, waste originating from infectious patients.
4. **Sharps:** Waste materials which could cause the person handling it, a cut or puncture of skin e.g. needles, broken glass, saws, nail, blades, scalpels.
5. **Pharmaceutical waste:** This includes pharmaceutical products, drugs, and chemicals that have been returned from wards, have been spilled, are outdated, or contaminated.
6. **Chemical waste:** This comprises discarded solid, liquid and gaseous chemicals e.g. cleaning, house keeping, and disinfecting product.
7. **Genotoxic Waste:** waste containing substance with genotosic properties e.g. waste containing cytostoxic durgs and genotoxic chemicals.
8. **Pressurised container waste:** gas cylinders, gas cartridges, aerosol cans, etc.
9. **Radioactive waste:** It includes solid, liquid, and gaseous waste that is contaminated with radionucleides generated from in-vitro analysis of body tissues and fluid, in-vivo body organ imaging and tumour localization and therapeutic procedures.